

Vector Algebra

Question1

Let $\hat{i} - 2\hat{j} + \hat{k}$, $\hat{i} + \hat{j} - 2\hat{k}$, $2\hat{i} - \hat{j} - \hat{k}$ and $\hat{i} + \hat{j} + \hat{k}$ be the position vectors of four points A, B, C and D respectively. If a point P divides AB in the ratio $2 : 1$ internally and a point Q divides CD in the ratio $1 : 2$ externally, then the ratio in which the point with position vectors $5\hat{i} - 6\hat{j} - 5\hat{k}$ divides PQ is

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Options:

A.

$2 : 1$

B.

$-2 : 1$

C.

$2 : 3$

D.

$-2 : 3$

Answer: B

Solution:

The position vectors are given as

$$a = \hat{i} - 2\hat{j} + \hat{k}, b = \hat{i} + \hat{j} - 2\hat{k}$$

$$c = 2\hat{i} - \hat{j} - \hat{k}, d = \hat{i} + \hat{j} + \hat{k}$$

$\therefore P$ divides AB internally at the ratio $2 : 1$

$$\begin{aligned}\therefore P &= \frac{2b + a}{2 + 1} = \frac{2(\hat{i} + \hat{j} - 2\hat{k}) + (\hat{i} - 2\hat{j} + \hat{k})}{3} \\ &= \hat{i} - \hat{k}\end{aligned}$$

and Q divides CD externally in the $1 : 2$

$$\begin{aligned}\therefore Q &= \frac{d - 2c}{1 - 2} = \frac{(\hat{i} + \hat{j} + \hat{k}) - 2(2\hat{i} - \hat{j} - \hat{k})}{-1} \\ &= 3(\hat{i} - \hat{j} - \hat{k})\end{aligned}$$

Let $\mathbf{r} = 5\hat{i} - 6\hat{j} - 5\hat{k}$ divides PQ in ratio $m : n$

$$\text{then } \mathbf{r} = \frac{n(\hat{i} - \hat{k}) + m(3\hat{i} - 3\hat{j} - 3\hat{k})}{m + n}$$

$$= 5\hat{\mathbf{i}} - 6\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$$

$$\text{then, } \frac{n+3m}{m+n} = 5, \frac{-3m}{m+n} = -6$$

$$\frac{-n-3m}{m+n} = -5$$

By solving these equations

We get, $m : n = -2 : 1$

Hence, the ratio in which R divides PQ is $-2 : 1$

Question2

If $\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$, $\mathbf{b} = 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$ are two vectors such that $\mathbf{r} \times \mathbf{a} = \mathbf{b} \times \mathbf{a}$ & $\mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$, then the unit vector in the direction of $\mathbf{r}$ is

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Options:

A.

$$\frac{1}{\sqrt{11}}(\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}})$$

B.

$$\frac{1}{\sqrt{11}}(\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}})$$

C.

$$\frac{1}{\sqrt{3}}(\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})$$

D.

$$\frac{1}{\sqrt{3}}(\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}})$$

Answer: A

Solution:

$$\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}} \text{ and } \mathbf{b} = 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$\mathbf{a} = \mathbf{b} \times \mathbf{a} \text{ and } \mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$$

So, $\mathbf{r} = \mathbf{a} + \mathbf{b}$ satisfies both equations Thus $\mathbf{r} = \mathbf{a} + \mathbf{b} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$

$$= \hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$\text{then, } |\mathbf{r}| = \sqrt{1^2 + 3^2 + (-1)^2} = \sqrt{11}$$

Therefore, the unit vector in the direction of

$$\mathbf{r} = \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{1}{\sqrt{11}}(\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}})$$

Question3

If $\mathbf{a} \cdot \mathbf{b} \cdot \mathbf{c}$ are three units vectors such that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \frac{\sqrt{3}}{2}\mathbf{b} + \frac{\mathbf{c}}{2}$ and α, β are the angles between $\mathbf{a}, \mathbf{c}$ and $\mathbf{a}, \mathbf{b}$ respectively, then $\alpha + \beta =$

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Options:

A.

$$\frac{\pi}{2}$$

B.

$$\frac{7\pi}{6}$$

C.

$$\frac{\pi}{6}$$

D.

$$\frac{5\pi}{6}$$

Answer: D

Solution:

$$\because \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \frac{\sqrt{3}}{2}\mathbf{b} + \frac{\mathbf{c}}{2}$$

$$\text{and } \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$$

$$\text{so, } (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} = \frac{\sqrt{3}}{2}\mathbf{b} + \frac{\mathbf{c}}{2}$$

$$\text{therefore, } \mathbf{a} \cdot \mathbf{c} = \frac{\sqrt{3}}{2}$$

Which gives $\alpha = \pi/6$ because $\mathbf{a}$ and $\mathbf{c}$ are unit vector

$$\text{and } \mathbf{a} \cdot \mathbf{b} = -\frac{1}{2}$$

$$\text{which gives } \beta = \frac{2\pi}{3}$$

$$\text{Hence, } \alpha + \beta = \frac{\pi}{6} + \frac{2\pi}{3} = \frac{5\pi}{6}$$

Question4

Let $(x, y) \in R \times R$ and $\mathbf{a} = x\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}, \mathbf{b} = 6\hat{\mathbf{i}} - y\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ be two vectors. If

$$|\mathbf{a} \times \mathbf{b}|^2 + |\mathbf{a} \cdot \mathbf{b}|^2 = f(x)g(y), \text{ then } f(x) + g(y) - 46 = 0$$

represents

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Options:

A.

a pair of line

B.

an ellipse

C.

a hyperbola

D.

a circle

Answer: D

Solution:

Step 1: List the Vectors

The two vectors are:

$$\mathbf{a} = x\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$\mathbf{b} = 6\hat{\mathbf{i}} - y\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

Step 2: Use the Given Relationship

We are given that:

$$|\mathbf{a} \times \mathbf{b}|^2 + |\mathbf{a} \cdot \mathbf{b}|^2 = f(x)g(y)$$

Step 3: Recognize the Identity

There is a formula: $|\mathbf{a} \times \mathbf{b}|^2 + |\mathbf{a} \cdot \mathbf{b}|^2 = |\mathbf{a}|^2 \cdot |\mathbf{b}|^2$

Step 4: Find the Lengths of the Vectors

For $\mathbf{a}$:

$$|\mathbf{a}| = \sqrt{x^2 + 2^2 + (-1)^2} = \sqrt{x^2 + 4 + 1} = \sqrt{x^2 + 5}$$

For $\mathbf{b}$:

$$|\mathbf{b}| = \sqrt{6^2 + y^2 + 2^2} = \sqrt{36 + y^2 + 4} = \sqrt{y^2 + 40}$$

Step 5: Substitute in the Formula

$$|\mathbf{a}|^2 \cdot |\mathbf{b}|^2 = (x^2 + 5)(y^2 + 40)$$

$$\text{So, } f(x)g(y) = (x^2 + 5)(y^2 + 40)$$

Step 6: Identify $f(x)$ and $g(y)$

$$\text{So, } f(x) = x^2 + 5$$

$$g(y) = y^2 + 40$$

Step 7: Use the Given Equation

$$f(x) + g(y) - 46 = 0$$

Step 8: Substitute for $f(x)$ and $g(y)$

$$x^2 + 5 + y^2 + 40 - 46 = 0$$

$$x^2 + y^2 - 1 = 0$$

Step 9: Final Form

So the equation is:

$$x^2 + y^2 = 1$$

Question 5

$\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are the position vectors of three non-collinear points on a plane. If

$\alpha = [\mathbf{abc}]$ and $\mathbf{r} = \mathbf{a} \times \mathbf{b} - \mathbf{c} \times \mathbf{b} - \mathbf{a} \times \mathbf{c}$, then $\left| \frac{\alpha}{\mathbf{r}} \right|$

represents

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Options:

A.

Ratio of areas of the triangles formed by $\mathbf{0}, \mathbf{a}, \mathbf{b}$ to $\mathbf{0}, \mathbf{b}, \mathbf{c}$

B.

Ratio of the numerical values of volume of the parallelepiped formed with $\mathbf{0}, \mathbf{a}, \mathbf{b}, \mathbf{c}$ and its height

C.

Ratio of lengths of the diagonals of the parallelepiped formed with $\mathbf{0}, \mathbf{a}, \mathbf{b}, \mathbf{c}$

D.

Length of the perpendicular from origin to the plane

Answer: D

Solution:

We have, $\mathbf{r} = \mathbf{a} \times \mathbf{b} - \mathbf{c} \times \mathbf{b} - \mathbf{a} \times \mathbf{c}$

$$\Rightarrow \mathbf{r} = \mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}$$

$$\Rightarrow \mathbf{r} \cdot \mathbf{a} = \alpha \cdot \mathbf{1}, \mathbf{r} \cdot \mathbf{b} = \alpha, \mathbf{r} \cdot \mathbf{c} = \alpha$$

Equation of plane passing through $A(\mathbf{a})$, $B(\mathbf{b})$ and $C(\mathbf{c})$ is

$$n = AB \times BC$$

$$= (\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{b}) = \mathbf{b} \times \mathbf{c} - \mathbf{a} \times \mathbf{c} + \mathbf{a} \times \mathbf{b}$$

$$= \mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} = \mathbf{r}$$

Equation of plane

$$(\mathbf{r}_1 - \mathbf{a}) \cdot \mathbf{n} = 0$$

$$\Rightarrow \mathbf{r}_1 \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} \Rightarrow \mathbf{r}_1 \cdot \mathbf{r} = \mathbf{a} \cdot \mathbf{r}$$

$$\mathbf{r}_1 \cdot \mathbf{r} = \alpha$$

$$\mathbf{r}_1 \cdot \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{|\alpha|}{|\mathbf{r}|} \Rightarrow \mathbf{r} \cdot \hat{\mathbf{n}} = p$$

$\therefore \frac{|\alpha|}{|\mathbf{r}|}$ is perpendicular distance from origin to plane.

Question6

If $P = (\mathbf{a} \times \hat{\mathbf{i}})^2 + (\mathbf{a} \times \hat{\mathbf{j}})^2 + (\mathbf{a} \times \hat{\mathbf{k}})^2$ and $Q = (\mathbf{a} \cdot \hat{\mathbf{i}})^2 + (\mathbf{a} \cdot \hat{\mathbf{j}})^2 + (\mathbf{a} \cdot \hat{\mathbf{k}})^2$, then

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Options:

A.

$$P = Q$$

B.

$$P = 2Q$$

C.

$$P = 3Q$$

D.

$$P = 4Q$$

Answer: B

Solution:

$$\text{Let } \mathbf{a} = a\hat{\mathbf{i}} + b\hat{\mathbf{j}} + c\hat{\mathbf{k}}$$

$$\therefore \mathbf{a} \times \hat{\mathbf{i}} = -b\hat{\mathbf{k}} + c\hat{\mathbf{j}} \Rightarrow \mathbf{a} \times \hat{\mathbf{j}} = a\hat{\mathbf{k}} - c\hat{\mathbf{i}}$$

$$\Rightarrow \mathbf{a} \times \hat{\mathbf{k}} = -a\hat{\mathbf{j}} + b\hat{\mathbf{i}}$$

$$\therefore (\mathbf{a} \times \hat{\mathbf{i}})^2 = b^2 + c^2$$

$$(\mathbf{a} \times \hat{\mathbf{j}})^2 = a^2 + c^2$$

$$(\mathbf{a} \times \hat{\mathbf{k}})^2 = a^2 + b^2$$

$$\therefore P = 2(a^2 + b^2 + c^2)$$

$$\text{Now, } Q = a^2 + b^2 + c^2$$

$$\therefore P = 2Q$$

Question7

$\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$, $\mathbf{c} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$ are three vectors. If $\mathbf{r}$ is a vector such that $\mathbf{r} \cdot \mathbf{a} = 0$, $\mathbf{r} \cdot \mathbf{c} = 3$ and $\begin{vmatrix} \mathbf{r} & \mathbf{a} \\ \mathbf{b} & \end{vmatrix} = 0$, then $|\mathbf{r}| =$

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Options:

A.

$$\sqrt{2}$$

B.

$$\sqrt{3}$$

C.

$$3$$

D.

$$7$$

Answer: A

Solution:

A Given,

$$\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}, \mathbf{b} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}},$$

$$\mathbf{c} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{r} \cdot \mathbf{a} = 0 \Rightarrow [\mathbf{r} \cdot \mathbf{ab}] = 0 \Rightarrow \mathbf{r} \cdot (\mathbf{a} \times \mathbf{b}) = 0$$

$$\therefore \mathbf{r} = \lambda(\mathbf{a} \times (\mathbf{a} \times \mathbf{b}))$$

$$= \lambda[(\mathbf{a} \cdot \mathbf{b})\mathbf{a} - \mathbf{a} \cdot \mathbf{a}(\mathbf{b})]$$

$$= \lambda(9\mathbf{a} - 6\mathbf{b}) = 3\lambda(3\mathbf{a} - 2\mathbf{b})$$

$$\mathbf{r} = k(3\mathbf{a} - 2\mathbf{b})$$

$$\mathbf{r} \cdot \mathbf{c} = k(3\mathbf{a} \cdot \mathbf{c} - 2\mathbf{b} \cdot \mathbf{c})$$

$$\Rightarrow 3 = k[3(2 - (-2) - 2(2 - 2 - 3))]$$

$$\Rightarrow 3 = k(-3 + 6) \Rightarrow k = 7$$

$$\therefore \mathbf{r} = \hat{\mathbf{i}} - \hat{\mathbf{j}} \Rightarrow |\mathbf{r}| = \sqrt{2}$$

Question8

In a right angled triangle, if the position vector of the vertex having the right angle is $-3\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and the position vector of the mid-point of its hypotenuse is $6\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$, then the position vector of its centroid is

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Options:

A.

$$3\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$$

B.

$$3\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

C.

$$\frac{3\hat{i}+7\hat{j}+7\hat{k}}{2}$$

D.

$$4\hat{j} + 3\hat{k}$$

Answer: A

Solution:

The position vector of **a** is

$$\mathbf{a} = -3\hat{i} + 5\hat{j} + 2\hat{k}$$

Let, M be the mid-point of the hypotenuse BC .

So, the position vector of **M** is

$$\mathbf{m} = 6\hat{i} + 2\hat{j} + 5\hat{k}$$

Let the centroid be G and its position vector be **g**.

$$\text{So, } \mathbf{g} = \frac{\mathbf{a}+\mathbf{b}+\mathbf{c}}{3} \quad \dots (i)$$

Since, the mid-point of the hypotenuse of a right triangle is equidistant from all three vertices.

$$\text{So, } M \text{ is the circumcenter of a triangle and } \mathbf{m} = \frac{\mathbf{b}+\mathbf{c}}{2}$$

$$\Rightarrow \mathbf{b} + \mathbf{c} = 2\mathbf{m}$$

Substituting this value in Eq. (i), we get

$$\begin{aligned} g &= \frac{\mathbf{a} + 2\mathbf{m}}{3} \\ \Rightarrow g &= \frac{(-3\hat{i} + 5\hat{j} + 2\hat{k}) + 2(6\hat{i} + 2\hat{j} + 5\hat{k})}{3} \\ &= \frac{-3\hat{i} + 5\hat{j} + 2\hat{k} + 12\hat{i} + 4\hat{j} + 10\hat{k}}{3} \\ &= \frac{9\hat{i} + 9\hat{j} + 12\hat{k}}{3} \\ &= 3\hat{i} + 3\hat{j} + 4\hat{k} \end{aligned}$$

So, the position vector of centroid is

$$3\hat{i} + 3\hat{j} + 4\hat{k}$$

Question9

If the position vectors of the vertices A, B, C of a triangle are

$3\hat{i} + 4\hat{j} - \hat{k}, \hat{i} + 3\hat{j} + \hat{k}, 5(\hat{i} + \hat{j} + \hat{k})$ respectively, then the magnitude of the altitude drawn from A on to the side BC is

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Options:

A.

$$\frac{4\sqrt{5}}{3}$$

B.

$$\frac{5\sqrt{5}}{3}$$

C.

$$\frac{7\sqrt{5}}{3}$$

D.

$$\frac{8\sqrt{5}}{3}$$

Answer: A

Solution:

Given,

$$\mathbf{A} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - \hat{\mathbf{k}}, \mathbf{B} = \hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \hat{\mathbf{k}} \text{ and}$$

$$\mathbf{C} = 5\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$$

$$\text{So, } \mathbf{BC} = \mathbf{C} - \mathbf{B}$$

$$= 5\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 5\hat{\mathbf{k}} - \hat{\mathbf{i}} - 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$= 4\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$$

$$\mathbf{BA} = \mathbf{A} - \mathbf{B} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - \hat{\mathbf{k}} - \hat{\mathbf{i}} - 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$= 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$$

$$\text{Now, } \mathbf{BA} \times \mathbf{BC} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & 1 & -2 \\ 4 & 2 & 4 \end{vmatrix}$$

$$= (4 + 4)\hat{\mathbf{i}} - (8 + 8)\hat{\mathbf{j}} + (4 - 4)\hat{\mathbf{k}}$$

$$= 8\hat{\mathbf{i}} - 16\hat{\mathbf{j}}$$

$$\text{And } |\mathbf{BA} \times \mathbf{BC}| = \sqrt{8^2 + (-16)^2}$$

$$= \sqrt{64 + 256}$$

$$= \sqrt{320} = 8\sqrt{5}$$

$$\text{So, area of triangle} = \frac{1}{2} |\mathbf{BA} \times \mathbf{BC}|$$

$$= \frac{1}{2} \times 8\sqrt{5} = 4\sqrt{5}$$

$$\text{Now, } |\mathbf{BC}| = \sqrt{4^2 + 2^2 + 4^2}$$

$$= \sqrt{16 + 4 + 16} = \sqrt{36} = 6$$

Let, the magnitude of the altitude h from A to BC is given by

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\Rightarrow 4\sqrt{5} = \frac{1}{2} \times |BC| \times h$$

$$\Rightarrow 4\sqrt{5} = \frac{1}{2} \times 6 \times h$$

$$\Rightarrow h = \frac{4\sqrt{5}}{3}$$

Question10

If the vectors $2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$, $-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ and $p\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$ are coplanar, then the unit vector in the direction of the vector $9p\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$ is

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Options:

A.

$$\frac{1}{6}(2\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}})$$

B.

$$\frac{1}{\sqrt{57}}(5\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}})$$

C.

$$\frac{1}{\sqrt{68}}(6\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}})$$

D.

$$\frac{1}{9}(-7\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}})$$

Answer: D

Solution:

$$\text{Let } \mathbf{a} = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 3\hat{\mathbf{k}},$$

$$\mathbf{b} = -\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}} \text{ and } \mathbf{c} = p\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\text{Now, } \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} 2 & 4 & -3 \\ -1 & 2 & 3 \\ p & -2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 2(2 + 6) - 4(-1 - 3p) - 3(2 - 2p) = 0$$

$$\Rightarrow 16 + 4 + 12p - 6 + 6p = 0$$

$$\Rightarrow 18p + 14 = 0$$

$$\Rightarrow p = \frac{-14}{18} = \frac{-7}{9}$$

$$\text{Now, } \mathbf{v} = 9p\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$$

$$= 9\left(\frac{-7}{9}\right)\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$$

$$= -7\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$$

Now, unit vector

$$\hat{\mathbf{u}} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{-7\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}}}{\sqrt{(-7)^2 + (-4)^2 + 4^2}}$$

$$\Rightarrow \hat{\mathbf{u}} = \frac{-7\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}}}{9}$$

$$= \frac{-7}{9}\hat{\mathbf{i}} - \frac{4}{9}\hat{\mathbf{j}} + \frac{4}{9}\hat{\mathbf{k}}$$

Question11

Let $\mathbf{a} = 4\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$ and $\mathbf{b}$ be two perpendicular vectors in the XOY -plane. A vector $\mathbf{c}$ in the same plane and having projections 1 and 2 respectively on $\mathbf{a}$ and $\mathbf{b}$ is

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Options:

A.

$$\hat{i} + 2\hat{j}$$

B.

$$2\hat{i} + \hat{j}$$

C.

$$\hat{i} - 2\hat{j}$$

D.

$$2\hat{i} - \hat{j}$$

Answer: D

Solution:

Let $\mathbf{b} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$ and $\mathbf{a}$ is perpendicular to $\mathbf{b}$.

$$\therefore \mathbf{a} \cdot \mathbf{b} = 0$$

$$\Rightarrow (4\hat{\mathbf{i}} + 3\hat{\mathbf{j}}) \cdot (x\hat{\mathbf{i}} + y\hat{\mathbf{j}}) = 0$$

$$\Rightarrow 4x + 3y = 0$$

$$\Rightarrow \frac{x}{3} = \frac{y}{-4} = \lambda$$

$$\Rightarrow x = 3\lambda, y = -4\lambda$$

$$\therefore \mathbf{b} = \lambda(3\hat{\mathbf{i}} - 4\hat{\mathbf{j}})$$

Let the required vector be $\mathbf{c} = c_1\hat{\mathbf{i}} + c_2\hat{\mathbf{j}}$ then the projection of $\mathbf{c}$ on $\mathbf{a}$ and $\mathbf{b}$ are $\frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{a}|}$ and $\frac{\mathbf{c} \cdot \mathbf{b}}{|\mathbf{b}|}$ respectively.

$$\therefore \frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{a}|} = 1 \text{ and } \frac{\mathbf{c} \cdot \mathbf{b}}{|\mathbf{b}|} = 2$$

$$\Rightarrow \frac{(c_1\hat{\mathbf{i}} + c_2\hat{\mathbf{j}}) \cdot (4\hat{\mathbf{i}} + 3\hat{\mathbf{j}})}{\sqrt{4^2 + 3^2}} = 1$$

$$\Rightarrow 4c_1 + 3c_2 = 5 \quad \dots (i)$$

$$\text{and } \frac{\mathbf{c} \cdot \mathbf{b}}{|\mathbf{b}|} = 2$$

$$\Rightarrow \frac{(c_1\hat{\mathbf{i}} + c_2\hat{\mathbf{j}}) \cdot (3\hat{\mathbf{i}} - 4\hat{\mathbf{j}})}{\sqrt{3^2 + (-4)^2}} = 2$$

$$\Rightarrow 3c_1 - 4c_2 = 10 \quad \dots (ii)$$

Solving Eq. (i) and (ii), we get

$$c_1 = 2, c_2 = -1$$

So, the required vector

$$= c_1\hat{\mathbf{i}} + c_2\hat{\mathbf{j}} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}}$$

Question12

If $\mathbf{a} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ and $\mathbf{b} = -\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ are two vectors, then the vector of magnitude 28 units in the direction of the vector $\mathbf{a} - \mathbf{b}$ is

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Options:

A.

$$3\hat{\mathbf{i}} + 6\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$$

B.

$$12\hat{\mathbf{i}} - 24\hat{\mathbf{j}} + 8\hat{\mathbf{k}}$$

C.

$$3\hat{\mathbf{i}} - 6\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$$

D.

$$12\hat{\mathbf{i}} + 24\hat{\mathbf{j}} - 8\hat{\mathbf{k}}$$

Answer: B

Solution:

$$\mathbf{a} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}, \mathbf{b} = -\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

$$\text{So, } \mathbf{a} - \mathbf{b} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}} + \hat{\mathbf{i}} - 3\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$$

$$= 3\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

$$\text{and } |\mathbf{a} - \mathbf{b}| = \sqrt{3^2 + (-6)^2 + 2^2} = 7$$

Unit vector in $(\mathbf{a} - \mathbf{b})$

$$= \frac{1}{7}(3\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$$

thus, the vector of magnitude 28 units

$$= 28 \left(\frac{1}{7}(3\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 2\hat{\mathbf{k}}) \right) = 4(3\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$$

$$= 12\hat{\mathbf{i}} - 24\hat{\mathbf{j}} + 8\hat{\mathbf{k}}$$

Question13

If $\bar{\mathbf{a}}$ is a unit vector, then

$$|\mathbf{a} \times \hat{\mathbf{i}}|^2 + |\mathbf{a} \times \hat{\mathbf{j}}|^2 + |\mathbf{a} \times \hat{\mathbf{k}}|^2 =$$

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Options:

A.

4

B.

1

C.

0

D.

2

Answer: D

Solution:

Let, $\mathbf{a} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$

$$\begin{aligned} \text{then, } \mathbf{a} \times \hat{\mathbf{i}} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ x & y & z \\ 1 & 0 & 0 \end{vmatrix} \\ &= \hat{\mathbf{i}}(0 - 0) - \hat{\mathbf{j}}(0 - z) + \hat{\mathbf{k}}(0 - y) \\ &= z\hat{\mathbf{j}} - y\hat{\mathbf{k}} \end{aligned}$$

So, $|\mathbf{a} \times \hat{\mathbf{i}}|^2 = z^2 + y^2$

Similarly, $|\mathbf{a} \times \hat{\mathbf{j}}|^2 = x^2 + z^2$

and $|\mathbf{a} \times \hat{\mathbf{k}}|^2 = x^2 + y^2$

thus, $|\mathbf{a} \times \hat{\mathbf{i}}|^2 + |\mathbf{a} \times \hat{\mathbf{j}}|^2 + |\mathbf{a} \times \hat{\mathbf{k}}|^2$

$$= 2(x^2 + y^2 + z^2)$$

$\therefore \mathbf{a}$ is unit vector

So, $x^2 + y^2 + z^2 = 1$

Hence, $|\mathbf{a} \times \hat{\mathbf{i}}|^2 + |\mathbf{a} \times \hat{\mathbf{j}}|^2 + |\mathbf{a} \times \hat{\mathbf{k}}|^2$

$$= 2 \times 1 = 2$$

Question14

If $\mathbf{a} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$, $\mathbf{b} = -2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$, $\mathbf{c} = 5\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ and $\mathbf{d} = 3\hat{\mathbf{i}} + \hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ are four vectors, then $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) =$

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Options:

A.

$$18\hat{i} + 6\hat{j} + 30\hat{k}$$

B.

$$8\hat{i} - 3\hat{j} + 8\hat{k}$$

C.

$$19\hat{i} - 5\hat{j} + 21\hat{k}$$

D.

$$27\hat{i} - 8\hat{j} + 29\hat{k}$$

Answer: A

Solution:

$$\begin{aligned}(\mathbf{a} \times \mathbf{b}) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & -3 \\ -2 & 3 & 4 \end{vmatrix} \\&= \hat{i}(-8 + 9) - \hat{j}(4 - 6) + \hat{k}(3 - 4) \\&= \hat{i} + 2\hat{j} - \hat{k} \\ \text{and } (\mathbf{c} \times \mathbf{d}) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & -4 & 3 \\ 3 & 1 & 5 \end{vmatrix} \\&= \hat{i}(-20 - 3) - \hat{j}(25 - 9) + \hat{k}(5 + 12) \\&= -23\hat{i} - 16\hat{j} + 17\hat{k} \\ \text{Thus, } (\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ -23 & -16 & 7 \end{vmatrix} \\&= \hat{i}(34 - 16) - \hat{j}(7 - 23) + \hat{k}(-16 + 4) \\&= 18\hat{i} + 6\hat{j} + 30\hat{k}\end{aligned}$$

Question 15

$3\hat{i} + \hat{j} + \hat{k}$, $2\hat{i} + \hat{k}$, $\hat{i} + 5\hat{j}$ are the position vectors of three non-collinear points A, B, C respectively. If the perpendicular drawn from C onto AB meets AB at the point $a\hat{i} + b\hat{j} + c\hat{k}$, then $a + b + c =$

Options:

A.

5

B.

3

C.

7

D.

9

Answer: C

Solution:

$$\mathbf{AB} = \mathbf{OB} - \mathbf{OA} = (2 - 3)\hat{\mathbf{i}} + (0 - 1)\hat{\mathbf{j}} + (1 - 1)\hat{\mathbf{k}} = -\hat{\mathbf{i}} - \hat{\mathbf{j}}$$

So, the direction vector of $\text{lin } \epsilon AB$ is

$$\mathbf{d} = -\hat{\mathbf{i}} - \hat{\mathbf{j}}$$

$$\text{let } \mathbf{P} = \mathbf{A} + \lambda \mathbf{d}$$

$$= (3\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) + \lambda(-\hat{\mathbf{i}} - \hat{\mathbf{j}})$$

$$= (3 - \lambda)\hat{\mathbf{i}} + (1 - \lambda)\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\text{and } \mathbf{PC} = \mathbf{C} - \mathbf{P} = (\hat{\mathbf{i}} + 5\hat{\mathbf{j}})$$

$$- [(3 - \lambda)\hat{\mathbf{i}} + (1 - \lambda)\hat{\mathbf{j}} + \hat{\mathbf{k}}]$$

$$= (\lambda - 2)\hat{\mathbf{i}} + (4 + \lambda)\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

As, given in the question

$$\therefore \mathbf{PC} \perp \mathbf{AB}$$

$$\Rightarrow \mathbf{PC} \cdot \mathbf{AB} = 0$$

$$\Rightarrow (\lambda - 2)(-1) + (4 + \lambda)(-1) + (-1)(0) = 0$$

$$\Rightarrow -\lambda + 2 - 4 - \lambda = 0$$

$$\Rightarrow \lambda = -1$$

$$\therefore \mathbf{P} = (3 - \lambda)\hat{\mathbf{i}} + (1 - \lambda)\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$= (3 + 1)\hat{\mathbf{i}} + (1 + 1)\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$= 4\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

therefore, $a = 4$, $b = 2$ and $c = 1$

Hence, $a + b + c = 4 + 2 + 1 = 7$

Question16

If the vectors $2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + l\hat{\mathbf{k}}$, $-3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 4l\hat{\mathbf{k}}$ and $\hat{\mathbf{i}} - \hat{\mathbf{j}} + 3/\hat{\mathbf{k}}$ form a right-angled triangle for a positive value of l , then the length of its hypotenuse is

Options:

A.

$$\sqrt{\frac{40}{3}}$$

B.

$$\sqrt{\frac{55}{3}}$$

C.

$$\sqrt{\frac{65}{3}}$$

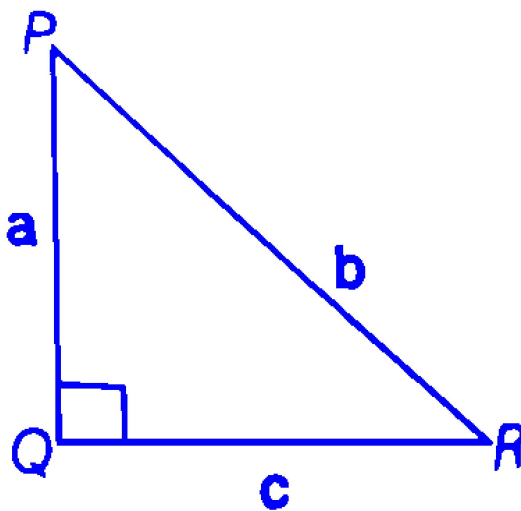
D.

$$\sqrt{\frac{59}{3}}$$

Answer: B

Solution:

$$\begin{aligned}\text{Let } \mathbf{a} &= 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + l\hat{\mathbf{k}} \\ \mathbf{b} &= -3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 4l\hat{\mathbf{k}} \\ \mathbf{c} &= \hat{\mathbf{i}} - \hat{\mathbf{j}} + 3l\hat{\mathbf{k}}\end{aligned}$$



For right angled Δ ;

$$\begin{aligned}|\mathbf{b}|^2 &= |\mathbf{a}|^2 + |\mathbf{c}|^2 \\ \Rightarrow (9 + 4 + 16l^2) &= (4 + 9 + l^2) + (1 + 1 + 9l^2)\end{aligned}$$

$$\Rightarrow l^2 = \frac{1}{3} \Rightarrow l = \pm \frac{1}{\sqrt{3}} \text{ but } l > 0$$

$$\therefore l = \frac{1}{\sqrt{3}}$$

$$\text{So, its hypotenuse } \mathbf{b} = -3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - \frac{4}{\sqrt{3}}\hat{\mathbf{k}}$$

$$\begin{aligned}\therefore |\mathbf{b}| &= \sqrt{9 + 4 + \frac{16}{3}} \\ &= \sqrt{\frac{27 + 12 + 16}{3}} = \sqrt{\frac{55}{3}}\end{aligned}$$

Question17

A unit vector that is perpendicular to the vector $2\hat{i} - \hat{j} + 2\hat{k}$ and coplanar with the vectors $\hat{i} + \hat{j} - \hat{k}$ and $2\hat{i} + 2\hat{j} - \hat{k}$ is

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Options:

A.

$$\frac{\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{6}}$$

B.

$$\frac{3\hat{i} + 2\hat{j} - 2\hat{k}}{\sqrt{17}}$$

C.

$$\frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}$$

D.

$$\frac{3\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{17}}$$

Answer: C

Solution:

Let $\mathbf{a} = 2\hat{i} - \hat{j} + 2\hat{k}$,

$\mathbf{b} = \hat{i} + \hat{j} - \hat{k}$, $\mathbf{c} = 2\hat{i} + 2\hat{j} - \hat{k}$

So, a vector which is perpendicular to $\mathbf{a}$ and coplanar with $\mathbf{b}$ and $\mathbf{c}$ is $\mathbf{b} \times \mathbf{c} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$

$$= (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$$

$$= 0 \cdot \mathbf{b} - (-1)\mathbf{c} = \mathbf{c} = 2\hat{i} + 2\hat{j} - \hat{k}$$

$$\begin{aligned}\text{So, its unit vector} &= \frac{2\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{2^2 + 2^2 + (-1)^2}} \\ &= \frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}\end{aligned}$$

Question18

If the vectors $2\hat{i} - \hat{j} + 3\hat{k}$, $\hat{i} + 4\hat{j} + \hat{k}$, $4\hat{i} + p\hat{j} + \hat{k}$ are coplanar, then $p =$

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Options:

A.

53

B.

37

C.

43

D.

59

Answer: C

Solution:

Since, given vectors are coplanar.

So, its determinant value = 0.

$$\Rightarrow \begin{vmatrix} 2 & -1 & 3 \\ 1 & 4 & 1 \\ 4 & p & 1 \end{vmatrix} = 0$$
$$\Rightarrow 2(4 - p) + 1(1 - 4) + 3(p - 16) = 0$$
$$\Rightarrow p = 43$$

Question19

If the magnitudes of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{a} + \mathbf{b}$ are respectively 3, 4 and 5, then the magnitude of $\mathbf{a} - \mathbf{b}$ is

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Options:

A.

3

B.

4

C.

6

D.

5

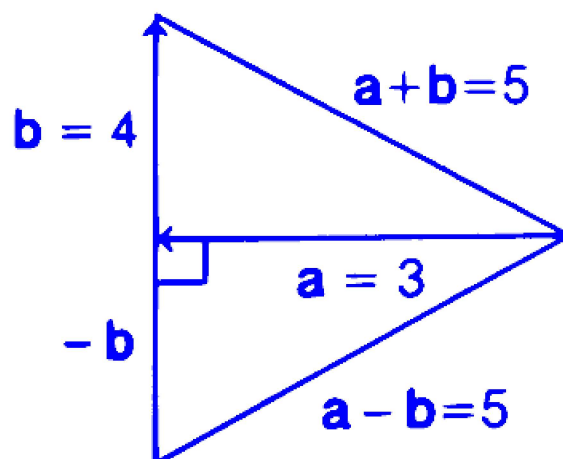
Answer: D

Solution:

(Logically) given that $|\mathbf{a}| = 3$,

$|\mathbf{b}| = 4, |\mathbf{a} + \mathbf{b}| = 5$

So, $|\mathbf{a} - \mathbf{b}| = 5$



Question20

If $\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}, -\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}, \hat{\mathbf{j}} + 2\hat{\mathbf{k}}, 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ are the position vectors of four points A, B, C, D respectively, then the shortest distance between the lines AB and CD is

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Options:

A.

$$\frac{1}{6}$$

B.

$$\frac{7}{3}$$

C.

$$\frac{1}{3}$$

D.

$$\frac{7}{6}$$

Answer: C

Solution:

Direction vector of line AB is

$$\mathbf{AB} = \mathbf{OB} - \mathbf{OA} = (-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}) - (\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}) = -2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

Similarly, direction vector of line CD is

$$\begin{aligned}\mathbf{CD} &= \mathbf{OD} - \mathbf{OC} = (2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}) - (\hat{\mathbf{j}} + 2\hat{\mathbf{k}}) \\ &= 2\hat{\mathbf{i}} - 2\hat{\mathbf{j}}\end{aligned}$$

$$\text{and } \mathbf{AC} = \mathbf{OC} - \mathbf{OA} = -\hat{\mathbf{i}} + 0 \cdot \hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

∴ Shortest distance (S.D)

$$\begin{aligned}&= \left| \frac{\mathbf{AC} \cdot (\mathbf{AB} \times \mathbf{CD})}{|\mathbf{AB} \times \mathbf{CD}|} \right| \\ \mathbf{AB} \times \mathbf{CD} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -2 & 1 & 2 \\ 2 & -2 & 0 \end{vmatrix} \\ &= 4\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}\end{aligned}$$

$$\text{and } \mathbf{AC} \cdot (\mathbf{AB} \times \mathbf{CD}) = 2$$

$$\text{Also, } |\mathbf{AB} \times \mathbf{CD}| = \sqrt{16 + 16 + 4} = 4$$

$$\text{Hence, S.D.} = \left| \frac{2}{4} \right| = \frac{1}{2}$$

Question 21

A line segment PQ has the length 63 and direction ratios $(3, -2, 6)$. If this line makes an obtuse angle with X -axis, then the components of the vector PQ are

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Options:

A.

7, 8, -4

B.

-7, 8, -4

C.

27, -18, 54

D.

-27, 18, -54

Answer: D

Solution:

Logically, the vector must be proportional to $(3, -2, 6)$ with magnitude 63.

So, $PQ = 9(3, -2, 6) = (27, -18, 54) \rightarrow$ Component form.

The line makes an obtuse angle with the X -axis.

∴ Components of the vector PQ are -27, 18, -54

Question22

The points in the argand plane represented by the complex numbers $4\hat{i} + \hat{j} + 3\hat{k}$, $6\hat{i} - 2\hat{j} - 3\hat{k}$ and $\hat{i} - \hat{j} - 3\hat{k}$ form

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Options:

A.

a right-angled triangle

B.

a right-angled isosceles triangle

C.

an equilateral triangle

D.

an isosceles triangle

Answer: D

Solution:

$$4\hat{i} + \hat{j} + 3\hat{k}, 6\hat{i} - 2\hat{j} - 3\hat{k}, \hat{i} - \hat{j} - 3\hat{k}$$

$\Rightarrow$ points are $P(4, 1, 3)$, $Q(6, -2, -3)$ and $R(1, -1, -3)$

$$\begin{aligned}\Rightarrow PQ &= \sqrt{(6-4)^2 + (-3)^2 + (-6)^2} \\ &= \sqrt{4 + 9 + 36} = \sqrt{49} = 7\end{aligned}$$

$$\begin{aligned}\Rightarrow QR &= \sqrt{(1-6)^2 + (-1+2)^2 + (-3+3)^2} \\ &= \sqrt{(-5)^2 + (1)^2 + 0} = \sqrt{26}\end{aligned}$$

$$\begin{aligned}\Rightarrow PR &= \sqrt{(1-4)^2 + (-1-1)^2 + (-3-3)^2} \\ &= \sqrt{9 + 4 + 36} = \sqrt{49} = 7\end{aligned}$$

$\Rightarrow PQ = PR = 7$ units

$\Rightarrow \triangle PQR$ is an isosceles triangle.

Question23

If the vector $\hat{i} - 7\hat{j} + 2\hat{k}$ is along the internal bisector of the angle between the vectors **a** and $-2\hat{i} - \hat{j} + 2\hat{k}$ and the unit vector along **a** is $x\hat{i} + y\hat{j} + z\hat{k}$ then, $x =$

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Options:

A.

$$0$$

B.

$$\frac{7}{9}$$

C.

$$-\frac{1}{9}$$

D.

$$\frac{5}{3}$$

Answer: B

Solution:

Magnitude of the vector $-2\hat{i} - \hat{j} + 2\hat{k}$

$$\Rightarrow |\mathbf{b}| = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

$$\Rightarrow \text{Unit vector } \hat{\mathbf{b}} = \frac{-2\hat{i}}{3} - \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}$$

Now, let $\mathbf{c} = \hat{i} - 7\hat{j} + 2\hat{k}$

$$|\mathbf{c}| = \sqrt{54} = 3\sqrt{6}$$

$$\hat{\mathbf{c}} = \frac{1}{3\sqrt{6}}\hat{i} - \frac{7}{3\sqrt{6}}\hat{j} + \frac{2}{3\sqrt{6}}\hat{k}$$

$$\Rightarrow \mathbf{c} = \mathbf{a} + \mathbf{b} \Rightarrow \mathbf{a} = \mathbf{c} - \mathbf{b}$$

$$= \frac{(1 + 2\sqrt{6})\hat{i}}{3\sqrt{6}} + \frac{(-7 + \sqrt{6})\hat{j}}{3\sqrt{6}} + \frac{(2 - 2\sqrt{6})\hat{k}}{3\sqrt{6}}$$

Compare with $\mathbf{a} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\Rightarrow x = \frac{1 + 2\sqrt{6}}{3\sqrt{6}} = \frac{\sqrt{6} + 12}{18} \cong \frac{7}{9}$$

$$\Rightarrow x = \frac{7}{9}$$

Question 24

If $\mathbf{a} = 2\hat{i} - \hat{j} + 6\hat{k}$; $\mathbf{b} = \hat{i} - \hat{j} + \hat{k}$ and $\mathbf{c} = 3\hat{j} - \hat{k}$, then $\mathbf{a} \times \mathbf{b} \times \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} =$

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Options:

A.

$$20\hat{i} + 3\hat{j} - 4\hat{k}$$

B.

$$20\hat{i} - 3\hat{j} + 4\hat{k}$$

C.

$$3\hat{i} + 20\hat{j} - 4\hat{k}$$

D.

$$4\hat{i} + 20\hat{j} - 3\hat{k}$$

Answer: A

Solution:

$$\mathbf{a} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 6\hat{\mathbf{k}}, \mathbf{b} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}},$$

$$\mathbf{c} = 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & -1 & 6 \\ 1 & -1 & 1 \end{vmatrix}$$

$$= \hat{\mathbf{i}}(-1 + 6) - \hat{\mathbf{j}}(2 - 6) + \hat{\mathbf{k}}(-2 + 1)$$

$$= 5\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & -1 & 1 \\ 0 & 3 & -1 \end{vmatrix}$$

$$= \hat{\mathbf{i}}(1 - 3) - \hat{\mathbf{j}}(-1 - 0) + \hat{\mathbf{k}}(3 - 0)$$

$$= -2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

$$\mathbf{c} \times \mathbf{a} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 3 & -1 \\ 2 & -1 & 6 \end{vmatrix}$$

$$= \hat{\mathbf{i}}(18 - 1) - \hat{\mathbf{j}}(+2) + \hat{\mathbf{k}}(-6)$$

$$= 17\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 6\hat{\mathbf{k}}$$

$$\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}$$

$$= 5\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - \hat{\mathbf{k}} + (-2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 3\hat{\mathbf{k}}) + 17\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 6\hat{\mathbf{k}}$$

$$= 20\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$$

Question25

Let $\mathbf{a} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$ and $\mathbf{b} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$ be two vectors. If $\mathbf{c}$ is a vector such that $\mathbf{a} \cdot \mathbf{c} = |\mathbf{c}|$, $|\mathbf{c} - \mathbf{a}| = 2\sqrt{2}$ and the angle between $\mathbf{a} \times \mathbf{b}$ and $\mathbf{c}$ is 30° , then $|(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}| =$

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Options:

A.

$$\frac{2}{3}$$

B.

$$\frac{3}{2}$$

C.

2

D.

3

Answer: B

Solution:

$$\mathbf{a} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}} \text{ and } \mathbf{b} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$$

$$\mathbf{a} \cdot \mathbf{c} = |\mathbf{c}|, |\mathbf{c} - \mathbf{a}| = 2\sqrt{2}$$

$$\mathbf{a} \times \mathbf{b} \text{ and } \mathbf{c} \text{ is } 30^\circ.$$

$$|(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}| = |\mathbf{a} \times \mathbf{b}| |\mathbf{c}| \sin \theta$$

$$= |\mathbf{a} \times \mathbf{b}| |\mathbf{c}| \times \frac{1}{2}$$

$$\mathbf{a} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}, \mathbf{b} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = \hat{\mathbf{i}}(2) - \hat{\mathbf{j}}(2) + \hat{\mathbf{k}}(1)$$

$$= 2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$|\mathbf{a} \times \mathbf{b}| = \sqrt{4 + 4 + 1} = 3$$

$$\mathbf{a} \cdot \mathbf{c} = |\mathbf{c}|$$

$$\Rightarrow |\mathbf{a}| |\mathbf{c}| \cos \alpha = |\mathbf{c}|$$

$$\cos \alpha = \frac{1}{3}$$

$$|\mathbf{c} - \mathbf{a}| = 2\sqrt{2}$$

$$\Rightarrow |\mathbf{c} - \mathbf{a}|^2 = 8$$

$$\Rightarrow |\mathbf{c}|^2 + |\mathbf{a}|^2 - 2|\mathbf{c}||\mathbf{a}| \cos \alpha = 8$$

$$\Rightarrow |\mathbf{c}|^2 + 9 - 2|\mathbf{c}|3 \times \frac{1}{3} = 8$$

$$\Rightarrow |\mathbf{c}|^2 - 2|\mathbf{c}| + 1 = 0$$

$$\Rightarrow (|\mathbf{c}| - 1)^2 = 0 \Rightarrow |\mathbf{c}| = 1$$

$$|(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}| = |\mathbf{a} \times \mathbf{b}| |\mathbf{c}| \sin \theta$$

$$= 3 \times 1 \times \sin 30^\circ$$

$$= 3 \times 1 \times \frac{1}{2} = \frac{3}{2}$$

Question26

$$(\mathbf{a} + 2\mathbf{b} - \mathbf{c}) \cdot (\mathbf{a} - \mathbf{b}) \times (\mathbf{a} - \mathbf{b} - \mathbf{c}) =$$

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Options:

A.

[abc]

B.

$$3[abc]$$

C.

$$[abc]^2$$

D.

$$2[abc]$$

Answer: B

Solution:

$$\begin{aligned} & (a + 2b - c)\{(a - b) \times (a - b - c)\} \\ &= (a + 2b - c)[a \times a - a \times b - a \times c - b \times a + b \times b + b \times c] \\ &= (\mathbf{a} + 2\mathbf{b} - \mathbf{c})[-\mathbf{a} \times \mathbf{b} - \mathbf{a} \times \mathbf{c} - \mathbf{b} \times \mathbf{a} + \mathbf{b} \times \mathbf{c}] \\ &= (\mathbf{a} + 2\mathbf{b} - \mathbf{c})[\mathbf{b} \times \mathbf{a} + \mathbf{c} \times \mathbf{a} + \mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c}] \\ &= \mathbf{a} \cdot (\mathbf{b} \times \mathbf{a}) + \mathbf{a} \cdot (\mathbf{c} \times \mathbf{a}) + \mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) \\ &\quad + \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \\ &\quad + 2\mathbf{b} \cdot (\mathbf{b} \times \mathbf{a}) + 2\mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) + 2\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) \\ &\quad + 2\mathbf{b} \cdot (\mathbf{b} \times \mathbf{c}) \\ &\quad - \mathbf{c} \cdot (\mathbf{b} \times \mathbf{a}) - \mathbf{c} \cdot (\mathbf{c} \times \mathbf{a}) - \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) - \mathbf{c} \cdot (\mathbf{b} \times \mathbf{c}) \\ &= [\mathbf{abc}] + 2[\mathbf{bca}] - [\mathbf{cba}] - [\mathbf{cab}] \\ &= [\mathbf{abc}] + 2[\mathbf{abc}] + [\mathbf{abc}] - [\mathbf{abc}] = 3[\mathbf{abc}] \end{aligned}$$

Question27

Points P and Q are given by $\mathbf{OP} = \hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}}$ and $\mathbf{OQ} = -\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$. A line along the vector $\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$ passes through the point P and another line along the vector $\mathbf{b} = \hat{\mathbf{j}} - \hat{\mathbf{k}}$ passes through the point Q . If a line along the vector $\mathbf{c} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$ intersects both the lines along the vectors $\mathbf{a}$ and $\mathbf{b}$ at L and M , respectively, then $\mathbf{PM} =$

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Options:

A.

$$\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

B.

$$4\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$$

C.

$$-2\hat{\mathbf{i}} + 10\hat{\mathbf{j}} - 6\hat{\mathbf{k}}$$

D.

$$3\hat{i} - 2\hat{j} + \hat{k}$$

Answer: C

Solution:

Given that

$$OP = \hat{i} - \hat{j} - \hat{k}, OQ = -\hat{i} + \hat{j} + \hat{k}$$

$$\mathbf{a} = \hat{i} + \hat{j}, \mathbf{b} = \hat{j} - \hat{k}$$

$$l_1 = \mathbf{r}_1 = \hat{i} - \hat{j} + \hat{k} + \lambda(\hat{i} + \hat{j}) \quad \dots (i)$$

$$l_2 = \mathbf{r}_2 = -\hat{i} + \hat{j} + \hat{k} + \mu(\hat{j} - \hat{k}) \quad \dots (ii)$$

Let point of intersection of first line.

$$\mathbf{L} = \mathbf{P} + \lambda\mathbf{a} = (1 + \lambda)\hat{i} + (-1 + \lambda)\hat{j} - \hat{k}$$

$$M = \mathbf{Q} + \mu\mathbf{b} = -\hat{i} + (1 + \mu)\hat{j} + (1 - \mu)\hat{k}$$

Line passing through L and M

$$\therefore \mathbf{LM} \parallel \mathbf{C}$$

$$\mathbf{LM} = (-2 - \lambda)\hat{i} + (2 + \mu - \lambda)\hat{j} + (2 - \mu)\hat{k}$$

$$\mathbf{LM} = t(\hat{i} - \hat{j} + \hat{k})$$

$$-2 - \lambda = t \quad \dots (iii)$$

$$\Rightarrow \lambda = -2 - t$$

$$2 + \mu - \lambda = -t \quad \dots (iv)$$

$$2 - \mu = t \quad \dots (v)$$

$$\Rightarrow \mu = 2 - t$$

$$t = -6, \lambda = 4$$

$$\mu = 8$$

$$\mathbf{M} = -\hat{i} + 9\hat{j} - 7\hat{k}$$

$$\mathbf{PM} = -2\hat{i} + 10\hat{j} - 6\hat{k}$$

Question 28

For $a \in R$, if the vectors $\mathbf{p} = (a + 1)\hat{i} + a\hat{j} + a\hat{k}$, $\mathbf{q} = a\hat{i} + (a + 1)\hat{j} + a\hat{k}$ and $\mathbf{r} = a\hat{i} + a\hat{j} + (a + 1)\hat{k}$ are coplanar and $3(\mathbf{p} \cdot \mathbf{q})^2 - \lambda|\mathbf{r} \times \mathbf{q}|^2 = 0$, then the value of λ is

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Options:

A.

$$\frac{2}{3}$$

B.

$$\frac{3}{2}$$

C.

2

D.

1

Answer: D

Solution:

$$\mathbf{p} = (a+1)\hat{\mathbf{i}} + a\hat{\mathbf{j}} + a\hat{\mathbf{k}}$$

$$\mathbf{q} = a\hat{\mathbf{i}} + (a+1)\hat{\mathbf{j}} + a\hat{\mathbf{k}}$$

$$\mathbf{r} = a\hat{\mathbf{i}} + a\hat{\mathbf{j}} + (a+1)\hat{\mathbf{k}}$$

$\mathbf{p}, \mathbf{q}$ and $\mathbf{r}$ are coplanar

$$\begin{vmatrix} a+1 & a & a \\ a & a+1 & a \\ a & a & a+1 \end{vmatrix} = 0$$

$$\Rightarrow (3a+1) \begin{vmatrix} 1 & 1 & 1 \\ a & a+1 & a \\ a & a & a+1 \end{vmatrix} = 0$$

$$\Rightarrow (3a+1) [(a+1)^2 - a^2 - 1(a^2 + a - a^2) + 1(a^2 - a^2 - a)] = 0$$

$$\Rightarrow (3a+1)[2a+1-a-a] \Rightarrow a = -\frac{1}{3}$$

$$\mathbf{p} \cdot \mathbf{q} = -\frac{2}{9} - \frac{2}{9} + \frac{1}{9} = \frac{-3}{9} = \frac{-1}{3}$$

$$\Rightarrow |\mathbf{r} \times \mathbf{q}| = \sqrt{\left(\frac{4}{9} - \frac{1}{9}\right)^2 - \left(\frac{-2}{9} - \frac{1}{9}\right)^2 + \left(\frac{1}{9} + \frac{2}{9}\right)^2}$$

$$= \sqrt{\frac{9-9+9}{81}} = \frac{1}{3}$$

$$3(\mathbf{p} \cdot \mathbf{q})^2 - \lambda |\mathbf{r} \times \mathbf{q}|^2 = 0 \Rightarrow \lambda = 1$$

Question29

If $\mathbf{a} = \hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$, $\mathbf{b} = -2\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$ and $\mathbf{c} = 3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$ are three vectors such that $(\mathbf{b} \times \mathbf{c}) \times \mathbf{a} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$, then $x + y - z =$

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Options:

A.

75

B.

-89

C.

125

D.

-389

Answer: A

Solution:

$$\mathbf{a} = \hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$$

$$\mathbf{b} = -2\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$$

$$\mathbf{c} = 3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$$

$$\Rightarrow (\mathbf{b} \times \mathbf{c}) \times \mathbf{a} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

$$\Rightarrow (\mathbf{b} \cdot \mathbf{a})\mathbf{c} - (\mathbf{c} \cdot \mathbf{a})\mathbf{b} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

$$\Rightarrow (-2 + 20 + 8)\mathbf{c} - (3 - 8 + 16)\mathbf{b}$$

$$= x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

$$\Rightarrow 100\hat{\mathbf{i}} + 107\hat{\mathbf{j}} - 82\hat{\mathbf{k}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

$$\Rightarrow x + y - z = 100 - 107 + 82 = 75$$

Question30

If $A = (0, 4, -3)$, $B = (5, 0, 12)$ and $C = (7, 24, 0)$, then $\sqrt{BAC} =$

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Options:

A.

$$60^\circ$$

B.

$$\cos^{-1}\left(\frac{16}{\sqrt{13}}\right)$$

C.

$$\cos^{-1}\left(\frac{13}{38}\right)$$

D.

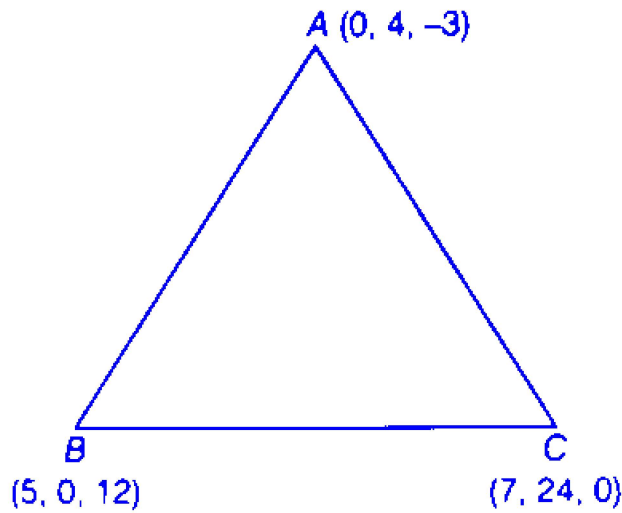
$$90^\circ$$

Answer: D

Solution:

$$\mathbf{AB} = 5\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 15\hat{\mathbf{k}}$$

$$\mathbf{AC} = 7\hat{\mathbf{i}} + 20\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$



$$\cos \theta = \frac{35 - 80 + 45}{|\mathbf{AB}||\mathbf{AC}|} = 0$$

$$\theta = 90^\circ$$

Question31

Let the position vectors of the vertices of a $\triangle ABC$ be $\mathbf{a}, \mathbf{b}, \mathbf{c}$. If on the plane of the triangle, P is a point having position vector $\mathbf{x}$ such that $\mathbf{x} \cdot (\mathbf{c} - \mathbf{b}) = \mathbf{a} \cdot \mathbf{c} - \mathbf{a} \cdot \mathbf{b}$ and $\mathbf{x} \cdot (\mathbf{a} - \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} - \mathbf{b} \cdot \mathbf{c}$, then for the $\triangle ABC$, P is the

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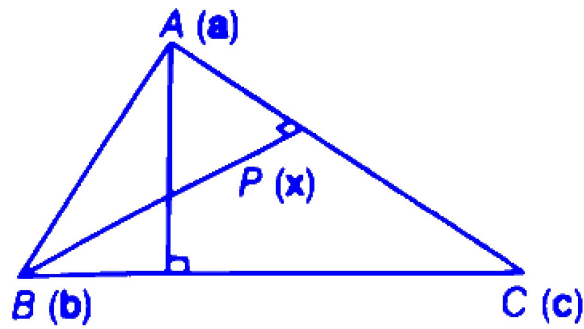
Options:

- A.
Centroid
- B.
Circumcentre
- C.
Incentre
- D.
Orthocentre

Answer: D

Solution:

We have, $\mathbf{x} \cdot (\mathbf{c} - \mathbf{b}) = \mathbf{a} \cdot \mathbf{c} - \mathbf{a} \cdot \mathbf{b}$



$$\begin{aligned}
 &= \mathbf{a}(\mathbf{c} - \mathbf{b}) \\
 (\mathbf{x} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{b}) &= 0 \\
 \mathbf{AP} \cdot \mathbf{BC} &= 0 \\
 \mathbf{AP} &\perp \mathbf{BC} \\
 \text{and } \mathbf{x} \cdot (\mathbf{a} - \mathbf{c}) &= \mathbf{a} \cdot \mathbf{b} - \mathbf{b} \cdot \mathbf{c} = (\mathbf{a} - \mathbf{c}) \cdot \mathbf{b} \\
 \Rightarrow (\mathbf{x} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{c}) &= 0 \\
 \mathbf{BP} \cdot \mathbf{AC} &= 0 \\
 \therefore \mathbf{BP} &\perp \mathbf{AC}
 \end{aligned}$$

$\therefore$ Point of intersection of altitude is called orthocentre.

$\therefore P$ is orthocentre of $\triangle ABC$.

Question32

$\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three vectors such that $|\mathbf{a}| = 2, |\mathbf{b}| = 3, |\mathbf{c}| = 5, |\mathbf{a} + \mathbf{b} + \mathbf{c}| = \sqrt{69}$. If $(\mathbf{a} \cdot \mathbf{b}) = (\mathbf{b} \cdot \mathbf{c}) = \frac{\pi}{3}$, then $(\mathbf{c}, \mathbf{a}) =$

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Options:

A.

$$\frac{\pi}{6}$$

B.

$$\frac{\pi}{4}$$

C.

$$\frac{\pi}{3}$$

D.

$$\frac{\pi}{2}$$

Answer: C

Solution:

Given $|\mathbf{a}| = 2, |\mathbf{b}| = 3, |\mathbf{c}| = 5$

$$(a \cdot b) = \frac{\pi}{3}$$

$$a \cdot b = 3 \text{ and } b \cdot c = \frac{15}{2}$$

$$\text{Given, } |\mathbf{a} + \mathbf{b} + \mathbf{c}| = \sqrt{69}$$

On squaring both sides, we get

$$|\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = 69$$

$$\Rightarrow 4 + 9 + 25 + 2\left(3 + \frac{15}{2} + \mathbf{c} \cdot \mathbf{a}\right) = 69$$

$$\Rightarrow 2\left(\frac{21}{2} + \mathbf{c} \cdot \mathbf{a}\right) = 31$$

$$\Rightarrow 21 + 2 \cdot \mathbf{c} = 31$$

$$\Rightarrow 2(\mathbf{c} \cdot \mathbf{a}) = 10$$

$$\Rightarrow \mathbf{c} \cdot \mathbf{a} = 5$$

$$\Rightarrow |\mathbf{c}| |\mathbf{a}| \cos \theta = 5$$

$$\Rightarrow 5 \times 2 \cos \theta = 5$$

$$\Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

Question33

If the points A, B, C, D with positions vectors

$\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}, \hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}, \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}, 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ respectively form a tetrahedron, then the angle between the faces ABC and ABD of the tetrahedron is

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Options:

A.

$$\cos^{-1}\left(\frac{-4}{\sqrt{29}}\right)$$

B.

$$\cos^{-1}\left(\frac{-4}{5}\right)$$

C.

$$\cos^{-1}\left(\frac{3}{5}\right)$$

D.

$$\cos^{-1}\left(\frac{\sqrt{29}}{\sqrt{33}}\right)$$

Answer: A

Solution:

We have, $\mathbf{OA} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}},$

$$\mathbf{OB} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

$$\mathbf{OC} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{OD} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$$

Let $\mathbf{n}_1$ is the vector normal to the face ABC

$$\therefore \mathbf{n}_1 = \mathbf{AB} \times \mathbf{AC}$$

$$\begin{aligned}\mathbf{n}_1 &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & -2 & 3 \\ 0 & -3 & 2 \end{vmatrix} \\ &= \hat{\mathbf{i}}(-4 + 9) + \hat{\mathbf{j}}(0) + \hat{\mathbf{k}}(0) = 5\hat{\mathbf{i}}\end{aligned}$$

Let $\mathbf{n}_2$ be the vector normal to face ABD

$$\therefore \mathbf{n}_2 = \mathbf{AB} \times \mathbf{AD}$$

$$\mathbf{n}_2 = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & -2 & 3 \\ 1 & 0 & 2 \end{vmatrix} = \hat{\mathbf{i}}(-4) - \hat{\mathbf{j}}(-3) + \hat{\mathbf{k}}(2)$$

$$\mathbf{n}_2 = -4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

Now, angle between $\mathbf{n}_1$ and $\mathbf{n}_2$

$$\begin{aligned}\cos \theta &= \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1| |\mathbf{n}_2|} = \frac{-20}{5\sqrt{29}} \\ \theta &= \cos^{-1} \left(\frac{-4}{\sqrt{29}} \right)\end{aligned}$$

Question34

$\mathbf{a}, \mathbf{b}, \mathbf{c}$ are unit vectors. If $\mathbf{a}, \mathbf{b}$ are perpendicular vectors, $(\mathbf{a} - \mathbf{c}) \cdot (\mathbf{b} + \mathbf{c}) = 0$ and $\mathbf{c} = l\mathbf{a} + m\mathbf{b} + n(\mathbf{a} \times \mathbf{b})$; (l, m, n are scalars), then $n^2 =$

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Options:

A.

$$l^2 + m^2$$

B.

$$-21m$$

C.

$$2l - 2m$$

D.

$$lm + l + m$$

Answer: B

Solution:

Given, $|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = 1$

$$\mathbf{a} \cdot \mathbf{b} = 0$$

$$(\mathbf{a} - \mathbf{c}) \cdot (\mathbf{b} + \mathbf{c}) = 0$$

$$\mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c} - \mathbf{c} \cdot \mathbf{b} - |\mathbf{c}|^2 = 0$$

$$\mathbf{a} \cdot \mathbf{c} - \mathbf{b} \cdot \mathbf{c} = 1$$

$$\text{Also, } \mathbf{c} = l\mathbf{a} + m\mathbf{b} + n(\mathbf{a} \times \mathbf{b})$$

$$\mathbf{a} \cdot \mathbf{c} = l(\mathbf{a} \cdot \mathbf{a}) + m(\mathbf{a} \cdot \mathbf{b}) + n(\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}))$$

$$\mathbf{a} \cdot \mathbf{c} = l$$

$$\text{Similarly, } \mathbf{b} \cdot \mathbf{c} = m$$

$$\therefore l - m = 1 \Rightarrow l^2 + m^2 - 2lm = 1$$

$$\text{Now, } (\mathbf{c} - l\mathbf{a} - m\mathbf{b})^2 = (n(\mathbf{a} \times \mathbf{b}))^2$$

$$\Rightarrow \mathbf{c}^2 + l^2\mathbf{a}^2 + m^2\mathbf{b}^2 - 2m\mathbf{b} \cdot \mathbf{c} + 2lm\mathbf{a} \cdot \mathbf{b} - 2l\mathbf{a} \cdot \mathbf{c} = n^2|\mathbf{a}|^2|\mathbf{b}|^2$$

$$\Rightarrow 1 + l^2 + m^2 - 2m^2 - 2l^2 = n^2$$

$$\Rightarrow 1 - l^2 - m^2 = n^2$$

$$\Rightarrow n^2 = -2lm \quad (\because -2lm = 1 - l^2 - m^2)$$

Question 35

If $O(0, 0, 0)$, $A(1, 2, 1)$, $B(2, 1, 3)$ and $C(-1, 1, 2)$ are the vertices of a tetrahedron, then the acute angle between its face OAB and edge BC is

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Options:

A.

$$\cos^{-1} \left(\frac{6\sqrt{2}}{5\sqrt{7}} \right)$$

B.

$$\sin^{-1} \left(\frac{6\sqrt{2}}{5\sqrt{7}} \right)$$

C.

$$\tan^{-1} \left(\frac{6\sqrt{2}}{5\sqrt{7}} \right)$$

D.

$$\frac{\pi}{2}$$

Answer: B

Solution:

Let $\mathbf{n}$ is the normal to the face OAB

$$\therefore \mathbf{n} = \mathbf{OA} \times \mathbf{OB}$$

$$\mathbf{n} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 2 & 1 \\ 2 & 1 & 3 \end{vmatrix}$$

$$\mathbf{n} = \hat{\mathbf{i}}(5) - \hat{\mathbf{j}}(1) + \hat{\mathbf{k}}(-3)$$

$$\mathbf{n} = 5\hat{\mathbf{i}} - \hat{\mathbf{j}} - 3\hat{\mathbf{k}}$$

Direction ratio of edge $\mathbf{BC} = -3, 0, -1$

Now, angle between $\mathbf{n}$ and edge $\mathbf{BC}$ is

$$\sin \theta = \left| \frac{-15 - 0 + 3}{\sqrt{35}\sqrt{10}} \right| = \frac{-12}{\sqrt{350}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$\therefore \sin \theta = \frac{6\sqrt{2}}{5\sqrt{7}}$$

$$\Rightarrow \theta = \sin^{-1} \left(\frac{6\sqrt{2}}{5\sqrt{7}} \right)$$

Question36

If the angles between the sides of the $\triangle ABC$ formed by $A(2, 3, 5)$, $B(-1, 3, 2)$ and $C(3, 5, -2)$ are α, β and γ , then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma =$

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Options:

A.

1

B.

2

C.

$\frac{3}{2}$

D.

$\frac{1}{2}$

Answer: B

Solution:

$$\mathbf{AB} = -3\hat{\mathbf{i}} - 3\hat{\mathbf{k}}$$

$$\mathbf{BC} = 4\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$$

$$\mathbf{CA} = -\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 7\hat{\mathbf{k}}$$

Given α, β and γ are angles between the sides

$$\begin{aligned}
\therefore \cos \alpha &= \frac{\mathbf{AB} \cdot \mathbf{BC}}{|\mathbf{AB}||\mathbf{BC}|} = 0 \\
\cos \beta &= \frac{\mathbf{BC} \cdot \mathbf{CA}}{|\mathbf{BC}||\mathbf{CA}|} = \frac{-36}{6\sqrt{54}} = -\frac{2}{\sqrt{6}} \\
\cos \gamma &= \frac{\mathbf{CA} \cdot \mathbf{AB}}{|\mathbf{CA}||\mathbf{AB}|} = \frac{-18}{6\sqrt{54}} = \frac{2}{\sqrt{12}} \\
\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma &= \sqrt{1 - \cos^2 \alpha} \\
&\quad + \left(\sqrt{1 - \cos^2 \beta} \right) + \left(\sqrt{1 - \cos^2 \gamma} \right) \\
&= 1 + \frac{2}{6} + \frac{8}{12} = 1 + 1 = 2
\end{aligned}$$

Question 37

Let $2\hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}}$, $5\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$, $-13\hat{\mathbf{i}} - 11\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$ be the position vectors of three points A , B and C , respectively. If $\mathbf{AB} = \lambda \mathbf{BC}$ and $\mathbf{AC} = \mu \mathbf{CB}$, then $\lambda + \mu =$

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Options:

A.

1

B.

-1

C.

2

D.

-2

Answer: B

Solution:

We have, $A(2\hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}})$, $B(5\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}})$ and $C(-13\hat{\mathbf{i}} - 11\hat{\mathbf{j}} + 4\hat{\mathbf{k}})$,

$$\mathbf{AB} = \lambda \mathbf{BC}$$

$$3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}} = \lambda(-18\hat{\mathbf{i}} - 12\hat{\mathbf{j}} + 6\hat{\mathbf{k}})$$

$$\therefore -18\lambda = 3$$

$$\lambda = -\frac{3}{18}$$

Given, $\mathbf{AC} = \mu \mathbf{CB}$

$$-15\hat{\mathbf{i}} - 10\hat{\mathbf{j}} + 5\hat{\mathbf{k}} = \mu(18\hat{\mathbf{i}} + 12\hat{\mathbf{j}} - 6\hat{\mathbf{k}})$$

$$\mu = -\frac{15}{18}$$

$$\therefore \lambda + \mu = -\frac{3}{18} - \frac{15}{18} = -\frac{18}{18} = -1$$

Question38

$\mathbf{a}$, $\mathbf{b}$ are position vectors of the point A and B respectively, C and D are points on the line AB such that $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{BD}$, $\overrightarrow{BA}$ are two pairs of like vectors. If $\overrightarrow{AC} = 3\overrightarrow{AB}$ and $\overrightarrow{BD} = 2\overrightarrow{BA}$, then $\overrightarrow{CD}$

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Options:

A.

$$3\mathbf{b} - 4\mathbf{a}$$

B.

$$4\mathbf{a} - 4\mathbf{b}$$

C.

$$4\mathbf{a} - 3\mathbf{b}$$

D.

$$3\mathbf{b} - 3\mathbf{a}$$

Answer: B

Solution:

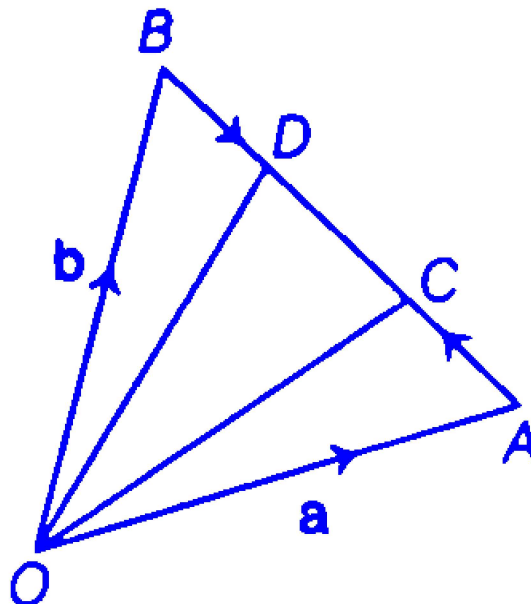
$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA}$$

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC}$$

$$\overrightarrow{OC} = \mathbf{a} + 3\overrightarrow{AB}$$

$$\overrightarrow{OC} = \mathbf{a} + 3(\mathbf{b} - \mathbf{a})$$

$$\overrightarrow{OC} = 3\mathbf{b} - 2\mathbf{a}$$



$$\begin{aligned}
&\text{Similarly, } \mathbf{BD} = \mathbf{OD} - \mathbf{OB} \\
&\mathbf{OD} = \mathbf{OB} + \mathbf{BD} \\
&\quad = \mathbf{b} + 2\mathbf{BA} \\
&\quad = \mathbf{b} + 2(\mathbf{a} - \mathbf{b}) \\
&\mathbf{OD} = 2\mathbf{a} - \mathbf{b} \\
&\mathbf{CD} = \mathbf{OD} - \mathbf{OC} \\
&\quad = (2\mathbf{a} - \mathbf{b}) - (3\mathbf{b} - 2\mathbf{a}) \\
&\mathbf{CD} = 4\mathbf{a} - 4\mathbf{b}
\end{aligned}$$

Question 39

If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three unit vectors such that $|\mathbf{a} - \mathbf{b}|^2 + |\mathbf{b} - \mathbf{c}|^2 + |\mathbf{c} - \mathbf{a}|^2 = 15$, then $|\mathbf{a} - \mathbf{b} - \mathbf{c}|^2 - 4(\mathbf{b} \cdot \mathbf{c}) =$

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Options:

A.

6

B.

15

C.

12

D.

10

Answer: C

Solution:

$$\begin{aligned}
&\text{Given,} \\
&|\mathbf{a} - \mathbf{b}|^2 + |\mathbf{b} - \mathbf{c}|^2 + |\mathbf{c} - \mathbf{a}|^2 = 15 \\
\Rightarrow &|\mathbf{a}|^2 + |\mathbf{b}|^2 - 2\mathbf{a} \cdot \mathbf{b} + |\mathbf{b}|^2 + |\mathbf{c}|^2 \\
&\quad - 2\mathbf{b} \cdot \mathbf{c} + |\mathbf{c}|^2 + |\mathbf{a}|^2 - 2\mathbf{c} \cdot \mathbf{a} = 15 \\
\Rightarrow &6 - 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = 15 \\
\Rightarrow &\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a} = -\frac{9}{2} \\
&|\mathbf{a} - \mathbf{b} - \mathbf{c}|^2 - 4(\mathbf{b} \cdot \mathbf{c}) \\
\Rightarrow &|\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2 \\
&\quad + 2(-\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} - \mathbf{c} \cdot \mathbf{a}) - 4(\mathbf{b} \cdot \mathbf{c}) \\
&= 3 - 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) \\
&= 3 - 2\left(-\frac{9}{2}\right) \\
&= 3 + 9 = 12
\end{aligned}$$

Question40

If $\mathbf{a} = \hat{\mathbf{i}} + p\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$, $\mathbf{b} = p\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{c} = -3\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ are three vectors such that $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a} \times \mathbf{c}|$, then $p =$

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Options:

A.

-2

B.

-1

C.

1

D.

2

Answer: D

Solution:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & p & -3 \\ p & -3 & 1 \end{vmatrix}$$

$$\hat{\mathbf{i}}(p-9) - \hat{\mathbf{j}}(1+3p) + \hat{\mathbf{k}}(-3-p^2)$$

$$|\mathbf{a} \times \mathbf{b}|^2 = (p-9)^2 + (3p+1)^2 + (p^2+3)^2$$

$$\mathbf{a} \times \mathbf{c} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & p & -3 \\ -3 & 1 & 2 \end{vmatrix}$$

$$= \hat{\mathbf{i}}(2p+3) - \hat{\mathbf{j}}(2-9) + \hat{\mathbf{k}}(1+3p)$$

$$|\mathbf{a} \times \mathbf{c}|^2 = (2p+3)^2 + 49 + (3p+1)^2$$

$$\text{Given, } |\mathbf{a} \times \mathbf{b}| = |\mathbf{a} \times \mathbf{c}|$$

$$\therefore (p-9)^2 + (3p+1)^2 + (p^2+3)^2 = (2p+3)^2 + 49 + (3p+1)^2$$

$$\Rightarrow p^2 + 81 - 18p + (p^2 + 3 + 2p + 3)(p^2 + 3 - 2p - 3) = 49$$

$$\Rightarrow p^2 + 81 - 18p + (p^2 + 2p + 6)(p^2 - 2p) = 49$$

$$\Rightarrow p^2 + 81 - 18p + (p^2 + 2p + 6)(p^2 - 2p) = 49$$

$$\Rightarrow p^4 + 3p^2 - 30p + 32 = 0$$

$$\text{Now put } p = 2.16 + 12 - .60 + 32 = 0$$

$$\therefore p = 2$$

Question41

If $\mathbf{a} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$, $\mathbf{c} = 3\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and $\mathbf{d} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ are four vectors, then $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) =$

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Options:

A.

$$17\hat{\mathbf{i}} - 15\hat{\mathbf{j}} + 9\hat{\mathbf{k}}$$

B.

$$31\hat{\mathbf{i}} - \hat{\mathbf{j}} + 23\hat{\mathbf{k}}$$

C.

$$17\hat{\mathbf{i}} - \hat{\mathbf{j}} + 23\hat{\mathbf{k}}$$

D.

$$31\hat{\mathbf{i}} - 15\hat{\mathbf{j}} + 9\hat{\mathbf{k}}$$

Answer: B

Solution:

$$(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d})$$

$$= [\mathbf{abd}]\mathbf{c} - [\mathbf{abc}]\mathbf{d}$$

$$[\mathbf{abd}]\mathbf{c} = 8(3\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}})$$

$$= 24\hat{\mathbf{i}} - 8\hat{\mathbf{j}} + 16\hat{\mathbf{k}}$$

$$[\mathbf{abc}]\mathbf{d} = -7(\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})$$

$$= -7\hat{\mathbf{i}} - 7\hat{\mathbf{j}} - 7\hat{\mathbf{k}}$$

$$[\mathbf{abd}]\mathbf{c} - [\mathbf{abc}]\mathbf{d} = 31\hat{\mathbf{i}} - \hat{\mathbf{j}} + 23\hat{\mathbf{k}}$$

Question42

If the vectors $a\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\hat{\mathbf{i}} + b\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\hat{\mathbf{i}} + \hat{\mathbf{j}} + c\hat{\mathbf{k}}$ ($a \neq b \neq c \neq 1$) are coplanar, then $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is equal to

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Options:

A. 0

B. 2

C. 1

D. -1

Answer: C

Solution:

Given the vectors $a\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\hat{\mathbf{i}} + b\hat{\mathbf{j}} + \hat{\mathbf{k}}$, and $\hat{\mathbf{i}} + \hat{\mathbf{j}} + c\hat{\mathbf{k}}$, we need to verify that they are coplanar. When vectors are coplanar, it means they can be expressed as a linear combination of the other vectors.

Thus, we can write:

$$a\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}} = x(\hat{\mathbf{i}} + b\hat{\mathbf{j}} + \hat{\mathbf{k}}) + y(\hat{\mathbf{i}} + \hat{\mathbf{j}} + c\hat{\mathbf{k}})$$

where x and y are scalars, and not both zero.

Expanding both sides, we get:

$$a\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}} = (x + y)\hat{\mathbf{i}} + (bx + y)\hat{\mathbf{j}} + (x + cy)\hat{\mathbf{k}}$$

Equating the coefficients of $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$, we derive:

$$a = x + y$$

$$1 = bx + y$$

$$1 = x + cy$$

Now, from Equation (1), we have:

$$1 - a = 1 - (x + y)$$

Substitute the rearranged form of Equation (2):

$$1 - b = 1 - (bx + y) = \frac{x - 1 + y}{x}$$

Substitute from Equation (3):

$$1 - c = 1 - (x + cy) = \frac{y - 1 + x}{y}$$

Calculate the desired expression:

$$\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = \frac{1}{1-(x+y)} + \frac{x}{x-1+y} + \frac{y}{y-1+x}$$

By simplifying:

$$= \frac{1-x-y}{1-(x+y)} = 1$$

Thus, the expression simplifies to 1.

Question43

If $\mathbf{AB} = 2\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$, $\mathbf{BC} = 6\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ are the vectors along two sides of a $\triangle ABC$. Then, perimeter of $\triangle ABC$ is

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Options:

A. 21

B. $\sqrt{74} + 14$

C. $\sqrt{74} + 19$

D. $\sqrt{74} + 3$

Answer: B

Solution:

$$\text{Given, } \mathbf{AB} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 6\hat{\mathbf{k}}$$

$$\mathbf{BC} = 6\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

$$\text{We know that } \mathbf{AB} + \mathbf{BC} + \mathbf{CA} = 0$$

$$\Rightarrow (8\hat{\mathbf{i}} + \hat{\mathbf{j}} - 3\hat{\mathbf{k}}) = \mathbf{AC}$$

$$\therefore |\mathbf{AB}| = \sqrt{4 + 9 + 36} = 7$$

$$|\mathbf{BC}| = \sqrt{36 + 4 + 9} = 7$$

$$\text{and } |\mathbf{AC}| = \sqrt{64 + 1 + 9} = \sqrt{74}$$

$\therefore$ Perimeter of $\triangle ABC$ is

$$7 + 7 + \sqrt{74} = 14 + \sqrt{74}$$

Question44

The orthogonal projection vector of $\mathbf{a} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ on $\mathbf{b} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$ is

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Options:

A. $-\frac{1}{6}(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 3\hat{\mathbf{k}})$

B. $\frac{1}{6}(-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}})$

C. $\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$

D. $-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$

Answer: B

Solution:

To find the orthogonal projection of vector $\mathbf{a} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ on vector $\mathbf{b} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$, we use the formula for the projection of $\mathbf{a}$ onto $\mathbf{b}$:

$$\text{Projection}_{\mathbf{b}}(\mathbf{a}) = \frac{(\mathbf{a} \cdot \mathbf{b})\mathbf{b}}{|\mathbf{b}|^2}$$

Step-by-step, this involves:

Calculate the dot product $\mathbf{a} \cdot \mathbf{b}$:

$$\mathbf{a} \cdot \mathbf{b} = (2)(1) + (3)(-2) + (3)(1) = 2 - 6 + 3 = -1$$

Find the magnitude squared of $\mathbf{b}$, $|\mathbf{b}|^2$:

$$|\mathbf{b}|^2 = (1)^2 + (-2)^2 + (1)^2 = 1 + 4 + 1 = 6$$

Substitute these values into the projection formula:

$$\text{Projection}_{\mathbf{b}}(\mathbf{a}) = \frac{-1(\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}})}{6} = \frac{-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}}{6}$$

Thus, the orthogonal projection vector of $\mathbf{a}$ on $\mathbf{b}$ is:

$$\frac{-\hat{i}+2\hat{j}-\hat{k}}{6}$$

Question45

If $\mathbf{a} = -4\hat{i} + 2\hat{j} + 4\hat{k}$ and $\mathbf{b} = \sqrt{2}\hat{i} - \sqrt{2}\hat{j}$ are two vectors, then angle between the vectors $2\mathbf{a}$ and $\frac{\mathbf{b}}{2}$ is

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Options:

A. 30°

B. 135°

C. 90°

D. 0°

Answer: B

Solution:

Given vectors are:

$$\mathbf{a} = -4\hat{i} + 2\hat{j} + 4\hat{k}$$

$$\mathbf{b} = \sqrt{2}\hat{i} - \sqrt{2}\hat{j}$$

First, we compute $2\mathbf{a}$ and $\frac{\mathbf{b}}{2}$:

$$2\mathbf{a} = -8\hat{i} + 4\hat{j} + 8\hat{k}$$

$$\frac{\mathbf{b}}{2} = \frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}$$

To find the angle between $2\mathbf{a}$ and $\frac{\mathbf{b}}{2}$, we use the dot product formula:

$$\cos \theta = \frac{2\mathbf{a} \cdot \frac{\mathbf{b}}{2}}{|2\mathbf{a}| |\frac{\mathbf{b}}{2}|}$$

This simplifies to:

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}$$

Calculating the dot product $\mathbf{a} \cdot \mathbf{b}$:

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= (-4)(\sqrt{2}) + (2)(-\sqrt{2}) + (4)(0) \\ &= -4\sqrt{2} - 2\sqrt{2} \\ &= -6\sqrt{2} \end{aligned}$$

Next, we find the magnitudes of $\mathbf{a}$ and $\mathbf{b}$:

$$|\mathbf{a}| = \sqrt{(-4)^2 + 2^2 + 4^2} = \sqrt{16 + 4 + 16} = \sqrt{36} = 6$$

$$|\mathbf{b}| = \sqrt{(\sqrt{2})^2 + (-\sqrt{2})^2} = \sqrt{2 + 2} = \sqrt{4} = 2$$

Substitute these into the cosine formula:

$$\cos \theta = \frac{-6\sqrt{2}}{6 \times 2} = \frac{-6\sqrt{2}}{12} = -\frac{1}{\sqrt{2}}$$

The angle θ is calculated as:

$$\theta = \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$$

Since $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$ corresponds to $\pi - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$:

$$\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4} = 135^\circ$$

Question46

A unit vector perpendicular to the vectors $\mathbf{a} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$ and $\mathbf{b} = 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ is

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Options:

A. $\frac{3\hat{\mathbf{i}}+2\hat{\mathbf{j}}-2\hat{\mathbf{k}}}{\sqrt{22}}$

B. $\frac{3\hat{\mathbf{i}}+2\hat{\mathbf{j}}-3\hat{\mathbf{k}}}{\sqrt{22}}$

C. $\frac{3\hat{\mathbf{i}}-2\hat{\mathbf{j}}+3\hat{\mathbf{k}}}{\sqrt{22}}$

D. $\frac{3\hat{\mathbf{i}}+2\hat{\mathbf{j}}+3\hat{\mathbf{k}}}{\sqrt{22}}$

Answer: B

Solution:

Given vectors $\mathbf{a} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$ and $\mathbf{b} = 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$, we need to find a unit vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. This can be found by determining the cross product $\mathbf{a} \times \mathbf{b}$.

First, let's compute the cross product $\mathbf{a} \times \mathbf{b}$:

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & 3 & 4 \\ 0 & 3 & 2 \end{vmatrix} \\ &= \hat{\mathbf{i}}(3 \cdot 2 - 4 \cdot 3) - \hat{\mathbf{j}}(2 \cdot 2 - 0 \cdot 4) + \hat{\mathbf{k}}(2 \cdot 3 - 0 \cdot 3) \\ &= \hat{\mathbf{i}}(6 - 12) - \hat{\mathbf{j}}(4 - 0) + \hat{\mathbf{k}}(6 - 0) \\ &= -6\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 6\hat{\mathbf{k}}\end{aligned}$$

Next, we need to find the magnitude of the cross product $\mathbf{a} \times \mathbf{b}$:

$$|\mathbf{a} \times \mathbf{b}| = \sqrt{(-6)^2 + (-4)^2 + 6^2} = \sqrt{36 + 16 + 36} = \sqrt{88} = 2\sqrt{22}$$

Thus, the unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ is derived by normalizing the cross product:

$$\begin{aligned}\text{Unit vector} &= \pm \frac{-6\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 6\hat{\mathbf{k}}}{2\sqrt{22}} \\ &= \frac{-3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}}{\sqrt{22}} \quad \text{or} \quad \frac{3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}}{\sqrt{22}}\end{aligned}$$

Therefore, the required unit vectors are $\frac{-3\hat{\mathbf{i}}-2\hat{\mathbf{j}}+3\hat{\mathbf{k}}}{\sqrt{22}}$ or $\frac{3\hat{\mathbf{i}}+2\hat{\mathbf{j}}-3\hat{\mathbf{k}}}{\sqrt{22}}$.

Question47

If the vectors $a\hat{i} + \hat{j} + 3\hat{k}$, $4\hat{i} + 5\hat{j} + \hat{k}$ and $4\hat{i} + 2\hat{j} + 6\hat{k}$ are coplanar, then a is equal to

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Options:

A. 2

B. 1

C. 3

D. 4

Answer: A

Solution:

Given that the vectors $a\hat{i} + \hat{j} + 3\hat{k}$, $4\hat{i} + 5\hat{j} + \hat{k}$, and $4\hat{i} + 2\hat{j} + 6\hat{k}$ are coplanar, we can determine the value of a by setting the determinant of their coefficients to zero.

The determinant is:

$$\begin{vmatrix} a & 1 & 3 \\ 4 & 2 & 6 \\ 4 & 5 & 1 \end{vmatrix} = 0$$

Expanding the determinant, we have:

$$a(2 \times 1 - 6 \times 5) - 1(4 \times 1 - 6 \times 4) + 3(4 \times 5 - 2 \times 4) = 0$$

Calculating each term gives:

$$a(2 - 30) - 1(4 - 24) + 3(20 - 8) = 0$$

$$a(-28) + 20 + 36 = 0$$

This simplifies to:

$$-28a + 56 = 0$$

Solving for a , we get:

$$28a = 56 \Rightarrow a = 2$$

Thus, the value of a is 2.

Question48

Let $|\hat{a}| = 2 = |\hat{b}| = 3$ and the angle between $\hat{a}$ and $\hat{b}$ be $\frac{\pi}{3}$. If a parallelogram is constructed with adjacent sides $2\hat{a} + 3\hat{b}$ and $\hat{a} - \hat{b}$, then its shorter diagonal is of length

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Options:

A. 108

B. 172

C. $6\sqrt{3}$

D. $2\sqrt{43}$

Answer: C

Solution:

Given:

$$|\mathbf{a}| = 2$$

$$|\mathbf{b}| = 3$$

$$\text{Angle } \theta = \frac{\pi}{3}$$

The parallelogram has adjacent sides:

$$\mathbf{p} = 2\mathbf{a} + 3\mathbf{b}$$

$$\mathbf{q} = \mathbf{a} - \mathbf{b}$$

The diagonals of the parallelogram are described by $\mathbf{p} + \mathbf{q}$ and $\mathbf{p} - \mathbf{q}$.

Calculate $\mathbf{p} + \mathbf{q}$:

$$\mathbf{p} + \mathbf{q} = (2\mathbf{a} + 3\mathbf{b}) + (\mathbf{a} - \mathbf{b}) = 3\mathbf{a} + 2\mathbf{b}$$

Calculate $|\mathbf{p} + \mathbf{q}|^2$:

$$\begin{aligned} |\mathbf{p} + \mathbf{q}|^2 &= |3\mathbf{a} + 2\mathbf{b}|^2 \\ &= 9|\mathbf{a}|^2 + 4|\mathbf{b}|^2 + 12(\mathbf{a} \cdot \mathbf{b}) \\ &= 9 \times 4 + 4 \times 9 + 12 \times |\mathbf{a}||\mathbf{b}| \cos \theta \\ &= 36 + 36 + 12 \times 2 \times 3 \times \frac{1}{2} \\ &= 36 + 36 + 36 \\ &= 108 \end{aligned}$$

Therefore, the length of the shorter diagonal, $|\mathbf{p} + \mathbf{q}|$, is:

$$\sqrt{108} = 6\sqrt{3}$$

Question49

The values of x for which the angle between the vectors $x^2\hat{\mathbf{i}} + 2x\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + x\hat{\mathbf{k}}$ is obtuse lie in the interval

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Options:

A. $(-\infty, 0) \cup (3, \infty)$

B. $(0, 3)$

C. $[0, 3]$

D. $(-\infty, 0) \cup 3, \infty)$

Answer: B

Solution:

Given,

and

$$\mathbf{v}_1 = x^2\hat{\mathbf{i}} + 2x\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{v}_2 = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + x\hat{\mathbf{k}}$$

The angle between $\mathbf{v}_1$ and $\mathbf{v}_2$ is obtuse.

Thus, $\mathbf{v}_1 \cdot \mathbf{v}_2 < 0$

$$\Rightarrow (x^2 - 4x + x) < 0 \Rightarrow x^2 - 3x < 0$$

$$\Rightarrow x(x - 3) < 0$$



$$x \in (0, 3)$$

Question50

Let $\hat{\mathbf{a}} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$, $\hat{\mathbf{b}} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$. The projection of the sum of the vectors $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ on the vector perpendicular to the plane of $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$, is

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Options:

A. 0

B. $4\sqrt{2}$

C. $7\sqrt{2}$

D. $\frac{1}{\sqrt{2}}$

Answer: A

Solution:

To find the projection of the sum of vectors $\mathbf{a}$ and $\mathbf{b}$ on a vector perpendicular to their plane, follow these steps:

Calculate $\mathbf{a} + \mathbf{b}$:

$$\mathbf{a} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 5\hat{\mathbf{k}}, \quad \mathbf{b} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$$

$$\mathbf{a} + \mathbf{b} = (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 5\hat{\mathbf{k}}) + (2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}})$$

$$= 5\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 7\hat{\mathbf{k}}$$

Find $\mathbf{a} \times \mathbf{b}$ (the cross product):

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 & 4 & -5 \\ 2 & 1 & -2 \end{vmatrix}$$

$$= \hat{\mathbf{i}}(-8 + 9) - \hat{\mathbf{j}}(-6 + 10) + \hat{\mathbf{k}}(3 - 8)$$

$$= \hat{\mathbf{i}} - 4\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$$

Normalize the perpendicular vector $\mathbf{n} = -3\hat{\mathbf{i}} - 4\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$:

First, find the magnitude $|\mathbf{n}|$:

$$|\mathbf{n}| = \sqrt{(-3)^2 + (-4)^2 + (-5)^2} = \sqrt{9 + 16 + 25} = \sqrt{50} = 5\sqrt{2}$$

The unit vector $\hat{\mathbf{n}}$ is:

$$\hat{\mathbf{n}} = \frac{\mathbf{n}}{|\mathbf{n}|} = \frac{-3\hat{\mathbf{i}} - 4\hat{\mathbf{j}} - 5\hat{\mathbf{k}}}{5\sqrt{2}}$$

$$= \frac{-3}{5\sqrt{2}}\hat{\mathbf{i}} - \frac{4}{5\sqrt{2}}\hat{\mathbf{j}} - \frac{5}{5\sqrt{2}}\hat{\mathbf{k}}$$

Projection of $\mathbf{a} + \mathbf{b}$ on $\hat{\mathbf{n}}$:

$$(\mathbf{a} + \mathbf{b}) \cdot \hat{\mathbf{n}} = (5\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 7\hat{\mathbf{k}}) \cdot \left(\frac{-3}{5\sqrt{2}}\hat{\mathbf{i}} - \frac{4}{5\sqrt{2}}\hat{\mathbf{j}} - \frac{5}{5\sqrt{2}}\hat{\mathbf{k}} \right)$$

Calculate the dot product, which results in 0. Thus, the projection of the sum of the vectors $\mathbf{a}$ and $\mathbf{b}$ on the vector perpendicular to the plane of $\mathbf{a}$ and $\mathbf{b}$ is 0.

Question 51

In $\triangle PQR$, $(4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}})$, $(2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}})$ and $(3\hat{\mathbf{i}} + \hat{\mathbf{j}} + 3\hat{\mathbf{k}})$ are the position vectors of the vertices P , Q and R respectively then, the position vector for the point of intersection of the angle bisector of P and QR is

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Options:

A. $(6\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 9\hat{\mathbf{k}})$

B. $(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 3\hat{\mathbf{k}})$

C. $(5\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 2\hat{\mathbf{k}})$

D. $\left(\frac{5}{2}\hat{\mathbf{i}} + \frac{3}{2}\hat{\mathbf{j}} + 3\hat{\mathbf{k}} \right)$

Answer: D

Solution:

In triangle $\triangle PQR$, the vertices have the following position vectors:

$$P(4\hat{i} + 3\hat{j} + 6\hat{k})$$

$$Q(2\hat{i} + 2\hat{j} + 3\hat{k})$$

$$R(3\hat{i} + \hat{j} + 3\hat{k})$$

The angle bisector of a vertex in a triangle divides the opposite side into segments proportional to the lengths of the other two sides. To find the position vector I of the intersection of the angle bisector from vertex P with side QR , we use the formula:

$$I = \frac{|PR| \cdot Q + |PQ| \cdot R}{|PR| + |PQ|}$$

First, calculate the lengths of PQ and PR :

$$|PQ| = \sqrt{(4-2)^2 + (3-2)^2 + (6-3)^2} = \sqrt{4+1+9} = \sqrt{14}$$

$$|PR| = \sqrt{(4-3)^2 + (3-1)^2 + (6-3)^2} = \sqrt{1+4+9} = \sqrt{14}$$

Since $|PQ| = |PR|$, the intersection point I is the midpoint of Q and R :

$$I = \frac{Q+R}{2} = \frac{(2\hat{i}+2\hat{j}+3\hat{k})+(3\hat{i}+\hat{j}+3\hat{k})}{2}$$

$$I = \frac{5\hat{i}+3\hat{j}+6\hat{k}}{2}$$

$$I = \frac{5}{2}\hat{i} + \frac{3}{2}\hat{j} + 3\hat{k}$$

Question 52

If $\hat{f} = \hat{i} + \hat{j} + \hat{k}$ and $\hat{g} = 2\hat{i} - \hat{j} + 3\hat{k}$, then the projection vector of $\hat{f}$ on $\hat{g}$ is

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Options:

A. $\frac{2}{7}(\hat{i} + \hat{j} + \hat{k})$

B. $\frac{2}{7}(2\hat{i} - \hat{j} + 3\hat{k})$

C. $\frac{1}{3}(\hat{i} + \hat{j} + \hat{k})$

D. $\frac{1}{14}(2\hat{i} - \hat{j} + 3\hat{k})$

Answer: B

Solution:

To find the projection vector of $\hat{f}$ onto $\hat{g}$, we start with the given vectors:

$$\hat{f} = \hat{i} + \hat{j} + \hat{k}$$

$$\hat{g} = 2\hat{i} - \hat{j} + 3\hat{k}$$

The formula for the orthogonal projection of $\hat{f}$ on $\hat{g}$ is:

$$\text{Projection of } \hat{f} \text{ on } \hat{g} = \frac{\hat{f} \cdot \hat{g}}{|\hat{g}|^2} \cdot \hat{g}$$

First, calculate the dot product $\hat{f} \cdot \hat{g}$:

$$\hat{\mathbf{f}} \cdot \hat{\mathbf{g}} = (1)(2) + (1)(-1) + (1)(3) = 2 - 1 + 3 = 4$$

Next, find the magnitude squared of $\hat{\mathbf{g}}$:

$$|\hat{\mathbf{g}}|^2 = (2)^2 + (-1)^2 + (3)^2 = 4 + 1 + 9 = 14$$

Now use these results to find the projection:

$$\text{Projection of } \hat{\mathbf{f}} \text{ on } \hat{\mathbf{g}} = \frac{4}{14}(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 3\hat{\mathbf{k}})$$

Simplify the fraction:

$$= \frac{2}{7}(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 3\hat{\mathbf{k}})$$

Thus, the projection vector of $\hat{\mathbf{f}}$ on $\hat{\mathbf{g}}$ is:

$$\frac{2}{7}(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 3\hat{\mathbf{k}})$$

Question 53

If θ is the angle between $\hat{\mathbf{f}} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ and $\hat{\mathbf{g}} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + a\hat{\mathbf{k}}$ and $\sin \theta = \sqrt{\frac{24}{28}}$, then $7a^2 + 24a =$

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Options:

A. 10

B. 12

C. 36

D. 15

Answer: A

Solution:

Given the vectors $\mathbf{f} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ and $\mathbf{g} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + a\hat{\mathbf{k}}$, we know that the sine of the angle θ between them is $\sin \theta = \sqrt{\frac{24}{28}}$.

Since $\sin^2 \theta + \cos^2 \theta = 1$, we can find $\cos \theta$ as follows:

$$\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{\frac{1}{7}}$$

The cosine of the angle θ is also given by:

$$\cos \theta = \frac{\hat{\mathbf{f}} \cdot \hat{\mathbf{g}}}{|\hat{\mathbf{f}}| \cdot |\hat{\mathbf{g}}|}$$

Calculating the dot product:

$$\hat{\mathbf{f}} \cdot \hat{\mathbf{g}} = (1 \cdot 2) + (2 \cdot -3) + (-3 \cdot a) = 2 - 6 - 3a = -4 - 3a$$

Calculating the magnitudes:

$$|\hat{\mathbf{f}}| = \sqrt{1^2 + 2^2 + (-3)^2} = \sqrt{14}$$

$$|\hat{\mathbf{g}}| = \sqrt{2^2 + (-3)^2 + a^2} = \sqrt{13 + a^2}$$

Thus, we have:

$$\frac{1}{\sqrt{7}} = \frac{-4-3a}{\sqrt{14}\sqrt{13+a^2}}$$

Simplifying, we find:

$$2(13 + a^2) = (4 + 3a)^2$$

Expanding and rearranging gives:

$$26 + 2a^2 = 16 + 9a^2 + 24a$$

This simplifies to the quadratic equation:

$$7a^2 + 24a - 10 = 0$$

Solving for a using the quadratic formula:

$$a = \frac{-24 \pm \sqrt{24^2 - 4 \times 7 \times (-10)}}{2 \times 7}$$

$$a = \frac{-24 \pm \sqrt{576 + 280}}{14}$$

$$a = \frac{-24 \pm \sqrt{856}}{14}$$

Finally, calculating $7a^2 + 24a$:

Substitute back into the quadratic result:

$$7a^2 + 24a = 10$$

Question54

If $\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}, 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}, -3\hat{\mathbf{i}} - \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$ are the position vectors of three points, A, B, C respectively, then A, B, C

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Options:

- A. are collinear point
- B. form an isosceles triangle which is not equilateral
- C. form an equilateral triangle
- D. form a scalene triangle

Answer: C

Solution:

To determine the relationship between the points A, B , and C given their position vectors:

$$\mathbf{r}_1 = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}} \text{ for point } A$$

$$\mathbf{r}_2 = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}} \text{ for point } B$$

$$\mathbf{r}_3 = -3\hat{\mathbf{i}} - \hat{\mathbf{j}} - 2\hat{\mathbf{k}} \text{ for point } C$$

We can calculate the distances between these points:

Distance between A and B :

$$\begin{aligned} |\mathbf{AB}| &= |\mathbf{r}_2 - \mathbf{r}_1| \\ &= |(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}) - (\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}})| \\ &= |\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 4\hat{\mathbf{k}}| \\ &= \sqrt{1^2 + 5^2 + 4^2} \\ &= \sqrt{1 + 25 + 16} \\ &= \sqrt{42} \end{aligned}$$

Distance between B and C :

$$\begin{aligned} |\mathbf{BC}| &= |\mathbf{r}_3 - \mathbf{r}_2| \\ &= |(-3\hat{\mathbf{i}} - \hat{\mathbf{j}} - 2\hat{\mathbf{k}}) - (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}})| \\ &= | -5\hat{\mathbf{i}} - 4\hat{\mathbf{j}} - \hat{\mathbf{k}}| \\ &= \sqrt{5^2 + 4^2 + 1^2} \\ &= \sqrt{25 + 16 + 1} \\ &= \sqrt{42} \end{aligned}$$

Distance between C and A :

Since

$$\mathbf{CA} = \mathbf{r}_1 - \mathbf{r}_3 = (\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}) - (-3\hat{\mathbf{i}} - \hat{\mathbf{j}} - 2\hat{\mathbf{k}})$$

you find

$$|\mathbf{CA}| = \sqrt{42}.$$

All three distances are equal,

$$|\mathbf{AB}| = |\mathbf{BC}| = |\mathbf{CA}| = \sqrt{42}$$

Thus, points A , B , and C form an equilateral triangle because all three sides are of equal length.

Question 55

If $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{d}$ are position vectors of 4 points such that $2\mathbf{a} + 3\mathbf{b} + 5\mathbf{c} - 10\mathbf{d} = \mathbf{0}$, then the ratio in which the line joining $\mathbf{c}$ and $\mathbf{d}$ divides the line segment joining $\mathbf{a}$ and $\mathbf{b}$ is

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Options:

- A. 2 : 3
- B. -1 : 2
- C. 2 : 1
- D. 3 : 2

Answer: D

Solution:

Given, the equation,

$$\begin{aligned} 2\mathbf{a} + 3\mathbf{b} + 5\mathbf{c} - 10\mathbf{d} &= \mathbf{0} \Rightarrow 5\mathbf{c} = 10\mathbf{d} - 2\mathbf{a} - 3\mathbf{b} \\ \mathbf{c} &= 2\mathbf{d} - \frac{2}{5}\mathbf{a} - \frac{3\mathbf{b}}{5} \quad \dots (i) \end{aligned}$$

Now, consider the points **a** and **b** with the line joining them divided by point **c** and **d** the position vector of any point dividing **a** and **b** internally in the ratio m : n given by

$$P = \frac{m\mathbf{b} + n\mathbf{a}}{m+n}$$

Similarly, let the point **c** dividing the line joining **a** and **b** in the ratio $K : 1$.

$$\mathbf{c} = \frac{K\mathbf{b} + \mathbf{a}}{K+1} \Rightarrow \frac{K\mathbf{b} + \mathbf{a}}{K+1} = 2\mathbf{d} - \frac{2}{5}\mathbf{a} - \frac{3}{5}\mathbf{b}$$

$$\Rightarrow K\mathbf{b} + \mathbf{a} = (K+1)\left(2\mathbf{d} - \frac{2}{5}\mathbf{a} - \frac{3}{5}\mathbf{b}\right)$$

$$\Rightarrow K\mathbf{b} + \mathbf{a} = 2(K+1)\mathbf{d} - \frac{2(K+1)}{5}\mathbf{a} - \frac{3(K+1)}{5}\mathbf{b}$$

$$\Rightarrow \left(K + \frac{3(K+1)}{5}\right)\mathbf{b} + \left(1 + \frac{2(K+1)}{5}\right)\mathbf{a} = \frac{3(K+1)}{5}\mathbf{d}$$

$$\Rightarrow \left(\frac{8K+3}{5}\right)\mathbf{b} + \left(\frac{7+2K}{5}\right)\mathbf{a} = 3(K+1)\mathbf{d}$$

On comparing the coefficients and solve for K , we get

$$K+1 = \frac{5}{2}$$

Thus, $\frac{K}{1}$ given the ratio in which the point C divides a and b . So, $K : 1 = 3 : 2$

Question 56

If **a**, **b**, **c** are 3 vectors such that $|\mathbf{a}| = 5$, $|\mathbf{b}| = 8$, $|\mathbf{c}| = 11$ and $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, then the angle between the vectors **a** and **b** is

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Options:

A. $\cos^{-1} \frac{2}{5}$

B. $\cos^{-1} \frac{10}{11}$

C. $\cos^{-1} \frac{41}{55}$

D. $\frac{\pi}{3}$

Answer: A

Solution:

Given three vectors **a**, **b**, and **c** with magnitudes $|\mathbf{a}| = 5$, $|\mathbf{b}| = 8$, and $|\mathbf{c}| = 11$, and given that they satisfy the equation $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, we need to determine the angle between vectors **a** and **b**.

First, from the equation $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, we can derive:

$$\mathbf{a} + \mathbf{b} = -\mathbf{c}$$

By squaring both sides, we have:

$$(\mathbf{a} + \mathbf{b})^2 = (-\mathbf{c})^2$$

This can be expanded to:

$$\mathbf{a}^2 + \mathbf{b}^2 + 2\mathbf{a} \cdot \mathbf{b} = \mathbf{c}^2$$

Substituting the magnitudes of the vectors, we get:

$$|\mathbf{a}|^2 + |\mathbf{b}|^2 + 2\mathbf{a} \cdot \mathbf{b} = |\mathbf{c}|^2$$

$$5^2 + 8^2 + 2\mathbf{a} \cdot \mathbf{b} = 11^2$$

$$25 + 64 + 2\mathbf{a} \cdot \mathbf{b} = 121$$

Solving for $2\mathbf{a} \cdot \mathbf{b}$, we find:

$$2\mathbf{a} \cdot \mathbf{b} = 121 - 89 = 32$$

$$\mathbf{a} \cdot \mathbf{b} = \frac{32}{2} = 16$$

The cosine of the angle θ between $\mathbf{a}$ and $\mathbf{b}$ is given by:

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{16}{5 \times 8} = \frac{2}{5}$$

Thus, the angle between vectors $\mathbf{a}$ and $\mathbf{b}$ is:

$$\theta = \cos^{-1} \left(\frac{2}{5} \right)$$

Question 57

$\mathbf{a} = \alpha \hat{\mathbf{i}} + \beta \hat{\mathbf{j}} + 3\hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and $\mathbf{c} = 3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$ are linearly dependent vectors and magnitude of $\alpha\sqrt{14}$. If α, β are integers, then $\alpha + \beta =$

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Options:

A. 3

B. -3

C. 5

D. -5

Answer: A

Solution:

$$\mathbf{a} = \alpha \hat{\mathbf{i}} + \beta \hat{\mathbf{j}} + 3\hat{\mathbf{k}}, \mathbf{b} = \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

$$\mathbf{c} = 3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}, |\mathbf{a}| = \sqrt{14}$$

Since, $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are linearly dependent the determinant of the matrix formed by these vector must be zero.

$$[abc] = 0$$

$$\begin{vmatrix} \alpha & \beta & 3 \\ 0 & 1 & 2 \\ 3 & 2 & 1 \end{vmatrix} = 0$$

$$\alpha(1 - 4) - \beta(0 - 6) + 3(0 - 3) = 0$$

$$-3\alpha + 6\beta - 9 = 0$$

$$-\alpha + 2\beta = 3 \Rightarrow \alpha = 2\beta - 3$$

$$|\alpha| = \sqrt{14}$$

$$\alpha^2 + \beta^2 + 9 = 14$$

$$\alpha^2 + \beta^2 = 14 - 9$$

$$\alpha^2 + \beta^2 = 5$$

$$(2\beta - 3)^2 + \beta^2 = 5$$

$$4\beta^2 + 9 - 12\beta + \beta^2 = 5$$

$$5\beta^2 - 12\beta + 4 = 0$$

$$5\beta^2 - (10 + 2)\beta + 4 = 0$$

$$5\beta^2 - 10\beta - 2\beta + 4 = 0$$

$$5\beta(\beta - 2) - 2(\beta - 2) = 0$$

$$(5\beta - 2)(\beta - 2) = 0$$

$$\beta = 22/5$$

On substitute into $-\alpha + 2\beta = 3$

$$\beta = 2 - \alpha + 4 = 3$$

$$\alpha = 1$$

$$\alpha + \beta = 1 + 2 = 3$$

$$\beta = \frac{2}{5} \Rightarrow -\alpha + \frac{4}{5} = 3$$

On substitute $-\alpha = 3 - \frac{4}{5} = \frac{15-4}{5}$

Neglecting $\alpha = \frac{-11}{5}$

Question 58

c is a vector along the bisector of the internal angle between the vectors

a = $4\hat{i} + 7\hat{j} - 4\hat{k}$ and **b** = $12\hat{i} - 3\hat{j} + 4\hat{k}$. If the magnitude of **c** is $3\sqrt{13}$, then **c**=

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Options:

A. $5\hat{i} - 8\hat{j} + 2\sqrt{2}\hat{k}$

B. $10\hat{i} + 4\hat{j} - \hat{k}$

C. $\hat{i} - 10\hat{j} + 4\hat{k}$

D. $2\sqrt{2}\hat{i} + 5\hat{j} - 8\hat{k}$

Answer: B

Solution:

Given, **a** = $4\hat{i} + 7\hat{j} - 4\hat{k}$ and

b = $12\hat{i} - 3\hat{j} + 4\hat{k}$, $|\mathbf{c}| = 3\sqrt{13}$

The required vectors $\mathbf{c}$ is given by

$$\begin{aligned}
 \mathbf{c} &= \lambda(\mathbf{a} + \mathbf{b}) \\
 &= \lambda \left(\frac{\mathbf{a}}{|\mathbf{a}|} + \frac{\mathbf{b}}{|\mathbf{b}|} \right) \\
 &= \lambda \left(\frac{4\hat{\mathbf{i}} + 7\hat{\mathbf{j}} - 4\hat{\mathbf{k}}}{\sqrt{16 + 49 + 16}} + \frac{12\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}}{\sqrt{144 + 9 + 16}} \right) \\
 &= \lambda \left[\frac{4\hat{\mathbf{i}} + 7\hat{\mathbf{j}} - 4\hat{\mathbf{k}}}{9} + \frac{12\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}}{13} \right] \\
 \mathbf{c} &= \lambda \left[\frac{52\hat{\mathbf{i}} + 91\hat{\mathbf{j}} - 52\hat{\mathbf{k}} + 108\hat{\mathbf{i}} - 27\hat{\mathbf{j}} + 36\hat{\mathbf{k}}}{9 \times 13} \right] \\
 \mathbf{c} &= \lambda \left[\frac{160\hat{\mathbf{i}} + 64\hat{\mathbf{j}} - 16\hat{\mathbf{k}}}{9 \times 13} \right] \\
 \mathbf{c} &= \frac{16\lambda}{9 \times 13} [10\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - \hat{\mathbf{k}}] \\
 |\mathbf{c}| &= \frac{16\lambda}{13 \times 9} \sqrt{100 + 16 + 1} = \frac{16\lambda}{13 \times 9} \sqrt{117} \\
 \mathbf{c} &= 3\sqrt{13} \quad \quad \quad (\text{given}) \\
 3\sqrt{13} &= \frac{16\lambda}{13 \times 9} \sqrt{117} \Rightarrow 3\sqrt{13} = \frac{16\lambda}{13 \times 9} \sqrt{13 \times 9} \\
 3\sqrt{13} &= \frac{16\lambda}{13 \times 9} 3\sqrt{13} \\
 \lambda &= \frac{13 \times 9}{16}
 \end{aligned}$$

On putting in Eq.(i), we get

$$\text{Hence, } \mathbf{c} = 10\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - \hat{\mathbf{k}}.$$

Question 59

$\mathbf{a} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{b} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ are two vectors and $\mathbf{c}$ is a unit vectors lying in the plane of $\mathbf{a}$ and $\mathbf{b}$. If $\mathbf{c}$ is perpendicular to $\mathbf{b}$, then $\mathbf{c}(\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}) =$

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Options:

- A. 0
- B. 5
- C. $\frac{1}{\sqrt{21}}$
- D. $\frac{2}{\sqrt{21}}$

Answer: B

Solution:

The vector $\mathbf{c}$ is a unit vector that is also perpendicular to $\mathbf{b}$, as expressed by:

$$\mathbf{b} \cdot \mathbf{c} = 0$$

Additionally, the magnitude of $\mathbf{c}$ is 1:

$$|\mathbf{c}| = 1$$

To find $\mathbf{c}$, which lies in the plane formed by the vectors $\mathbf{a}$ and $\mathbf{b}$, we can use the cross product of these vectors:

$$\mathbf{c} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & -1 & 1 \\ 2 & 1 & 1 \end{vmatrix}$$

Calculating the determinant gives us:

$$= \hat{\mathbf{i}}(-1 - 1) - \hat{\mathbf{j}}(1 - 2) + \hat{\mathbf{k}}(1 + 2)$$

This simplifies to:

$$= -2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

Finally, we compute the dot product of $\mathbf{c}$ with the vector $\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$:

$$\mathbf{c} \cdot (\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}) = (-2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 3\hat{\mathbf{k}}) \cdot (\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}})$$

This yields:

$$= 5$$

Thus, $\mathbf{c}(\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}) = 5$.

Question60

If $\mathbf{a} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$, $\mathbf{c} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$. $\mathbf{d} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ are four vector, then $(\mathbf{a} \times \mathbf{c}) \times (\mathbf{b} \times \mathbf{d}) =$

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Options:

A. $2\hat{\mathbf{i}} + 19\hat{\mathbf{j}} - 11\hat{\mathbf{k}}$

B. $-8\hat{\mathbf{i}} + 19\hat{\mathbf{j}} - 29\hat{\mathbf{k}}$

C. $2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 11\hat{\mathbf{k}}$

D. $-8\hat{\mathbf{i}} + \hat{\mathbf{j}} - 29\hat{\mathbf{k}}$

Answer: D

Solution:

To solve for $(\mathbf{a} \times \mathbf{c}) \times (\mathbf{b} \times \mathbf{d})$, we will begin by calculating the cross products $\mathbf{a} \times \mathbf{c}$ and $\mathbf{b} \times \mathbf{d}$ separately.

We have the vectors defined as:

$$\mathbf{a} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{b} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$$

$$\mathbf{c} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$\mathbf{d} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$$

Step 1: Calculate $\mathbf{a} \times \mathbf{c}$

The cross product $\mathbf{a} \times \mathbf{c}$ is determined using the determinant:

$$\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & -1 & 1 \\ 2 & -3 & -1 \end{vmatrix}$$

Calculating the determinant, we have:

$$\hat{\mathbf{i}}[(-1)(-1) - (1)(-3)] - \hat{\mathbf{j}}[(1)(-1) - (1)(2)] + \hat{\mathbf{k}}[(1)(-3) - (-1)(2)]$$

This simplifies to:

$$\hat{\mathbf{i}}(1 + 3) - \hat{\mathbf{j}}(-1 - 2) + \hat{\mathbf{k}}(-3 + 2)$$

Resulting in:

$$4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

Step 2: Calculate $\mathbf{b} \times \mathbf{d}$

Similarly, compute the cross product $\mathbf{b} \times \mathbf{d}$ using the determinant:

$$\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 1 & -2 \\ 2 & 1 & 1 \end{vmatrix}$$

Calculating this determinant, we find:

$$\hat{\mathbf{i}}[(1)(1) - (-2)(1)] - \hat{\mathbf{j}}[(1)(1) - (-2)(2)] + \hat{\mathbf{k}}[(1)(1) - (1)(2)]$$

This gives:

$$\hat{\mathbf{i}}(1 + 2) - \hat{\mathbf{j}}(1 + 4) + \hat{\mathbf{k}}(1 - 2)$$

Resulting in:

$$3\hat{\mathbf{i}} - 5\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

Step 3: Calculate $(\mathbf{a} \times \mathbf{c}) \times (\mathbf{b} \times \mathbf{d})$

Now use the result from both cross products to find their cross product:

$$\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 4 & 3 & -1 \\ 3 & -5 & -1 \end{vmatrix}$$

This determinant is computed as follows:

$$\hat{\mathbf{i}}[(3)(-1) - (-1)(-1)] - \hat{\mathbf{j}}[(4)(-1) - (-1)(-1)] + \hat{\mathbf{k}}[(4)(-5) - (3)(3)]$$

Which results in:

$$\hat{\mathbf{i}}(-3 + 1) - \hat{\mathbf{j}}(-4 - 1) + \hat{\mathbf{k}}(-20 - 9)$$

Simplifying, we find:

$$-2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 29\hat{\mathbf{k}}$$

Thus, the final result of $(\mathbf{a} \times \mathbf{c}) \times (\mathbf{b} \times \mathbf{d})$ is:

$$-2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 29\hat{\mathbf{k}}$$

Question 61

The angle between the diagonals of the parallelogram whose adjacent sides are

$$2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 5\hat{\mathbf{k}}, \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

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Options:

A. $\cos^{-1}\left(\frac{7}{\sqrt{69}}\right)$

B. $\cos^{-1}\left(\frac{1}{7\sqrt{69}}\right)$

C. $\cos^{-1}\left(\frac{1}{7}\right)$

D. $\cos^{-1}\left(\frac{31}{7\sqrt{69}}\right)$

Answer: D

Solution:

To find the angle between the diagonals of a parallelogram, we start with the given vectors representing the adjacent sides of the parallelogram:

Let:

$$\mathbf{a} = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$$

$$\mathbf{b} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

The diagonal vectors of the parallelogram can be calculated as follows:

First diagonal, c:

$$\mathbf{c} = \mathbf{a} + \mathbf{b} = (2+1)\hat{\mathbf{i}} + (4+2)\hat{\mathbf{j}} + (-5+3)\hat{\mathbf{k}} = 3\hat{\mathbf{i}} + 6\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$$

Second diagonal, d:

$$\mathbf{d} = \mathbf{a} - \mathbf{b} = (2-1)\hat{\mathbf{i}} + (4-2)\hat{\mathbf{j}} + (-5-3)\hat{\mathbf{k}} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 8\hat{\mathbf{k}}$$

Next, we calculate the cosine of the angle θ between the diagonals using the dot product formula:

$$\cos \theta = \frac{\mathbf{c} \cdot \mathbf{d}}{|\mathbf{c}||\mathbf{d}|}$$

Dot Product:

$$\mathbf{c} \cdot \mathbf{d} = (3)(1) + (6)(2) + (-2)(-8) = 3 + 12 + 16 = 31$$

Magnitudes of the diagonals:

$$|\mathbf{c}| = \sqrt{3^2 + 6^2 + (-2)^2} = \sqrt{9 + 36 + 4} = \sqrt{49} = 7$$

$$|\mathbf{d}| = \sqrt{1^2 + 2^2 + (-8)^2} = \sqrt{1 + 4 + 64} = \sqrt{69}$$

Substituting these into the formula for $\cos \theta$:

$$\cos \theta = \frac{31}{7\sqrt{69}}$$

Thus, the angle θ between the diagonals is:

$$\theta = \cos^{-1}\left(\frac{31}{7\sqrt{69}}\right)$$

Therefore, the angle between the diagonals of the parallelogram is $\cos^{-1}\left(\frac{31}{7\sqrt{69}}\right)$.

Question 62

If the points having the position vectors $-i + 4j - 4k$, $3i + 2j - 5k$, $-3i + 8j - 5k$ and $-3i + 2j + \lambda k$ are coplanar, then $\lambda =$

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Options:

- A. 1
- B. 2
- C. -2
- D. -3

Answer: C

Solution:

To determine if the points with the given position vectors are coplanar, we need to ensure the determinant of the matrix formed by vectors $\mathbf{AB}$, $\mathbf{AC}$, and $\mathbf{AD}$ is zero.

Given:

$$\mathbf{OA} = -\hat{i} + 4\hat{j} - 4\hat{k}$$

$$\mathbf{OB} = 3\hat{i} + 2\hat{j} - 5\hat{k}$$

$$\mathbf{OC} = -3\hat{i} + 8\hat{j} - 5\hat{k}$$

$$\mathbf{OD} = -3\hat{i} + 2\hat{j} + \lambda\hat{k}$$

Calculate the vectors:

$$\mathbf{AB} = \mathbf{OB} - \mathbf{OA} = (3 + 1)\hat{i} + (2 - 4)\hat{j} + (-5 + 4)\hat{k} = 4\hat{i} - 2\hat{j} - \hat{k}$$

$$\mathbf{AC} = \mathbf{OC} - \mathbf{OA} = (-3 + 1)\hat{i} + (8 - 4)\hat{j} + (-5 + 4)\hat{k} = -2\hat{i} + 4\hat{j} - \hat{k}$$

$$\mathbf{AD} = \mathbf{OD} - \mathbf{OA} = (-3 + 1)\hat{i} + (2 - 4)\hat{j} + (\lambda + 4)\hat{k} = -2\hat{i} - 2\hat{j} + (\lambda + 4)\hat{k}$$

The determinant for coplanarity is:

$$\begin{vmatrix} 4 & -2 & -1 \\ -2 & 4 & -1 \\ -2 & -2 & \lambda + 4 \end{vmatrix} = 0$$

Expanding the determinant:

$$= 4((-2)(\lambda + 4) - (-2)(-1)) - (-2)((-2)(\lambda + 4) - (-2)(-1)) - 1((-2)(-2) - (4)(-2))$$

$$= 4(4\lambda + 16 - 2) - 2(2\lambda + 8 - 2) - (4 + 8)$$

Simplify:

$$= 16\lambda + 56 - 4\lambda - 20 - 12 = 0$$

Solving:

$$12\lambda + 24 = 0$$

$$12\lambda = -24$$

$$\lambda = -2$$

Therefore, the value of λ is -2 .

Question63

If $|f| = 10$, $|g| = 14$ and $|f - g| = 15$, then $|f + g| =$

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Options:

A. 367

B. $\sqrt{367}$

C. 400

D. 20

Answer: B

Solution:

Given, $|f| = 10$, $|g| = 14$ and

$$|f - g| = 15$$

We know that

$$|f - g|^2 = |f|^2 + |g|^2 - 2\operatorname{Re}f\bar{g}$$

$$\Rightarrow 225 = 100 + 196 - 2\operatorname{Re}f\bar{g}$$

$$\Rightarrow 2\operatorname{Re}f\bar{g} = 296 - 225 = 71$$

$$\text{Now, } |f + g|^2 = |f|^2 + |g|^2 + 2\operatorname{Re}f\bar{g}$$

$$= 100 + 196 + 71$$

$$= 367$$

$$\Rightarrow |f + g| = \sqrt{367}$$

Question64

If $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are three vectors such that $|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = \sqrt{3}$ and $(a + b - c)^2 + (b + c - a)^2 + (c + a - b)^2 = 36$, then $|2a - 3b + 2c| =$

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Options:

A. 15

B. 25

C. 147

D. 75

Answer: D

Solution:

Given, $|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = \sqrt{3}$ and

$$\begin{aligned}
& (a+b-c)^2 + (b+c-a)^2 + (c+a-b)^2 = 36 \\
& |a|^2 + |b|^2 + |c|^2 + 2a \cdot b - 2a \cdot c - 2b \cdot c \\
& + |b|^2 + |c|^2 + |a|^2 + 2b \cdot c - 2b \cdot a - 2c \cdot a \\
& + |c|^2 + |a|^2 + |b|^2 + 2c \cdot a - 2a \cdot b \\
& - 2b \cdot c = 36 \\
& 3(|a|^2 + |b|^2 + |c|^2) - 2a \cdot b - 2a \cdot c - 2b \cdot c = 36 \\
& - 2b \cdot c = 36 \\
& 3(3+3+3) - 2a \cdot b - 2a \cdot c - 2b \cdot c = 36 \\
& (\mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{c}) = \frac{27-36}{2} = \frac{-9}{2} \quad \dots (i)
\end{aligned}$$

$$\begin{aligned}
& \text{Now, } |2\mathbf{a} - 3\mathbf{b} + 2\mathbf{c}|^2 \\
& = 4|a|^2 + 4|b|^2 + 4|c|^2 - 12\mathbf{a} \cdot \mathbf{b} + 8\mathbf{a} \cdot \mathbf{c} - 12\mathbf{b} \cdot \mathbf{c}
\end{aligned}$$

$$\text{Also, } |\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 \geq 0$$

$$\begin{aligned}
& \Rightarrow |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) \geq 0 \\
& \Rightarrow \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a} \geq -\frac{9}{2}
\end{aligned}$$

$$\text{Since, } \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a} = -\frac{9}{2}$$

$$\begin{aligned}
& \text{So, } |\mathbf{a} + \mathbf{b} + \mathbf{c}| = 0 \\
& \Rightarrow a + b + c \quad \dots (ii) \\
& \text{So, } |2a - 3b + 2c| = |(2a + 2c) - 3b| \\
& \quad |-2b - 3b|, d = |-5b| \\
& \quad = 5|b| = 5\sqrt{3}
\end{aligned}$$

$$\therefore |2a - 3b + 2c|^2 = (5\sqrt{3})^2 = 75$$

Question65

a, b, c are non-coplanar vectors. If $\alpha \mathbf{d} = \mathbf{a} + \mathbf{b} + \mathbf{c}$ and $\beta \mathbf{a} = \mathbf{b} + \mathbf{c} + \mathbf{d}$, then $|\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d}| =$

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Options:

- A. 1
- B. 2
- C. $|a - b - c|$
- D. 0

Answer: D

Solution:

Given the non-coplanar vectors **a**, **b**, and **c**, we start with the equations:

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = \alpha \mathbf{d}$$

From this equation, we can express **b + c** as:

$$\mathbf{b} + \mathbf{c} = \alpha \mathbf{d} - \mathbf{a} \quad (\text{Equation i})$$

We are also given:

$$\mathbf{b} + \mathbf{c} + \mathbf{d} = \beta \mathbf{a}$$

From which we derive:

$$\mathbf{b} + \mathbf{c} = \beta \mathbf{a} - \mathbf{d} \quad (\text{Equation ii})$$

By equating both expressed forms of $\mathbf{b} + \mathbf{c}$ from Equations (i) and (ii), we have:

$$\alpha \mathbf{d} - \mathbf{a} = \beta \mathbf{a} - \mathbf{d}$$

Rearranging terms gives:

$$\alpha \mathbf{d} + \mathbf{d} = \beta \mathbf{a} + \mathbf{a}$$

Thus, we have:

$$(\alpha + 1)\mathbf{d} = (\beta + 1)\mathbf{a}$$

Since $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are non-coplanar, they are linearly independent. Therefore, the coefficients of $\mathbf{a}$ and $\mathbf{d}$ must separately satisfy the equality:

$$1 + \beta = 0 \text{ implies } \beta = -1$$

$$1 + \alpha = 0 \text{ implies } \alpha = -1$$

Therefore:

$$\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} = 0$$

Thus, the magnitude is:

$$|\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d}| = 0$$

Question 66

$\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are three unit vectors. Let $\hat{\mathbf{p}} = \hat{\mathbf{u}} + \hat{\mathbf{v}} + \hat{\mathbf{w}}$ and $\hat{\mathbf{q}} = \hat{\mathbf{u}} \times (\hat{\mathbf{v}} \times \hat{\mathbf{w}})$. If $\hat{\mathbf{p}} \cdot \hat{\mathbf{u}} = \frac{3}{2}$ and $\hat{\mathbf{p}} \cdot \hat{\mathbf{v}} = \frac{7}{4}$, then $K =$

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Options:

A. -1

B. 2

C. 3

D. -2

Answer: B

Solution:

Given that $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are unit vectors, we define:

$$\hat{\mathbf{p}} = \hat{\mathbf{u}} + \hat{\mathbf{v}} + \hat{\mathbf{w}}$$

$$\hat{\mathbf{q}} = \hat{\mathbf{u}} \times (\hat{\mathbf{v}} \times \hat{\mathbf{w}})$$

Additionally, we know:

$$\hat{\mathbf{p}} \cdot \hat{\mathbf{u}} = \frac{3}{2}$$

$$\hat{\mathbf{p}} \cdot \hat{\mathbf{v}} = \frac{7}{4}$$

$$|\hat{\mathbf{p}}| = 2$$

Calculation Steps:

From the equation $\mathbf{p} \cdot \mathbf{u}$:

$$\mathbf{p} \cdot \mathbf{u} = |\mathbf{u}|^2 + \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w} = \frac{3}{2}$$

This implies:

$$1 + \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w} = \frac{3}{2}$$

$$\Rightarrow \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w} = \frac{1}{2}$$

From the equation $\mathbf{p} \cdot \mathbf{v}$:

$$\mathbf{p} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{v} + |\mathbf{v}|^2 + \mathbf{v} \cdot \mathbf{w} = \frac{7}{4}$$

Thus:

$$\mathbf{u} \cdot \mathbf{v} + 1 + \mathbf{v} \cdot \mathbf{w} = \frac{7}{4}$$

$$\Rightarrow \mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{w} = \frac{3}{4}$$

Equating results:

Solve the system:

$$\mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w} = \frac{1}{2}$$

$$\mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{w} = \frac{3}{4}$$

Subtract to find:

$$\mathbf{u} \cdot \mathbf{w} - \mathbf{v} \cdot \mathbf{w} = -\frac{1}{4}$$

Vector subtraction and dot product:

$$\mathbf{w} \cdot (\mathbf{u} - \mathbf{v}) = -\frac{1}{4}$$

Given also:

$$\mathbf{p} \cdot (\mathbf{u} - \mathbf{v}) = \frac{3}{2} - \frac{7}{4} = -\frac{1}{4}$$

Conclude for K :

This means:

$$\mathbf{p} = \mathbf{w} \quad \Rightarrow \quad |\mathbf{p}| = |\mathbf{w}| \quad \Rightarrow \quad \mathbf{u} + \mathbf{v} = 0$$

For:

$$\mathbf{q} = \mathbf{u} \times (\mathbf{v} \times \mathbf{w})$$

Expanding:

$$\mathbf{q} = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$$

Knowing $\mathbf{v} = K\mathbf{q}$:

$$\Rightarrow \mathbf{u} \cdot \mathbf{v} = 0$$

Then solve for K :

$$K = \frac{1}{\mathbf{u} \cdot \mathbf{w}}$$

From above, $\mathbf{u} \cdot \mathbf{v} = \frac{1}{2}$, hence:

$$K = 2$$

Question67

If $\mathbf{a}$ and $\mathbf{b}$ are the two non collinear vectors, then $|\mathbf{b}|\mathbf{a} + |\mathbf{a}|\mathbf{b}$ represents

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Options:

- A. a vector parallel to an angle bisector of $\mathbf{a}$, $\mathbf{b}$
- B. a vector along the difference of the $\mathbf{a}$, $\mathbf{b}$
- C. a vector along $\mathbf{a} + \mathbf{b}$
- D. a vector outside the triangle having $\mathbf{a}$, $\mathbf{b}$ as adjacent sides

Answer: A

Solution:

The expression $|\mathbf{b}|\mathbf{a} + |\mathbf{a}|\mathbf{b}$ represents a vector parallel to the angle bisector of the two non-collinear vectors $\mathbf{a}$ and $\mathbf{b}$.

To understand why, consider the angle bisector of the angle between vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\lambda(\hat{\mathbf{a}} + \hat{\mathbf{b}})$$

This can be rewritten as:

$$\lambda \left(\frac{\mathbf{a}}{|\mathbf{a}|} + \frac{\mathbf{b}}{|\mathbf{b}|} \right)$$

Simplifying further, we have:

$$\frac{\lambda}{|\mathbf{a}||\mathbf{b}|} (|\mathbf{b}|\mathbf{a} + |\mathbf{a}|\mathbf{b}) = k(|\mathbf{b}|\mathbf{a} + |\mathbf{a}|\mathbf{b})$$

This derivation shows that $|\mathbf{b}|\mathbf{a} + |\mathbf{a}|\mathbf{b}$ is indeed parallel to the angle bisector of $\mathbf{a}$ and $\mathbf{b}$.

Question68

If L, M, N are the mid-points of the sides PQ, QR and RP of $\triangle PQR$ respectively, then $\mathbf{QM} + \mathbf{LN} + \mathbf{ML} + \mathbf{RN} - \mathbf{MN} - \mathbf{QL} =$

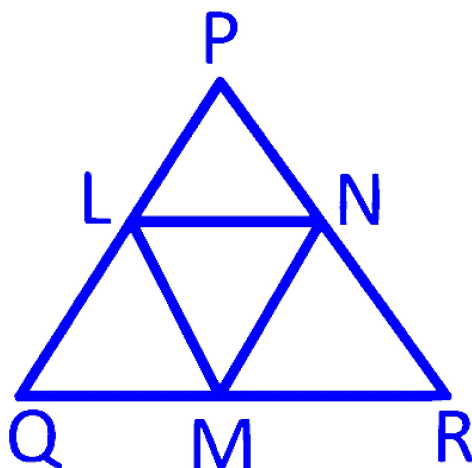
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Options:

- A. $\mathbf{PQ} + \mathbf{QR} + \mathbf{LM} + \mathbf{MN}$
- B. $\mathbf{LP} + \mathbf{PM} + \mathbf{MQ}$
- C. $\mathbf{PQ} + \mathbf{QR} - \mathbf{PR}$
- D. $\mathbf{LM} - \mathbf{MN} + \mathbf{NR}$

Answer: C

Solution:



By mid-point theorem,

$$LN = QM = \frac{1}{2}QR$$

$$LM = NR = \frac{1}{2}PR$$

$$MN = QL = \frac{1}{2}PQ$$

$$QM + LN + ML + RN - MN - QL$$

=

$$= \frac{1}{2}QR + \frac{1}{2}QR - \frac{1}{2}PR$$

$$- \frac{1}{2}PR + \frac{1}{2}PQ + \frac{1}{2}PQ$$

$$= QR - PR + PQ$$

Question69

Let $\mathbf{a} \times \mathbf{b} = 7\hat{\mathbf{i}} - 5\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$ and $\mathbf{a} = \hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$. If the length of projection of $\mathbf{b}$ on $\mathbf{a}$ is $\frac{8}{\sqrt{14}}$, then $|\mathbf{b}| =$

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Options:

A. 121

B. $\sqrt{11}$

C. $\sqrt{12}$

D. 144

Answer: B

Solution:

The given information states that the projection of vector $\mathbf{b}$ onto vector $\mathbf{a}$ is $\frac{8}{\sqrt{14}}$. Let's go through the calculation step-by-step to find $|\mathbf{b}|$.

First, we express the projection of $\mathbf{b}$ onto $\mathbf{a}$:

$$\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|} = \frac{8}{\sqrt{14}}$$

From this, we have:

$$\frac{\mathbf{a} \cdot \mathbf{b}}{\sqrt{14}} = \frac{8}{\sqrt{14}} \implies \mathbf{a} \cdot \mathbf{b} = 8$$

Next, the magnitudes of the vectors satisfy the following relation:

$$|\mathbf{a}|^2 |\mathbf{b}|^2 = (\mathbf{a} \cdot \mathbf{b})^2 + |\mathbf{a} \times \mathbf{b}|^2$$

Given that $\mathbf{a} \times \mathbf{b} = 7\hat{\mathbf{i}} - 5\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$, the magnitude $|\mathbf{a} \times \mathbf{b}|$ can be calculated as:

$$|\mathbf{a} \times \mathbf{b}| = \sqrt{7^2 + (-5)^2 + (-4)^2} = \sqrt{49 + 25 + 16} = \sqrt{90}$$

Now, substitute these values into the equation:

$$14|\mathbf{b}|^2 = 64 + 90$$

$$14|\mathbf{b}|^2 = 154$$

Therefore:

$$|\mathbf{b}|^2 = \frac{154}{14} = 11$$

Thus, the magnitude of $\mathbf{b}$ is:

$$|\mathbf{b}| = \sqrt{11}$$

Question70

Let ABC be an equilateral triangle of side a . M and N are two points on the sides AB and AC , respectively such that $\mathbf{AN} = K\mathbf{AC}$ and $\mathbf{AB} = 3\mathbf{AM}$. If the vectors $\mathbf{BN}$ and $\mathbf{CM}$ are perpendicular, then $K =$

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Options:

- A. $\frac{1}{5}$
- B. $\frac{2}{5}$
- C. $-\frac{1}{5}$
- D. $-\frac{2}{5}$

Answer: A

Solution:

Consider point A as the origin, denoted as O . Let $\mathbf{b}$ and $\mathbf{c}$ be the position vectors of points B and C respectively.

Since the vector $\mathbf{AM} = \frac{\mathbf{AB}}{3} = \frac{\mathbf{b}}{3}$, the position vector of M is $\frac{\mathbf{b}}{3}$.

The position vector of N is given by $\mathbf{AN} = K\mathbf{AC} = K\mathbf{c}$.

Since the vectors $\mathbf{BN}$ and $\mathbf{CM}$ are perpendicular, we use the dot product property for perpendicular vectors:

$$(K\mathbf{c} - \mathbf{b}) \cdot \left(\frac{\mathbf{b}}{3} - \mathbf{c}\right) = 0$$

Expanding this, we have:

$$\frac{K}{3} \mathbf{b} \cdot \mathbf{c} - \frac{|\mathbf{b}|^2}{3} - K|\mathbf{c}|^2 + \mathbf{b} \cdot \mathbf{c} = 0$$

Substituting, knowing that for an equilateral triangle, $\mathbf{b} \cdot \mathbf{c} = |\mathbf{b}||\mathbf{c}| \cos(60^\circ) = \frac{a^2}{2}$:

$$\frac{K}{3} \cdot \frac{a^2}{2} - \frac{a^2}{3} - Ka^2 + \frac{a^2}{2} = 0$$

Simplifying the equation:

$$|\mathbf{a}|^2 \left[\frac{K}{6} - \frac{1}{3} - K + \frac{1}{2} \right] = 0$$

Since $|\mathbf{a}| \neq 0$ (as it represents side length a), the term in square brackets must equal zero:

$$\frac{K}{6} - K - \frac{1}{3} + \frac{1}{2} = 0$$

This simplifies to:

$$\frac{-5K}{6} + \frac{1}{6} = 0$$

Thus:

$$-5K + 1 = 0 \quad \Rightarrow \quad K = \frac{1}{5}$$

Question 71

Let $\mathbf{a}$ and $\mathbf{b}$ be two non-collinear vector of unit modulus. If $\mathbf{u} = \mathbf{a} - (\mathbf{a} \cdot \mathbf{b})\mathbf{b}$ and $\mathbf{v} = \mathbf{a} \times \mathbf{b}$, then $|\mathbf{v}| =$

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Options:

A. $|\mathbf{u}| + |\mathbf{u} \cdot \mathbf{v}|$

B. $\frac{|\mathbf{u}|}{2}$

C. $|\mathbf{u}| + \frac{|\mathbf{u} \cdot \mathbf{b}|}{2}$

D. $\frac{|\mathbf{u}|}{5}$

Answer: A

Solution:

To clarify the problem, we start by considering two non-collinear unit vectors, $\mathbf{a}$ and $\mathbf{b}$.

We are given:

$$\mathbf{u} = \mathbf{a} - (\mathbf{a} \cdot \mathbf{b})\mathbf{b}$$

$$\mathbf{v} = \mathbf{a} \times \mathbf{b}$$

Let's analyze $|\mathbf{u}|$:

$$\mathbf{u} = \mathbf{a} - (\mathbf{a} \cdot \mathbf{b})\mathbf{b}$$

The dot product of $\mathbf{u}$ with itself yields:

$$|\mathbf{u}|^2 = \mathbf{u} \cdot \mathbf{u} = [\mathbf{a} - (\mathbf{a} \cdot \mathbf{b})\mathbf{b}] \cdot [\mathbf{a} - (\mathbf{a} \cdot \mathbf{b})\mathbf{b}]$$

Expanding gives:

$$|\mathbf{u}|^2 = \mathbf{a} \cdot \mathbf{a} - 2(\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{a}) + (\mathbf{a} \cdot \mathbf{b})^2(\mathbf{b} \cdot \mathbf{b})$$

Given $|\mathbf{a}| = |\mathbf{b}| = 1$, this simplifies to:

$$|\mathbf{u}|^2 = 1 - (\mathbf{a} \cdot \mathbf{b})^2$$

Since $(\mathbf{a} \cdot \mathbf{b}) = \cos \theta$, we have:

$$|\mathbf{u}|^2 = 1 - \cos^2 \theta = \sin^2 \theta$$

Thus, $|\mathbf{u}| = \sin \theta$.

Now, consider $|\mathbf{v}|$:

$$\mathbf{v} = \mathbf{a} \times \mathbf{b}$$

The magnitude of the cross product is:

$$|\mathbf{v}| = |\mathbf{a}||\mathbf{b}| \sin \theta = \sin \theta$$

Now let us verify the expression $|\mathbf{u}| + |\mathbf{u} \cdot \mathbf{v}|$:

$$\begin{aligned} |\mathbf{u} \cdot \mathbf{v}| &= |(\mathbf{a} - (\mathbf{a} \cdot \mathbf{b})\mathbf{b}) \cdot (\mathbf{a} \times \mathbf{b})| \\ &= |\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) - (\mathbf{a} \cdot \mathbf{b})\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b})| \end{aligned}$$

Using vector triple product identities, we know:

$$\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0$$

$$\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = 0$$

Therefore:

$$|\mathbf{u} \cdot \mathbf{v}| = 0$$

Thus, $|\mathbf{u}| + |\mathbf{u} \cdot \mathbf{v}| = \sin \theta + 0 = \sin \theta$.

Hence, $|\mathbf{v}| = |\mathbf{u}| + |\mathbf{u} \cdot \mathbf{v}|$.

Question 72

In a regular hexagon $ABCDEF$, $\mathbf{AB} = \mathbf{a}$ and $\mathbf{BC} = \mathbf{b}$, then $\mathbf{FA} =$

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Options:

A. $\mathbf{a} - \mathbf{b}$

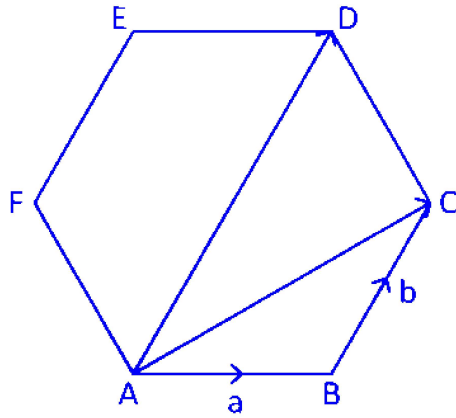
B. $\mathbf{a} + \mathbf{b}$

C. $\mathbf{b} - \mathbf{a}$

D. $2\mathbf{b} - \mathbf{a}$

Answer: A

Solution:



In $\triangle ABC$, $AB + BC = AC$

$$\Rightarrow AC = a + b$$

AD is parallel to BC and also

$$AD = 2BC$$

$$AD = 2b$$

$$CD = AD - AC = 2b - (a + b) = b - a$$

FA is parallel to CD, but is opposite direction.

$$FA = -CD = -(b - a) = a - b$$

Question73

If $\mathbf{f}$, $\mathbf{g}$, $\mathbf{h}$ be mutually orthogonal vectors of equal magnitudes, then the angle between the vectors $\mathbf{f} + \mathbf{g} + \mathbf{h}$ and $\mathbf{h}$ is

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Options:

A. $\cos^{-1} \left(\frac{\sqrt{3}}{4} \right)$

B. $\cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$

C. $\pi - \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$

D. $\pi - \cos^{-1} \left(\frac{\sqrt{3}}{4} \right)$

Answer: B

Solution:

To find the angle between the vectors $\mathbf{f} + \mathbf{g} + \mathbf{h}$ and $\mathbf{h}$, we start by noting that $\mathbf{f}$, $\mathbf{g}$, and $\mathbf{h}$ are mutually orthogonal vectors of equal magnitude. This means:

$$|\mathbf{f}| = |\mathbf{g}| = |\mathbf{h}|.$$

$$\mathbf{f} \cdot \mathbf{g} = \mathbf{g} \cdot \mathbf{h} = \mathbf{h} \cdot \mathbf{f} = 0.$$

The dot product of $(\mathbf{f} + \mathbf{g} + \mathbf{h})$ with $\mathbf{h}$ can be calculated as:

$$(\mathbf{f} + \mathbf{g} + \mathbf{h}) \cdot \mathbf{h} = \mathbf{f} \cdot \mathbf{h} + \mathbf{g} \cdot \mathbf{h} + \mathbf{h} \cdot \mathbf{h} = |\mathbf{h}|^2$$

Let θ be the angle between $\mathbf{f} + \mathbf{g} + \mathbf{h}$ and $\mathbf{h}$. We use the formula for the dot product in terms of angle:

$$|\mathbf{f} + \mathbf{g} + \mathbf{h}||\mathbf{h}| \cos \theta = |\mathbf{h}|^2$$

Since $\mathbf{f}$, $\mathbf{g}$, and $\mathbf{h}$ are orthogonal and of equal magnitude, the magnitude of $\mathbf{f} + \mathbf{g} + \mathbf{h}$ is:

$$|\mathbf{f} + \mathbf{g} + \mathbf{h}| = \sqrt{|\mathbf{f}|^2 + |\mathbf{g}|^2 + |\mathbf{h}|^2} = \sqrt{3}|\mathbf{h}|$$

Substitute this back into the equation:

$$\sqrt{3}|\mathbf{h}||\mathbf{h}| \cos \theta = |\mathbf{h}|^2$$

Simplifying, we find:

$$\cos \theta = \frac{1}{\sqrt{3}}$$

Therefore, the angle θ is:

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

Question 74

Let $\mathbf{a}$, $\mathbf{b}$ be two unit vectors. If $\mathbf{c} = \mathbf{a} + 2\mathbf{b}$ and $\mathbf{d} = 5\mathbf{a} - 4\mathbf{b}$ are perpendicular to each other, then the angle between $\mathbf{a}$ and $\mathbf{b}$ is

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Options:

A. $\frac{\pi}{6}$

B. $\frac{\pi}{4}$

C. $\frac{\pi}{3}$

D. $\frac{\pi}{8}$

Answer: C

Solution:

Given two unit vectors $\mathbf{a}$ and $\mathbf{b}$, we know that:

$$|\mathbf{a}| = |\mathbf{b}| = 1$$

Consider the vectors $\mathbf{c} = \mathbf{a} + 2\mathbf{b}$ and $\mathbf{d} = 5\mathbf{a} - 4\mathbf{b}$. These vectors are perpendicular, meaning:

$$\mathbf{c} \cdot \mathbf{d} = 0$$

Let's compute the dot product:

$$(\mathbf{a} + 2\mathbf{b}) \cdot (5\mathbf{a} - 4\mathbf{b}) = \mathbf{a} \cdot (5\mathbf{a} - 4\mathbf{b}) + 2\mathbf{b} \cdot (5\mathbf{a} - 4\mathbf{b})$$

Expanding this, we have:

$$= \mathbf{a} \cdot 5\mathbf{a} - \mathbf{a} \cdot 4\mathbf{b} + 2\mathbf{b} \cdot 5\mathbf{a} - 2\mathbf{b} \cdot 4\mathbf{b}$$

This simplifies to:

$$= 5|\mathbf{a}|^2 - 4\mathbf{a} \cdot \mathbf{b} + 10\mathbf{a} \cdot \mathbf{b} - 8|\mathbf{b}|^2$$

Simplifying further by substituting $|\mathbf{a}| = 1$ and $|\mathbf{b}| = 1$:

$$= 5(1) + 6\mathbf{a} \cdot \mathbf{b} - 8(1) = 0$$

Which simplifies to:

$$5 - 8 + 6(\mathbf{a} \cdot \mathbf{b}) = 0$$

$$-3 + 6(\mathbf{a} \cdot \mathbf{b}) = 0$$

Solving for $\mathbf{a} \cdot \mathbf{b}$:

$$6(\mathbf{a} \cdot \mathbf{b}) = 3$$

$$\mathbf{a} \cdot \mathbf{b} = \frac{3}{6} = \frac{1}{2}$$

The dot product $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$, so:

$$\cos \theta = \frac{1}{2}$$

Thus, the angle θ between $\mathbf{a}$ and $\mathbf{b}$ is:

$$\theta = \frac{\pi}{3}$$

Question 75

If the vectors $\mathbf{a} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$, $\mathbf{c} = 3\hat{\mathbf{i}} + p\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ are coplanar, then $p =$

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Options:

A. 4

B. 14

C. -4

D. 41

Answer: C

Solution:

To determine the value of p for which the vectors $\mathbf{a} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$, and $\mathbf{c} = 3\hat{\mathbf{i}} + p\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ are coplanar, we need to evaluate the condition of coplanarity. The vectors are coplanar if the determinant of the matrix formed by their components is zero:

$$\begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & -3 \\ 3 & p & 5 \end{vmatrix} = 0$$

Calculate the determinant:

$$= 2(2 \cdot 5 - (-3) \cdot p) - (-1)(1 \cdot 5 - (-3) \cdot 3) + 1(1 \cdot p - 2 \cdot 3)$$

Simplify each term:

$$\text{The first term: } 2(10 + 3p) = 20 + 6p$$

$$\text{The second term: } -1(5 + 9) = -14$$

$$\text{The third term: } 1(p - 6) = p - 6$$

Combining these terms gives:

$$20 + 6p - 14 + p - 6 = 0$$

Simplifying further:

$$20 + 6p - 14 + p - 6 = 0 \implies 7p = -28 \implies p = -4$$

Thus, the value of p is -4 .

Question 76

If (α, β, γ) are the direction cosines of an angular bisector of two lines whose direction ratios are $(2, 2, 1)$ and $(2, -1, -2)$, then $(\alpha + \beta + \gamma)^2 =$

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Options:

A. 3

B. 2

C. 4

D. 5

Answer: B

Solution:

Given the problem involves finding the value of $(\alpha + \beta + \gamma)^2$ when (α, β, γ) are the direction cosines of the angular bisector of two lines with direction ratios $(2, 2, 1)$ and $(2, -1, -2)$.

Finding Direction Cosines:

The direction ratios for line L_1 are $(2, 2, 1)$, and for line L_2 are $(2, -1, -2)$.

The direction cosines for L_1 are found as:

$$\left(\frac{2}{3}, \frac{2}{3}, \frac{1}{3}\right)$$

Similarly, for L_2 the direction cosines are:

$$\left(\frac{2}{3}, \frac{-1}{3}, \frac{-2}{3}\right)$$

Perpendicular Check:

The dot product of the two direction ratios confirms that the lines are perpendicular:

$$2 \times 2 + 2 \times (-1) + 1 \times (-2) = 0$$

Angle Bisector Direction Cosines:

The formulas for the direction cosines of the angle bisectors are:

$$\left(\frac{l_1+l_2}{2 \cos \frac{\theta}{2}}, \frac{m_1+m_2}{2 \cos \frac{\theta}{2}}, \frac{n_1+n_2}{2 \cos \frac{\theta}{2}}\right)$$

and

$$\left(\frac{l_1-l_2}{2 \sin \frac{\theta}{2}}, \frac{m_1-m_2}{2 \sin \frac{\theta}{2}}, \frac{n_1-n_2}{2 \sin \frac{\theta}{2}}\right)$$

Calculate the Direction Cosines:

For an angle bisector direction, calculate:

$$\left(\frac{4}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, \frac{-1}{3\sqrt{2}}\right)$$

Another possible set is:

$$\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

Calculation of $(\alpha + \beta + \gamma)^2$:

$$\text{Using } \alpha = \frac{4}{3\sqrt{2}}, \beta = \frac{1}{3\sqrt{2}}, \gamma = \frac{-1}{3\sqrt{2}}:$$

Sum:

$$\alpha + \beta + \gamma = \frac{4}{3\sqrt{2}} + \frac{1}{3\sqrt{2}} - \frac{1}{3\sqrt{2}} = \frac{4}{3\sqrt{2}}$$

Squaring:

$$(\alpha + \beta + \gamma)^2 = \left(\frac{4}{3\sqrt{2}}\right)^2 = \frac{16}{18} = \frac{8}{9}$$

$$\text{Alternatively, use } \alpha = 0, \beta = \frac{1}{\sqrt{2}}, \gamma = \frac{1}{\sqrt{2}}:$$

Sum:

$$\alpha + \beta + \gamma = 0 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

Squaring:

$$(\alpha + \beta + \gamma)^2 = (\sqrt{2})^2 = 2$$

Based on calculations with valid assumptions, the value of $(\alpha + \beta + \gamma)^2$ considering a correct scenario for calculation comes out to be 2.

Question77

a, b, c are non-coplanar vectors. If

$$\mathbf{a} + 3\mathbf{b} + 4\mathbf{c} = x(\mathbf{a} - 2\mathbf{b} + 3\mathbf{c}) + y(\mathbf{a} + 5\mathbf{b} - 2\mathbf{c}) + z(6\mathbf{a} + 14\mathbf{b} + 4\mathbf{c}), \text{ then } x + y + z =$$

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Options:

A. -5

B. -4

C. 4

D. 5

Answer: B

Solution:

$$\mathbf{a} + 3\mathbf{b} + 4\mathbf{c} = \mathbf{a}(x + y + 6z) + \mathbf{b}(-2x + 5y + 14z) + \mathbf{c}(3x - 2y + 4z)$$

(because a, b and c are non-coplanar vectors)

Comparing both sides,

$$\begin{aligned}
 x + y + 6z &= 1 \quad \dots (i) \\
 -2x + 5y + 14z &= 3 \quad \dots (ii) \\
 3x - 2y + 4z &= 4 \quad \dots (iii)
 \end{aligned}$$

By solving Eqs. (i), (ii) and (iii), we get

$$\begin{aligned}
 x &= -2, y = -3, z = 1 \\
 x + y + z &= -4
 \end{aligned}$$

Question78

Three vectors of magnitudes $a, 2a, 3a$ are along the directions of the diagonals of 3 adjacent faces of a cube that meet in a point. Then, the magnitude of the sum of those diagonals is

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Options:

- A. $4a$
- B. $5a$
- C. $6a$
- D. $8a$

Answer: B

Solution:

Let the vectors of magnitude $a, 2a, 3a$ are along OP, OQ, OR , respectively.

Then, vectors are OP, OQ, OR are $a \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right), 2a \left(\frac{\hat{j} + \hat{k}}{\sqrt{2}} \right), 3a \left(\frac{\hat{k} + \hat{i}}{\sqrt{2}} \right)$, respectively.

Their resultant say R is given by

$$\begin{aligned}
 \mathbf{R} &= a \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) + 2a \left(\frac{\hat{j} + \hat{k}}{\sqrt{2}} \right) + 3a \left(\frac{\hat{k} + \hat{i}}{\sqrt{2}} \right) \\
 &= \frac{a}{\sqrt{2}} (4\hat{i} + 3\hat{j} + 5\hat{k}) \\
 \therefore |\mathbf{R}| &= \sqrt{\frac{a^2}{2} (16 + 9 + 25)} = 5a
 \end{aligned}$$

Question79

If $\mathbf{a}$ is collinear with $\mathbf{b} = 3\hat{i} + 6\hat{j} + 6\hat{k}$ and $\mathbf{a} \cdot \mathbf{b} = 27$, then $|\mathbf{a}| =$

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Options:

- A. 1
- B. 2
- C. 3
- D. 4

Answer: C

Solution:

$$\text{Let } \mathbf{a} = x\hat{i} + y\hat{j} + z\hat{k}, \mathbf{b} = 3\hat{i} + 6\hat{j} + 6\hat{k} \\ \mathbf{a} \cdot \mathbf{b} = 27$$

According to question, a is a collinear with b, then

$$\mathbf{a} = \lambda \mathbf{b} \quad \dots (i)$$

$$\Rightarrow \mathbf{a} \cdot \mathbf{b} = 27$$

$$\Rightarrow \lambda \mathbf{b} \cdot \mathbf{b} = 27 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow \lambda |\mathbf{b}|^2 = 27$$

$$\lambda = \frac{27}{\left(\sqrt{(3)^2 + (6)^2 + (6)^2}\right)^2} = \frac{27}{(9)^2}$$

$$\lambda = \frac{1}{3}$$

$$\mathbf{a} = \frac{1}{3}(3\hat{i} + 6\hat{j} + 6\hat{k}) \Rightarrow \mathbf{a} = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\text{So, } |\mathbf{a}| = \sqrt{1 + (2)^2 + (2)^2} \\ = \sqrt{9} = 3 \text{ units}$$

Question80

Let a, b and c be unit vectors such that a is perpendicular to the plane containing b and c and angle between b and c is $\frac{\pi}{3}$. Then, $|\mathbf{a} + \mathbf{b} + \mathbf{c}| =$

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Options:

- A. 3
- B. 1
- C. 2
- D. 4

Answer: C

Solution:

As given $\mathbf{a}$ is perpendicular to $\mathbf{b}$ and $\mathbf{c}$

$$\Rightarrow \mathbf{a} \cdot \mathbf{b} = 0 \text{ and } \mathbf{a} \cdot \mathbf{c} = 0 \text{ and angle between } \mathbf{b} \text{ and } \mathbf{c} = \frac{\pi}{3}$$

$$\therefore \mathbf{b} \cdot \mathbf{c} = |\mathbf{b}||\mathbf{c}| \cos \frac{\pi}{3} = 1 \cdot 1 \cdot \frac{1}{2}$$

$$= \frac{1}{2} [\because \mathbf{b} \text{ and } \mathbf{c} \text{ are unit vectors}]$$

$$\text{Now, } |\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a})$$

$$= 1 + 1 + 1 + 2 \left(0 + \frac{1}{2} + 0 \right) = 1 + 1 + 1 + 1 = 4$$

$$|\mathbf{a} + \mathbf{b} + \mathbf{c}| = \sqrt{4} = 2 \text{ units}$$

Question81

Let $\mathbf{F} = 2\hat{i} + 2\hat{j} + 5\hat{k}$, $A = (1, 2, 5)$, $B = (-1, -2, -3)$ and $\mathbf{BA} \times \mathbf{F} = 4\hat{i} + 6\hat{j} + 2\lambda\hat{k}$, then $\lambda =$

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Options:

A. 0

B. 1

C. 2

D. -2

Answer: D

Solution:

Given,

$$\mathbf{F} = 2\hat{i} + 2\hat{j} + 5\hat{k}, A = (1, 2, 5)$$

$$B = (-1, -2, -3)$$

$$\mathbf{BA} = \mathbf{A} - \mathbf{B} = (\hat{i} + 2\hat{j} + 5\hat{k}) - (-\hat{i} - 2\hat{j} - 3\hat{k})$$

$$\mathbf{BA} = 2\hat{i} + 4\hat{j} + 8\hat{k}$$

$$\mathbf{BA} \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 4 & 8 \\ 2 & 2 & 5 \end{vmatrix}$$

$$= \hat{i}(20 - 16) - \hat{j}(10 - 16) + \hat{k}(4 - 8)$$

$$= 4\hat{i} + 6\hat{j} - 4\hat{k}$$

$$\text{So, } 4\hat{i} + 6\hat{j} - 4\hat{k} = 4\hat{i} + 6\hat{j} + 2\lambda\hat{k}$$

$$\therefore 2\lambda = -4$$

$$\lambda = -2$$

Question82

$OABC$ is a tetrahedron. If D, E are the mid-points of OA and BC respectively, then $\vec{DE} =$

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Options:

- A. $\frac{1}{2}(OA + OB + OC)$
- B. $\frac{1}{2}(OA + OB - OC)$
- C. $\frac{1}{2}(OA - OB + OC)$
- D. $\frac{1}{2}(-OA + OB + OC)$

Answer: D

Solution:

Given:

Tetrahedron $OABC$

- D is midpoint of OA

$$\Rightarrow \vec{OD} = \frac{1}{2}\vec{OA}$$

- E is midpoint of BC

$$\Rightarrow \vec{OE} = \frac{1}{2}(\vec{OB} + \vec{OC})$$

Find $\vec{DE}$

$$\vec{DE} = \vec{OE} - \vec{OD}$$

Substitute:

$$\vec{DE} = \frac{1}{2}(\vec{OB} + \vec{OC}) - \frac{1}{2}\vec{OA}$$

Simplify:

$$\vec{DE} = \frac{1}{2}(-\vec{OA} + \vec{OB} + \vec{OC})$$

Final Answer:

$$\vec{DE} = \frac{1}{2}(-\vec{OA} + \vec{OB} + \vec{OC})$$

Question83

If $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$ and $|\mathbf{a}| = 7, |\mathbf{b}| = 5, |\mathbf{c}| = 3$ then the angle between $\mathbf{b}$ and $\mathbf{c}$ is

AP EAPCET 2022 - 4th July Evening Shift

Options:

A. 30°

B. 45°

C. 60°

D. 90°

Answer: C

Solution:

$$\begin{aligned} \text{Given, } \mathbf{a} + \mathbf{b} + \mathbf{c} &= \mathbf{0} \text{ and} \\ |\mathbf{a}| &= 7, |\mathbf{b}| = 5, |\mathbf{c}| = 3 \\ \Rightarrow \mathbf{b} + \mathbf{c} &= -\mathbf{a} \Rightarrow |\mathbf{b} + \mathbf{c}|^2 = |-\mathbf{a}|^2 \\ \Rightarrow |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2 \cdot \mathbf{b} \cdot \mathbf{c} &= |\mathbf{a}|^2 \\ \Rightarrow 25 + 9 + 2 \cdot \mathbf{b} \cdot \mathbf{c} &= 49 \\ \Rightarrow 2\mathbf{b} \cdot \mathbf{c} &= 15 \Rightarrow 2|\mathbf{b}| \cdot |\mathbf{c}| \cos \theta = 15 \\ \Rightarrow \cos \theta &= \frac{1}{2} = \cos 60^\circ \\ \therefore \theta &= 60^\circ \end{aligned}$$

Question 84

If P and Q are two points on the curve $y = 2^{x+2}$ in the rectangular cartesian coordinate system such that $\mathbf{OP} \cdot \hat{i} = -1$, $\mathbf{OQ} \cdot \hat{i} = 2$, then $\mathbf{OQ} - 4\mathbf{OP} =$

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Options:

A. $3\hat{i} + 8\hat{j}$

B. $4\hat{i} + 6\hat{j}$

C. $6\hat{i} + 8\hat{j}$

D. $4\hat{i} + 3\hat{j}$

Answer: C

Solution:

$$\text{Given, } \mathbf{OP} \cdot \hat{i} = -1 \Rightarrow P_x = -1$$

$$P \text{ lies on } y = 2^{x+2}$$

$$\therefore y = 2^{-1+2} = 2$$

$$\text{Thus, } (Px, Py) = (-1, 2)$$

and also given

$$OQ \cdot \hat{i} = 2 \Rightarrow Q_x = 2$$

$$\text{Q lies on } y = 2^{x+2}$$

$$\therefore y = 16$$

$$\therefore (Q_x, Q_y) = (2, 16)$$

$$\text{Thus } \mathbf{OP} = -\hat{i} + 2\hat{j} \text{ and } \mathbf{OQ} = 2\hat{i} + 16\hat{j}$$

$$\mathbf{OQ} - 4\mathbf{OP} = 2\hat{i} + 16\hat{j} + 4\hat{i} - 8\hat{j} = 6\hat{i} + 8\hat{j}$$

Question 85

In quadrilateral $ABCD$, $\mathbf{AB} = \mathbf{a}$, $\mathbf{BC} = \mathbf{b}$. $\mathbf{DA} = \mathbf{a} - \mathbf{b}$, M is the mid-point of BC and X is a point on DM such that, $\mathbf{DX} = \frac{4}{5} \mathbf{DM}$. Then, the points A , X and C .

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Options:

- A. form an equilateral triangle.
- B. are collinear
- C. form an isosceles triangle
- D. form a right angled triangle

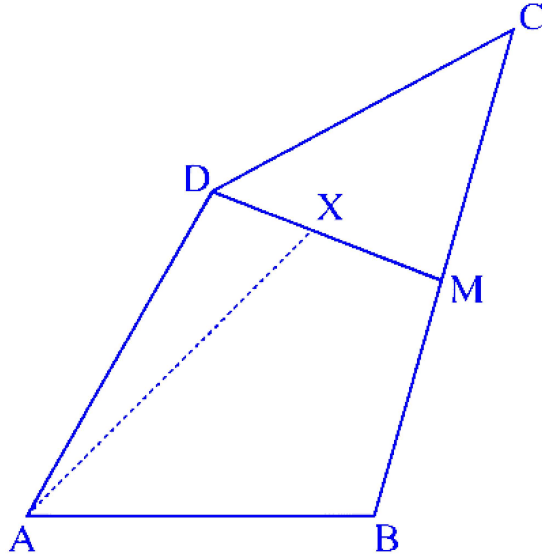
Answer: B

Solution:

$$\mathbf{AB} = \mathbf{a}, \mathbf{BC} = \mathbf{b}, \mathbf{DA} = \mathbf{a} - \mathbf{b}$$

In $\triangle ABC$, using vector concept, we have

$$\begin{aligned} \mathbf{AB} + \mathbf{BC} &= \mathbf{AC} \\ \Rightarrow \mathbf{AC} &= \mathbf{a} + \mathbf{b} \quad \dots (i) \end{aligned}$$



In $\triangle ADC$,

$$\begin{aligned} AD + DC &= AC \\ \Rightarrow DC &= AC - AD \\ &= a + b - (b - a) = 2a \end{aligned}$$

In $\triangle MDC$,

$$\begin{aligned} MD + DC &= MC \\ \Rightarrow MD &= \frac{b}{2} - 2a \end{aligned}$$

[$\because M$ is mid-point of BC]

In $\triangle ADX$,

$$\begin{aligned} \mathbf{AD} + \mathbf{DX} &= \mathbf{AX} \\ \mathbf{AD} + \frac{4}{5}\mathbf{DM} &= \mathbf{AX} \\ \Rightarrow \mathbf{AX} &= \mathbf{b} - \mathbf{a} + \frac{4}{5}\left(2\mathbf{a} - \frac{\mathbf{b}}{2}\right) \\ &= \mathbf{b} - \mathbf{a} + \frac{8}{5}\mathbf{a} - \frac{2}{5}\mathbf{b} = \frac{3}{5}(\mathbf{a} + \mathbf{b}) \\ \Rightarrow \mathbf{AX} &= \frac{3}{5}\mathbf{AC} \quad [\text{from Eq. (i)}] \end{aligned}$$

which says that $\mathbf{AX} \parallel \mathbf{AC}$ and one point A is common there.

Therefore, A , X and C are collinear.

Question 86

The vectors $3\mathbf{a} - 5\mathbf{b}$ and $2\mathbf{a} + \mathbf{b}$ are mutually perpendicular and the vectors $\mathbf{a} + 4\mathbf{b}$ and $-\mathbf{a} + \mathbf{b}$ are also mutually perpendicular, then the acute angle between $\mathbf{a}$ and $\mathbf{b}$ is

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Options:

A. $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

B. $\cos^{-1}\left(\frac{9}{5\sqrt{43}}\right)$

C. $\pi - \cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

D. $\pi - \cos^{-1}\left(\frac{9}{5\sqrt{43}}\right)$

Answer: A

Solution:

$$(3a - 5b) \perp (2a + b)$$

$$\Rightarrow (3a - 5b)(2a + b) = 0$$

$$\Rightarrow 6a \cdot a + 3a \cdot b - 10b \cdot a - 5b \cdot b = 0$$

$$\Rightarrow 6|a|^2 + 3a \cdot b - 10a \cdot b - 5|b|^2 = 0$$

$$\Rightarrow 6|a|^2 - 5|b|^2 = 7a \cdot b \quad \dots (i)$$

$$\text{Also, } (a + 4b) \perp (-a + b)$$

$$\Rightarrow (a + 4b) \cdot (-a + b) = 0$$

$$\Rightarrow -a \cdot a + a \cdot b - 4b \cdot a + 4b \cdot b = 0$$

$$\Rightarrow -|a|^2 + a \cdot b - 4a \cdot b + 4|b|^2 = 0$$

$$\Rightarrow -|a|^2 + 4|b|^2 = 3a \cdot b \quad \dots (ii)$$

$$\frac{1}{7}(6|a|^2 - 5|b|^2) = \frac{1}{3}(-|a|^2 + 4|b|^2)$$

$$\Rightarrow 18|a|^2 - 15|b|^2 = -7|a|^2 + 28|b|^2$$

$$\Rightarrow 25|a|^2 = 43|b|^2$$

$$\therefore 3a \cdot b = -|a|^2 + 4|b|^2 \quad [\text{from Eq. (ii)}]$$

$$3|a||b| \cos \theta = -\frac{43}{25}|b|^2 + 4|b|^2$$

$$\text{From Eqs. (i) and (ii), we have } \Rightarrow 3|a||b| \cos \theta = \frac{57}{25}|b|^2$$

$$\Rightarrow 3\sqrt{\frac{43}{25}}|b| \cos \theta = \frac{57}{25}|b|^2$$

$$\Rightarrow \frac{\sqrt{43}}{5} \cos \theta = \frac{19}{25} \Rightarrow \cos \theta = \frac{19}{5\sqrt{43}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$$

Question 87

Let $\mathbf{a} = x\hat{i} + y\hat{j} + z\hat{k}$ and $x = 2y$. If $|\mathbf{a}| = 5\sqrt{2}$ and $\mathbf{a}$ makes an angle of 135° with the Z-axis, then $\mathbf{a} =$

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Options:

A. $2\sqrt{3}\hat{i} + \sqrt{3}\hat{j} - 3\hat{k}$

B. $2\sqrt{6}\hat{i} + \sqrt{6}\hat{j} - 6\hat{k}$

C. $2\sqrt{5}\hat{i} + \sqrt{5}\hat{j} - 5\hat{k}$

D. $2\sqrt{5}\hat{i} + \sqrt{5}\hat{j} + 5\hat{k}$

Answer: C

Solution:

$$\mathbf{a} = x\hat{i} + y\hat{j} + z\hat{k},$$

$$x = 2y \text{ and } |\mathbf{a}| = 5\sqrt{2}$$

$$\therefore x = |a| \cos \alpha$$

$$\Rightarrow x = 5\sqrt{2} \cos \alpha,$$

$$y = 5\sqrt{2} \cos \beta$$

$$\text{and } z = 5\sqrt{2} \cos \gamma = 5\sqrt{2} \cos (135^\circ)$$

$$= 5\sqrt{2} \left(-\frac{1}{\sqrt{2}} \right) = -5$$

$$\text{Now, } x = 2y$$

$$\text{Now, } x = 2y$$

$$\Rightarrow 5\sqrt{2} \cos \alpha = 2 \times 5\sqrt{2} \cos \beta$$

$$\Rightarrow \cos \alpha = 2 \cos \beta$$

$$\text{As, } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\Rightarrow 4 \cos^2 \beta + \cos^2 \beta + \frac{1}{2} = 1$$

$$\Rightarrow \cos^2 \beta = \frac{1}{10}$$

$$\Rightarrow \cos \beta = \frac{1}{\sqrt{10}}$$

$$\therefore x = 2y = 2(5\sqrt{2} \cos \beta) = 2 \left(5\sqrt{2} \times \frac{1}{\sqrt{10}} \right) = 2\sqrt{5}$$

$$\therefore y = \frac{x}{2} = \sqrt{5}$$

$$\therefore a = 2\sqrt{5}\hat{i} + \sqrt{5}\hat{j} - 5\hat{k}$$

Question88

Let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be the position vectors of the vertices of a $\triangle ABC$. Through the vertices, lines are drawn parallel to the sides to form the $\triangle A'B'C'$. Then, the centroid of $\triangle A'B'C'$ is

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Options:

A. $\frac{a+b+c}{9}$

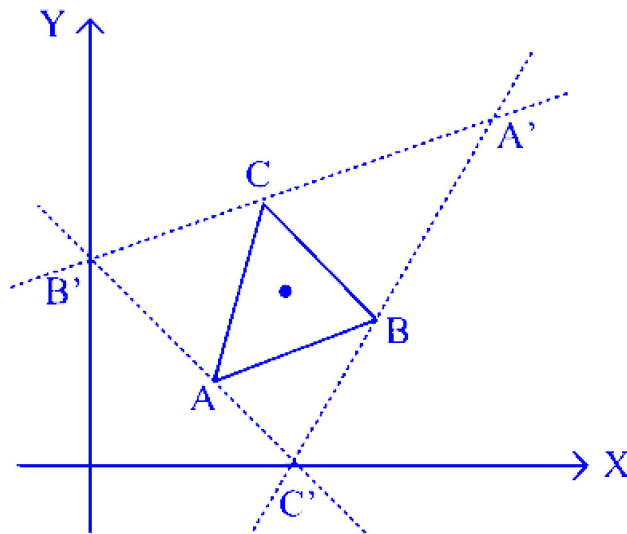
B. $\frac{a+b+c}{6}$

C. $\frac{a+b+c}{3}$

D. $\frac{2(a+b+c)}{3}$

Answer: C

Solution:



Position vectors of the vertices of the $\triangle ABC$ are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

$\therefore$ Centroid of $\triangle ABC$ is $\frac{\mathbf{a}+\mathbf{b}+\mathbf{c}}{3}$

We know, when through the vertices, lines are drawn parallel to the sides to form a new triangle, then obtained such triangle will have the same centroid as the original triangle, which is also clear from the above graphical figure.

$\therefore$ Centroid of $\triangle A'B'C'$ is $\frac{\mathbf{a}+\mathbf{b}+\mathbf{c}}{3}$.

Question89

If $\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and $\mathbf{c} = x\hat{\mathbf{i}} + (x - 2)\hat{\mathbf{j}} - \hat{\mathbf{k}}$ and if the vector $\mathbf{c}$ lies in the plane of vectors $\mathbf{a}$ and $\mathbf{b}$ and then x equals

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Options:

A. 0

B. 1

C. 2

D. -2

Answer: D

Solution:

$$\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}, \mathbf{b} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

$$\mathbf{c} = x\hat{\mathbf{i}} + (x - 2)\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

Given, $\mathbf{c}$ lies in the plane of $\mathbf{a}$ and $\mathbf{b}$.

Then, $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = 0$

$$\Rightarrow [\mathbf{a}, \mathbf{b}, \mathbf{c}] = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ x & x-2 & -1 \end{vmatrix} = 0$$

$$1(1 - 2(x - 2)) - 1(-1 - 2x) + 1(x - 2 + x) = 0$$

$$\Rightarrow 1 - 2x + 4 + 1 + 2x + x - 2 + x = 0$$

$$\Rightarrow 2x + 4 = 0$$

$$\Rightarrow x = -2$$

Question90

Let $u = 2\hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $v = 3\hat{\mathbf{i}} - 5\hat{\mathbf{j}}$. Consider three points P, Q and R having the position vectors $(\frac{5}{2})\hat{\mathbf{i}} - 2\hat{\mathbf{j}}$; $(\frac{7}{3})\hat{\mathbf{i}} - \hat{\mathbf{j}}$ and $(\frac{9}{4})\hat{\mathbf{i}}$ respectively. Among these, the points in the line passing through u and v are

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Options:

- A. Only P and Q
- B. Only P and R
- C. Only Q and R
- D. All P, Q and R

Answer: A

Solution:

The equation of line passing through u and v is $= \vec{u} + \lambda(\vec{v} - \vec{u})$

$$\begin{aligned} \mathbf{r} &= 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + \lambda(\hat{\mathbf{i}} - 6\hat{\mathbf{j}}) \\ &= (2 + \lambda)\hat{\mathbf{i}} + (1 - 6\lambda)\hat{\mathbf{j}} \end{aligned}$$

For the point P ,

$$\frac{5}{2}\hat{\mathbf{i}} - 2\hat{\mathbf{j}} = (2 + \lambda)\hat{\mathbf{i}} + (1 - 6\lambda)\hat{\mathbf{j}}$$

$$\Rightarrow 2 + \lambda = \frac{5}{2} \Rightarrow \lambda = \frac{1}{2}$$

$$\text{and } 1 - 6\lambda = 1 - 6 \times \frac{1}{2} = -2$$

Thus, point P lie in the line through u and v .

For the point Q

$$2 + \lambda = 7/3 \Rightarrow \lambda = 1/3$$

$$\text{and } 1 - 6\lambda = -1$$

Thus, point Q also lie in the line passes through u and v .

Question91

The point of intersection of the lines joining points $\hat{i} + 2\hat{j}$, $2\hat{i} - \hat{j}$ and $-\hat{i}$, $2\hat{i}$ is

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Options:

A. $\frac{5}{3}\hat{i}$

B. $\frac{3\hat{i}+\hat{j}}{5}$

C. $\frac{-3}{5}\hat{i}$

D. $\frac{2}{5}\hat{j}$

Answer: A

Solution:

Points $(\hat{i} + 2\hat{j})$, $(2\hat{i} - \hat{j})$ i.e. $(1, 2)$ and $(2, -1)$.

Line joining points $(1, 2)$ and $(2, -1)$ is

$$y - 2 = \frac{-1-2}{2-1}(x - 1)$$

or $y - 2 = -3x + 3$

$$y + 3x = 5 \dots (i)$$

Points $(-\hat{i})$, $(2\hat{i})$ i.e. $(-1, 0)$, $(2, 0)$

Line joining points $(-1, 0)$, $(2, 0)$ is

$$y - 0 = \frac{0-0}{2+1}(x + 1)$$

$$\Rightarrow y = 0 \dots (ii)$$

Solve Eqs. (i) and (ii) for x and y , we get

$$y = 0, x = 5/3$$

$\therefore$ Point of intersection $(\frac{5}{3}, 0)$,

$$\text{i.e. } \frac{5}{3}\hat{i} + 0\hat{j} = \frac{5}{3}\hat{i}$$

Question92

The value of $\frac{(\mathbf{a} \times \mathbf{b})^2 + (\mathbf{a} \cdot \mathbf{b})^2}{2(\mathbf{a})^2(\mathbf{b})^2}$ is

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Options:

- A. 0
- B. 1
- C. $\frac{1}{2}$
- D. $\frac{1}{4}$

Answer: C

Solution:

$$\begin{aligned} & \frac{(\mathbf{a} \times \mathbf{b})^2 + (\mathbf{a} \cdot \mathbf{b})^2}{2|\mathbf{a}|^2|\mathbf{b}|^2} \\ &= \frac{(|\mathbf{a}| \cdot |\mathbf{b}| \sin \theta \hat{\mathbf{n}})^2 + (|\mathbf{a}||\mathbf{b}| \cos \theta)^2}{2|\mathbf{a}|^2|\mathbf{b}|^2} \\ &= \frac{|\mathbf{a}|^2|\mathbf{b}|^2 \sin^2 \theta \hat{\mathbf{n}} \cdot \hat{\mathbf{n}} + |\mathbf{a}|^2|\mathbf{b}|^2 \cos^2 \theta}{2|\mathbf{a}|^2|\mathbf{b}|^2} \\ &= \frac{\sin^2 \theta + \cos^2 \theta}{2} = \frac{1}{2} \end{aligned}$$

Question93

Let $\mathbf{a} = \hat{\mathbf{i}} - \hat{\mathbf{j}}$, $\mathbf{b} = \hat{\mathbf{j}} - \hat{\mathbf{k}}$ and $\mathbf{c} = \hat{\mathbf{k}} - \hat{\mathbf{i}}$ if $\mathbf{d}$ is a unit vector such $\mathbf{a} \cdot \mathbf{b} = 0 = [\mathbf{bcd}]$, then $\mathbf{d}$ is

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Options:

- A. $\pm \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$
- B. $\pm \frac{\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{6}}$
- C. $\pm \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$
- D. $\pm \frac{\hat{i} + \hat{j} + 2\hat{k}}{\sqrt{6}}$

Answer: B

Solution:

$$\mathbf{a} = \hat{\mathbf{i}} - \hat{\mathbf{j}}, \mathbf{b} = \hat{\mathbf{j}} - \hat{\mathbf{k}}, \mathbf{c} = \hat{\mathbf{k}} - \hat{\mathbf{i}}$$

$$\text{Let } \mathbf{d} = p\hat{\mathbf{i}} + q\hat{\mathbf{j}} + r\hat{\mathbf{k}}$$

$$\text{Given, } \mathbf{a} \cdot \mathbf{d} = 0$$

$$\Rightarrow (\hat{\mathbf{i}} - \hat{\mathbf{j}}) \cdot (p\hat{\mathbf{i}} + q\hat{\mathbf{j}} + r\hat{\mathbf{k}}) = 0$$

$$\Rightarrow p - q = 0 \Rightarrow p = q$$

$$\text{Given, } [\mathbf{bcd}] = 0$$

$$\Rightarrow \begin{vmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ p & p & r \end{vmatrix} = 0$$

$$\Rightarrow (-1)(-r - p) - 1(-p) = 0$$

$$\Rightarrow r + p + p = 0$$

$$\Rightarrow r = -2p$$

$$\therefore \mathbf{d} = p\hat{\mathbf{i}} + p\hat{\mathbf{j}} - 2p\hat{\mathbf{k}}$$

$$\text{Also, } |\mathbf{d}| = 1$$

$$\Rightarrow \sqrt{p^2 + p^2 + (-2p)^2} = 1$$

$$\Rightarrow 6p^2 = 1$$

$$\Rightarrow p = \pm \frac{1}{\sqrt{6}}$$

$$\therefore \mathbf{d} = \frac{\pm \hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}}{\sqrt{6}}$$

Question94

Let $\mathbf{u}$ and $\mathbf{v}$ be two non-zero vectors in R^3 with the intermediate angle 45° . Then $|\mathbf{u} \times \mathbf{v}|$ is equal to

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Options:

A. $|u||v|$

B. $2|u||v|$

C. $\mathbf{u} \cdot \mathbf{v}$

D. $|u| + |v|$

Answer: C

Solution:

$$\begin{aligned} |\mathbf{u} \times \mathbf{v}| &= \|\mathbf{u}\| \cdot \|\mathbf{v}\| \sin \theta_{\mathbf{u}, \mathbf{v}} \\ &= \|\mathbf{u}\| \cdot \|\mathbf{v}\| \sin 45^\circ \\ &= \|\mathbf{u}\| \cdot \|\mathbf{v}\| \cdot \frac{1}{\sqrt{2}} \\ &= \|\mathbf{u}\| \cdot \|\mathbf{v}\| \cos 45^\circ = \mathbf{u} \cdot \mathbf{v} \end{aligned}$$

Question95

Given, $\mathbf{a} = 3\hat{\mathbf{i}} - \hat{\mathbf{j}}$, $\mathbf{b} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ and $\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$ where $\mathbf{b}_1$ is parallel to $\mathbf{a}$ and $\mathbf{b}_2$ is perpendicular to $\mathbf{a}$. Then, $\mathbf{b}_2$ is equal to

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Options:

A. $\frac{1}{2}\hat{\mathbf{i}} + \frac{3}{2}\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

B. $\frac{1}{2}\hat{\mathbf{i}} - \frac{3}{2}\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$

C. $\frac{1}{2}\hat{\mathbf{i}} + \frac{3}{2}\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$

D. $\frac{1}{2}\hat{\mathbf{i}} - \frac{3}{2}\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

Answer: A

Solution:

$$\mathbf{a} = 3\hat{\mathbf{i}} - \hat{\mathbf{j}}, \mathbf{b} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 3\hat{\mathbf{k}}$$

$$\text{Let } \mathbf{b}_1 = k(3\hat{\mathbf{i}} - \hat{\mathbf{j}}), \mathbf{b}_2 = p\hat{\mathbf{i}} + q\hat{\mathbf{j}} + r\hat{\mathbf{k}}$$

$$\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$$

$$2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 3\hat{\mathbf{k}} = k(3\hat{\mathbf{i}} - \hat{\mathbf{j}}) + (p\hat{\mathbf{i}} + q\hat{\mathbf{j}} + r\hat{\mathbf{k}})$$

$$2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 3\hat{\mathbf{k}} = (3k + p)\hat{\mathbf{i}} + (-k + q)\hat{\mathbf{j}} + r\hat{\mathbf{k}}$$

$$\Rightarrow 3k + p = 2, -k + q = 1$$

$$\text{and } r = -3 \quad \dots \quad (i)$$

$$\text{Given that, } \mathbf{a} \cdot \mathbf{b}_2 = 0$$

$$\Rightarrow 3p - q = 0 \quad \dots \quad (ii)$$

Using Eq. (ii) in Eq. (i) $3p = q$, we obtain

$$3k + p = 2$$

$$3p - k = 1$$

Solve for k and p , we get

$$k = \frac{1}{2}, p = \frac{1}{2}$$

$$\therefore q = 3p \Rightarrow q = \frac{3}{2}$$

$$\mathbf{b}_2 = \frac{1}{2}\hat{\mathbf{i}} + \frac{3}{2}\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$$

Question96

The position vectors of the points A and B with respect to O are $2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$. The length of the internal bisector of $\angle BOA$ of $\triangle AOB$ is (take proportionality constant is 2)

Options:

A. $\frac{\sqrt{136}}{9}$

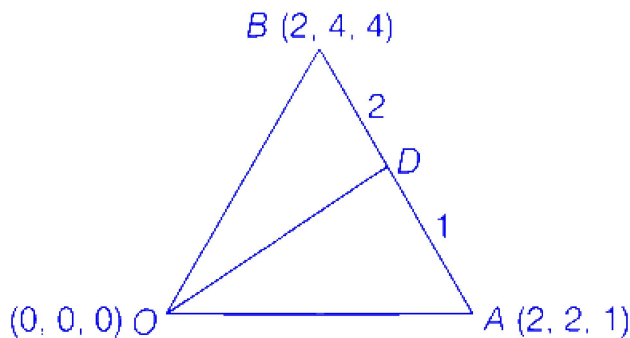
B. $\frac{\sqrt{136}}{3}$

C. $\frac{20}{3}$

D. $\frac{25}{3}$

Answer: B

Solution:



$$AO = \sqrt{2^2 + 2^2 + 1^2} = 3$$

$$OB = \sqrt{2^2 + 4^2 + 4^2} = 6$$

$$\frac{AO}{BO} = \frac{AD}{BD} = \frac{3}{6} \quad [\text{Angle bisector theorem}]$$

$$D = \frac{2(2, 2, 1) + 1(2, 4, 4)}{2 + 1} = \frac{(6, 8, 6)}{3}$$

$$= \left(2, \frac{8}{3}, 2\right)$$

$$(OD) = \sqrt{2^2 + \left(\frac{8}{3}\right)^2 + 2^2} = \frac{\sqrt{136}}{3}$$

Question97

Let $\mathbf{u} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{v} = -3\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ and $\mathbf{w} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + 4\hat{\mathbf{k}}$. Then which of the following statement is true?

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Options:

A. u is perpendicular to v but not w

- B. v is perpendicular to w but not u
- C. w is perpendicular to u but not v
- D. u is perpendicular to both v and w

Answer: A

Solution:

Given,

$$\mathbf{u} = \langle 2, 3, 1 \rangle$$

$$\mathbf{v} = \langle -3, 2, 0 \rangle$$

$$\mathbf{w} = \langle 1, -1, 4 \rangle$$

$$\mathbf{u} \cdot \mathbf{v} = -6 + 6 + 0 = 0$$

$$\mathbf{u} \cdot \mathbf{w} = 2 - 3 + 4 = 3 \neq 0$$

$\mathbf{u} \perp \mathbf{v}$ but $\mathbf{u}$ is not perpendicular to $\mathbf{w}$.

Question98

If $\mathbf{a} = (1, 1, 0)$ and $\mathbf{b} = (1, 1, 1)$, then unit vector in the plane of $\mathbf{a}$ and $\mathbf{b}$ and perpendicular to $\mathbf{a}$ is

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Options:

- A. $(0, 1, 0)$
- B. $(1, -1, 0)$
- C. $\mathbf{k}$
- D. $(1, 0, 1)$

Answer: C

Solution:

$$\mathbf{a} = (1, 1, 0),$$

$$\mathbf{b} = (1, 1, 1) \Rightarrow 0 = (\mathbf{a} + \lambda \mathbf{b}) \cdot \mathbf{a}$$

$$\Rightarrow \mathbf{a} \cdot \mathbf{a} + \lambda \mathbf{a} \cdot \mathbf{b} = 0$$

$$\Rightarrow (1 + 1 + 0) + \lambda(1 + 1 + 0) = 0$$

$$\Rightarrow \lambda = -1$$

$$\mathbf{c} = \mathbf{a} + \lambda \mathbf{b} = (0, 0, 1) = \mathbf{k}$$

Question99

Let $\mathbf{a} = \hat{\mathbf{i}}$ and $\mathbf{b} = \hat{\mathbf{j}}$, the point of intersection of the lines $\mathbf{r} \times \mathbf{a} = \mathbf{b} \times \mathbf{a}$ and $\mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$ is

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Options:

A. $\mathbf{r} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$

B. $\mathbf{r} = \hat{\mathbf{i}} - \hat{\mathbf{j}}$

C. $\mathbf{r} = \hat{\mathbf{k}}$

D. $\mathbf{r} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}}$

Answer: A

Solution:

$$\mathbf{a} = \hat{\mathbf{i}}, \mathbf{b} = \hat{\mathbf{j}}$$

$$(\mathbf{r} \times \mathbf{a}) = (\mathbf{b} \times \mathbf{a}) \Rightarrow (\mathbf{r} \times \mathbf{a}) - (\mathbf{b} \times \mathbf{a}) = 0$$

$$(\mathbf{r} - \mathbf{b}) \times \mathbf{a} = 0 \Rightarrow \mathbf{r} - \mathbf{b} = \lambda \mathbf{a}$$

$$\mathbf{r} = \mathbf{b} + \lambda \mathbf{a}$$

$$= \lambda \hat{\mathbf{i}} + \hat{\mathbf{j}} \quad \dots\dots\dots (i)$$

$$\mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow (\mathbf{r} - \mathbf{a}) \times \mathbf{b} = 0$$

$$\mathbf{r} = \mathbf{a} + \alpha \mathbf{b}$$

$$= \hat{\mathbf{i}} + \alpha \hat{\mathbf{j}} \quad \dots\dots\dots (ii)$$

Comparing Eqs. (i) and (ii), we get

$$\lambda \hat{\mathbf{i}} + \hat{\mathbf{j}} = \hat{\mathbf{i}} + \alpha \hat{\mathbf{j}} \Rightarrow \lambda = 1, \alpha = 1$$

$$\mathbf{r} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$$

Question100

Which of the following vector is equally inclined with the coordinate axes?

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Options:

A. $\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$

B. $2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$

C. $3\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

D. $4\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$

Answer: D

Solution:

In option (d) it is clearly that dr' s of $4\hat{i} + 4\hat{j} + 4\hat{k}$ is 4, 4, 4

Which makes equal angles with coordinate axes.

Question101

If $\hat{i} + 4\hat{j} + 3\hat{k}$, $\hat{i} + 2\hat{j} + 3\hat{k}$, and $3\hat{i} + 2\hat{j} + \hat{k}$ are position vectors of A , B and C respectively and if D and E are mid points of sides BC and AC , then DE is equal to

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Options:

A. $\hat{i} + \hat{j} + \hat{k}$

B. $\hat{i} + \hat{j}$

C. $\hat{j}$

D. $\hat{j} + \hat{k}$

Answer: C

Solution:

Position vector of $A = \mathbf{OA} = \hat{i} + 4\hat{j} + 3\hat{k}$

Position vector of $B = \mathbf{OB} = \hat{i} + 2\hat{j} + 3\hat{k}$

Position vector of $C = \mathbf{OC} = 3\hat{i} + 2\hat{j} + \hat{k}$

Position vector of $D = \mathbf{OD} = \text{mid point of } \mathbf{OB} \text{ and } \mathbf{OC} = \frac{\mathbf{OB} + \mathbf{OC}}{2}$

$= 2\hat{i} + 2\hat{j} + 2\hat{k}$

Position vector of $E = \mathbf{OE} = \text{mid point of } \mathbf{OA}$

$$\begin{aligned}\mathbf{OC} &= \frac{\mathbf{OA} + \mathbf{OC}}{2} \\ \text{and} \quad &= 2\hat{i} + 3\hat{j} + 2\hat{k} \\ \mathbf{DE} &= \mathbf{OE} - \mathbf{OD} = \hat{j}\end{aligned}$$

Question102

If $\mathbf{a}$ and $\mathbf{b}$ are two vectors such that $\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} < 0$ and $|\mathbf{a} \cdot \mathbf{b}| = |\mathbf{a} \times \mathbf{b}|$ then the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$ is

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Options:

- A. $\frac{\pi}{4}$
- B. $\sec^{-1}(-\sqrt{2})$
- C. $\tan^{-1}\left(\frac{-1}{2}\right)$
- D. $\sin^{-1}\left(\frac{1}{2}\right)$

Answer: B

Solution:

$$\because \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} < 0$$
$$\Rightarrow \mathbf{a} \cdot \mathbf{b} < 0$$

Let θ be the angle between $\mathbf{a}$ and $\mathbf{b}$, then

$$\begin{aligned} \Rightarrow \quad |\mathbf{a} \cdot \mathbf{b}| &= |\mathbf{a} \times \mathbf{b}| \\ \Rightarrow \quad -|\mathbf{a}| \cdot |\mathbf{b}| \cos \theta &= |\mathbf{a}| |\mathbf{b}| \sin \theta \\ \Rightarrow \quad \tan \theta &= -1 \\ \Rightarrow \quad \sec \theta &= -\sqrt{2} \\ \Rightarrow \quad \theta &= \sec^{-1}(-\sqrt{2}) \end{aligned}$$

Question 103

Let $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be three-unit vectors and $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c} = 0$. If the angle between $\mathbf{b}$ and $\mathbf{c}$ is $\frac{\pi}{3}$. Then $[\mathbf{abc}]^2$ is equal to

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Options:

- A. $\frac{3}{2}$
- B. $\frac{3}{4}$
- C. $\frac{2}{3}$
- D. $\frac{4}{3}$

Answer: B

Solution:

$\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are unit vectors

$$\Rightarrow |\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = 1$$

$$\because \mathbf{a} \cdot \mathbf{b} = 0 \text{ and } \mathbf{a} \cdot \mathbf{c} = 0$$

$$\Rightarrow \mathbf{a} \perp \mathbf{b} \text{ and } \mathbf{a} \perp \mathbf{c}$$

Angle between $\mathbf{b}$ and $\mathbf{c}$ is $\frac{\pi}{3}$

$$\therefore \mathbf{b} \times \mathbf{c} = |\mathbf{b}| \cdot |\mathbf{c}| \sin \frac{\pi}{3} \cdot \hat{\mathbf{a}}$$

[$\because \mathbf{a}$ is perpendicular to $\mathbf{b}$ and $\mathbf{c}$ and $\hat{\mathbf{a}} = 1$]

$$= 1 \cdot 1 \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

$$\text{Now, } [\mathbf{abc}]^2 = [\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})]^2$$

$$= \left[\mathbf{a} \cdot \frac{\sqrt{3}}{2} \right]^2 = |\mathbf{a}|^2 \cdot \left(\frac{\sqrt{3}}{2} \right)^2$$

$$= 1 \cdot \frac{3}{4} = \frac{3}{4}$$

Question104

Let x and y are real numbers. If $\mathbf{a} = (\sin x)\hat{\mathbf{i}} + (\sin y)\hat{\mathbf{j}}$ and $\mathbf{b} = (\cos x)\hat{\mathbf{i}} + (\cos y)\hat{\mathbf{j}}$, then $|\mathbf{a} \times \mathbf{b}|$ is

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Options:

A. 0

B. greater than one

C. less than or equal to 1

D. less than 1

Answer: C

Solution:

$$\mathbf{a} = (\sin x)\hat{\mathbf{i}} + (\sin y)\hat{\mathbf{j}}$$

$$\mathbf{b} = (\cos x)\hat{\mathbf{i}} + (\cos y)\hat{\mathbf{j}}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \sin x & \sin y & 0 \\ \cos x & \cos y & 0 \end{vmatrix}$$

$$= \hat{\mathbf{k}}(\sin x \cos y - \cos x \sin y)$$

$$\mathbf{a} \times \mathbf{b} = \hat{\mathbf{k}} \sin(x - y)$$

$$\text{or } |\mathbf{a} \times \mathbf{b}| = \sin(x - y) \leq 1 \quad [\because -1 < \sin x \leq 1, \forall x]$$

Question105

A vector makes equal angles α with X and Y -axis, and 90° with Z -axis. Then, α is equal to (c) 45° and 135° (d) 90°

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Options:

A. 60° or 120°

B. 30° or 150°

C. 45° or 135°

D. 90°

Answer: C

Solution:

$$l = \cos \alpha$$

$$m = \cos \alpha$$

$$n = \cos 90^\circ = 0$$

$$\therefore l^2 + m^2 + n^2 = 1$$

$$\Rightarrow 2 \cos^2 \alpha = 1$$

$$\Rightarrow \cos^2 \alpha = \frac{1}{2}$$

$$\Rightarrow \cos \alpha = \pm \frac{1}{\sqrt{2}}$$

$$\Rightarrow \alpha = 45^\circ \text{ or } 135^\circ$$

Question 106

Angle made by the position vector of the point $(5, -4, -3)$ with the positive direction of X -axis is

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Options:

A. $\frac{\pi}{2}$

B. $\frac{\pi}{6}$

C. $\frac{\pi}{4}$

D. $\frac{\pi}{3}$

Answer: C

Solution:

Let $A(5, -4, -3)$

Direction ratios of $\mathbf{OA} = 5, -4, -3$

Direction ratios of X -axis $= 1, 0, 0$

$$\begin{aligned}\text{Angle, } \cos \theta &= \frac{5 \cdot 1 + (-4) \cdot 0 + (-3) \cdot 0}{\sqrt{5^2 + (-4)^2 + (-3)^2} \sqrt{1^2 + 0^2 + 0^2}} \\ &= \frac{5}{\sqrt{50}} = \frac{1}{\sqrt{2}} \\ \therefore \theta &= \frac{\pi}{4}\end{aligned}$$

Question 107

If the volume of the parallelopiped formed by the vectors $\hat{i} + a\hat{j} + \hat{k}, \hat{j} + a\hat{k}$ and $a\hat{i} + \hat{k}$ becomes minimum, then a is equal to

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Options:

- A. $\frac{1}{3}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\frac{2}{\sqrt{3}}$
- D. $\frac{2}{3}$

Answer: B

Solution:

Let $\mathbf{p} = \hat{i} + a\hat{j} + \hat{k}$

$\mathbf{q} = \hat{j} + a\hat{k} \Rightarrow \mathbf{r} = a\hat{i} + \hat{k}$

$$\begin{aligned}&= \mathbf{p} \cdot (\mathbf{q} \times \mathbf{r}) \\ V &= \begin{vmatrix} 1 & a & 1 \\ 0 & 1 & a \\ a & 0 & 1 \end{vmatrix} \\ \Rightarrow V &= a^3 - a + 1\end{aligned}$$

Differentiating V with respect to a

$$\frac{dV}{da} = 3a^2 - 1$$

Again, differentiating w.r.t. a

$$\frac{d^2V}{da^2} = 6a$$

For stationary points

$$\frac{dV}{da} = 0$$

$$\Rightarrow 3a^2 - 1 = 0$$

or

$$a^2 = \frac{1}{3} \text{ or } a = \pm \frac{1}{\sqrt{3}}$$

At,

$$a = \frac{1}{\sqrt{3}},$$

$$\frac{d^2V}{da^2} = 6 \cdot \frac{1}{\sqrt{3}} > 0$$

$\therefore$ Volume is minimum at $a = \frac{1}{\sqrt{3}}$

Question108

If $\mathbf{a} = \frac{3}{2}\hat{\mathbf{k}}$ and $\mathbf{b} = \frac{2\hat{\mathbf{i}}+2\hat{\mathbf{j}}-\hat{\mathbf{k}}}{2}$, then angle between $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$ is

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Options:

- A. 45°
- B. 90°
- C. 30°
- D. 60°

Answer: B

Solution:

$$\mathbf{a} = \frac{3}{2}\hat{\mathbf{k}}, \mathbf{b} = \frac{2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}}{2}$$

$$\mathbf{a} + \mathbf{b} = \frac{3}{2}\hat{\mathbf{k}} + \frac{2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}}{2} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{a} - \mathbf{b} = \frac{3}{2}\hat{\mathbf{k}} - \frac{2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}}{2} = -\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

Let θ be the angle between $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$ then

$$\cos \theta = \frac{(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b})}{|\mathbf{a} + \mathbf{b}| |\mathbf{a} - \mathbf{b}|}$$

$$= \frac{1(-1) + 1(-1) + 1(2)}{\sqrt{1^2 + 1^2 + 1^2} \sqrt{(-1)^2 + (-1)^2 + 2^2}} = 0$$

$$\therefore \theta = 90^\circ$$

Question109

Let $\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ and $\mathbf{c} = 7\hat{\mathbf{i}} + 9\hat{\mathbf{j}} + 11\hat{\mathbf{k}}$, then the area of parallelogram having diagonals $\mathbf{a} + \mathbf{b}$ and $\mathbf{b} + \mathbf{c}$ is

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Options:

A. $4\sqrt{6}$ sq units

B. $2\sqrt{6}$ sq units

C. $\sqrt{6}$ sq units

D. $6\sqrt{6}$ sq units

Answer: A

Solution:

$$\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{b} = \hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$$

$$\mathbf{c} = 7\hat{\mathbf{i}} + 9\hat{\mathbf{j}} + 11\hat{\mathbf{k}}$$

Then, $\mathbf{a} + \mathbf{b} = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 6\hat{\mathbf{k}}$ and $\mathbf{b} + \mathbf{c} = 8\hat{\mathbf{i}} + 12\hat{\mathbf{j}} + 16\hat{\mathbf{k}} = 4(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}})$ Area of parallelogram $= \frac{1}{2}|(\mathbf{a} + \mathbf{b}) \times (\mathbf{b} + \mathbf{c})|$

$$= \frac{1}{2} |2(\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}) \times 4(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}})|$$

$$= \frac{8}{2} |(\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}) \times (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}})|$$

$$= 4 \left\| \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{vmatrix} \right\| = 4 |-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}|$$

$$= 4\sqrt{1 + 4 + 1} = 4\sqrt{6} \text{ sq units}$$

Question110

If $\mathbf{a}$ and $\mathbf{b}$ are two vectors such that $|\mathbf{a}| = 2$, $|\mathbf{b}| = 3$ and $\mathbf{a} + t\mathbf{b}$ and $\mathbf{a} - t\mathbf{b}$ are perpendicular, where t is a positive scalar, then

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Options:

A. $t = \pm \frac{2}{3}$

B. $t = \frac{4}{9}$

C. $t = \frac{2}{3}$

D. $t = \frac{2}{9}$

Answer: C

Solution:

$$|\mathbf{a}| = 2, |\mathbf{b}| = 3$$

$$\because (\mathbf{a} + t\mathbf{b}) \perp (\mathbf{a} - t\mathbf{b})$$

$$\Rightarrow (\mathbf{a} + t\mathbf{b}) \cdot (\mathbf{a} - t\mathbf{b}) = 0$$

$$\Rightarrow \mathbf{a} \cdot \mathbf{a} - t(\mathbf{a} \cdot \mathbf{b}) + t(\mathbf{b} \cdot \mathbf{a}) - t^2(\mathbf{b} \cdot \mathbf{b}) = 0$$

$$\Rightarrow |\mathbf{a}|^2 - t^2|\mathbf{b}|^2 = 0$$

$$\Rightarrow t^2 = \frac{|\mathbf{a}|^2}{|\mathbf{b}|^2}$$

$$\Rightarrow t = \frac{|\mathbf{a}|}{|\mathbf{b}|} = \frac{2}{3}$$
